

Bio 110 The Fundamentals of Physiology

Numbered Transcript

BIO 110 • Dr. Benjamin Bikman

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Lecture 1: Cells in Balance

At a glance — Bikman opens on cells in balance — the basics of human physiology.

- 1.1** Hello and welcome to Basics of Human Physiology. I am Professor Ben Bikman, and my goal in this lecture series is to help you understand the wonder of the human body. As a physiologist or cell biologist by training, we need to start at the level of the cell. After all, if you hope to understand the function of the body, you need to understand the fundamental building block. But that is, in fact, part of physiology.
- 1.2** Physiology, as a science, is the study of the system or the whole organism. It starts at the level of the cell and the molecules that make up the cell, and then how these molecules within the cells then constitute through these biological levels an organ or a tissue. Then the tissue has its own system. We're going to go through several systems in the time that we have together. We're going to go through things like the peripheral nervous system or the skeletal muscle system and so on.
- 1.3** Then even then, once we get to one of the systems within the body, I want you to leave this lecture series with an appreciation of how these systems then become part of a greater whole, allowing the body to maintain its normal and expected function. Perhaps underpinning all of this is my hope that you leave with some appreciation of just how remarkable the human body is. Dare I say, it's miraculous how it functions, and it's miraculous that it functions as well as it does. In fact, that might be as much of the miracle as anything, that it goes right as often as it does because these systems are so complex.
- 1.4** There's so much opportunity for things to get wrong that when they go right, which they do the overwhelming amount of time, it really is quite remarkable. This is the system's physiology, going from molecule to cell, from cell to organ, an organ system, and then the collective systems representing the entire body and its function. In the time that we have together, I mentioned the systems that we're going to be covering. We'll start with the cell, and we'll just discuss cell physiology. Then we will move into the peripheral nervous system, so the functioning and the interaction of the nervous system with the body as we think of it.
- 1.5** Then next, we will move into one of the most relevant systems, which is skeletal muscle, the part that is physically animating and moving the body. Then discuss the cardiopulmonary system, that being the interaction of heart and lung because they're both just so essential to each other's function. Then we'll move on to the gastrointestinal system and then wrap up by discussing my favorite, which is the endocrine system, all of the complexity of hormones, why they're made and how they're moving and what they're doing.
- 1.6** Let's start with cell physiology, the fundamental building block of systems physiology and the basics of human physiology. Now, cell physiology itself, unfortunately, is a

pretty complicated topic. Well, also, fortunately, because it makes it a little more fun. We're going to talk about three subtopics within this broader topic of cell physiology, specifically homeostasis. Then we're going to move on to the transport of molecules across the cell membrane, how the cell is receiving things and how a cell is giving things up to the rest of the system.

- 1.7 Then lastly, within this topic of cell physiology, we will discuss electrophysiology. How are signals being moved along cells in a way to animate the body so that the body is doing what it needs to and indeed what we want it to. All right, so let's start with homeostasis. Homeostasis is this idea that the body is going to constantly want to snap back to normal, whatever normal may be. Within this topic of homeostasis, we're going to discuss what homeostasis is and how the physiological systems are trying to always come back to normal, and then also discuss the feedback systems within the body, and then discuss its components, the components of the feedback system.
- 1.8 Now, it is no surprise that the human body is capable of tolerating a variety of environments. We are right now, and when you're watching this later, have the benefit of generally being in a pretty well-controlled, comfortable environment. But we're capable of areas of extreme heat, of extreme cold, of extreme altitude changes. All of these require the body to make some adjustments. For example, body temperature. The body wants to be regulated. It wants to keep its temperature within just a couple degrees Celsius, 36 to 38 degrees Celsius or so.
- 1.9 Now, that becomes a problem if we're in a very hot environment, a hot and humid environment, or if we're in a very cold, dry environment. The body has to make some adjustments in order to keep body temperature at that static level. Also, pH—you're familiar with acids. People will say the body is too acidic or it's too basic. In reality, it is exceptionally finely controlled into a very narrow pH range. The body will make adjustments accordingly in order to maintain an optimal balance of acid and base to ensure proper function.
- 1.10 Then for a scientist like me who studies diabetes, one of the most common health problems, there's also other variables, indeed, countless variables, but one being blood glucose levels. Blood glucose levels, which is something we'll discuss as an example in just a moment with feedback, the body wants to keep it in a very narrow range, that if we go too far outside of that range of blood glucose levels, it can become acutely lethal, as is the case with each of the variables I mentioned and many, many more.
- 1.11 The body wants to maintain a particular set point across myriad variables. Then if it starts to get outside of it, the body will make adjustments in order to bring it back. That is homeostasis. Now, I mentioned that I wanted to use blood glucose as an example. Typically, the body wants to maintain blood glucose levels in a range of about 70 to 110 milligrams per deciliter. Those are the units that we discuss when we talk about blood glucose levels. Now, within the body, that's a milligram per deciliter unit.

- 1.12** But if you look at the total amount of blood glucose, commonly referred to as blood sugar, in the blood, it's actually just a few grams, a teaspoon. Within the entire blood of my body, I have just a teaspoon of the equivalent of blood glucose. A very little amount when it's all said and done. Now, you can imagine how problematic that then becomes if I go and eat a candy bar, which just this one little candy bar could easily have 10 times the amount of sugar I have in my blood at any moment. I want to have one teaspoon, and then all of a sudden, I'm eating or drinking 10 teaspoons of sugar.
- 1.13** Well, if that all hits my bloodstream and stays there, I'm dead. It will kill me. It will not work. That is so far outside of homeostatic or homeostasis that the body has to make some adjustments. If my blood glucose levels start to climb and get higher and higher and higher, if there's not an adjustment, death will follow fairly soon. There must be a way to account, to measure this big, to detect this big rise in blood sugar levels and then have a response that will bring it back down in order to restore homeostasis. In the case of blood sugar, that's going to be done through a signal called insulin.
- 1.14** Insulin, the hormone, will act as this signal to try to take this big spike in blood sugar levels and bring it right back down to where the body wants it. In other words, helping to restore homeostasis. Now, within the broader sense, with this as an example and the other conditions I mentioned earlier, like body temperature changes or pH changes, in every instance, this is negative feedback, and this is a fundamental principle of physiology. Negative feedback is the system where the body can sense a change, then the change will elicit a response. The response will then help restore the body back to where it was.
- 1.15** Let's take, for example, something that you're all familiar with, especially in the heat and humidity of a place like Florida, where we are at the moment. We have thermostats within our homes or apartments. The thermostat is a perfect example of negative feedback. Within the thermostat, we, of course, have a thermometer. Then the thermometer is constantly sampling the temperature of the space. If the temperature starts to drop, the thermometer is detecting it. Then the thermometer will then activate the thermostat, which is controlling or comparing whatever the thermometer is telling it with what we've set it at.
- 1.16** So the thermostat has a set point that it wants to maintain a temperature at. As the thermometer is sensing temperature has gone low, that will tell the thermostat that the temperature has gone low, and the thermostat will turn on the furnace, not something that has to happen in an environment here. But in my native land of Canada, that's something that we rely on very, very regularly. We need that thermostat to be working, to kick on the furnace. The furnace will turn on, and that will begin producing heat, of course.
- 1.17** Then as the temperature in that space begins to increase, the thermometer will sense that, and then the thermometer will tell the thermostat, "Hey, it worked. It's time to turn it off." Thus, the furnace will stop, and then eventually, temperature will start to

drop again, in this case where we're fighting the cold. That would kick the process on again as needed. We had this really elegantly designed system of sensing and then responding and having the reaction be one that would get things back to normal.

1.18 There are three components that are essential within a negative feedback system, the first being the sensor. This is going to be something that is measuring whatever variable we're interested in. It's sensing the variable. In this case, that would be temperature. Then there is a control center, which is going to compare whatever the sensor is saying with what we've told it it should be. We'll move to the body in just a moment. Within the thermostat, that's whatever we set the temperature at. The control center is saying, "We want it to be set here," but the sensor, the thermometer, is telling me that it's here or here.

1.19 Then lastly, there will be the third and final component of the negative feedback system, an effector. The effector is going to be the part of the system that actually makes the response in order to correct, in order to get the variable back to where it was. So, within the thermostat, the sensor, the first part of it, which is constantly sampling whatever the variable is, in this case temperature, that would be the thermometer. Then the control center is going to be the thermostat itself, what we have set the temperature to.

1.20 Then lastly, it was the effector. This would be the furnace or the air conditioner, the system that would kick on to bring the temperature back up or to bring it back down depending on where we were going. Were we moving up or were we moving down? It will return it back to where it was. Now, of course, the human body is not a thermostat. We don't have a furnace and an air conditioner built in. Let's move it back to human physiology. In fact, let's get back to blood glucose.

1.21 So blood glucose was the variable that I mentioned earlier. It's one of the countless variables within the body that the body is trying to keep within a normal range. As I elaborated, it's a pretty narrow range. If the blood sugar levels are getting too high, which, let's face it, we do often, given how we eat these days, and that's a topic for another time, but we're constantly combating these spikes in blood sugar or blood glucose. Then the system will correct. As I mentioned, insulin is going to be part of what corrects this.

1.22 Now, within this, the human body, this is an interesting example, the control of blood sugar, because we actually have all three components of the negative feedback being the sensor, the control center, and then lastly, the effector. It's actually all encompassed within one single entity within the body, and that is the beta cells of the pancreas. So very, very briefly, because this is not a lecture about diabetes, but rather cell physiology and homeostasis, it is the beta cell that is sensing the change in the blood sugar levels. If blood glucose levels are going up, the beta cell—which is, of course, what's happening often—the beta cells sense that change, and they get turned on as that change happens.

- 1.23** So they're also the control center. But it is also the effector, or at least giving rise to the effector, because it's the beta cell of the pancreas that then starts pumping out the insulin to spread through the system. And then insulin will open doors to cells throughout the body to allow the glucose to come into cells, including the beta cell, and thus it will start to lower the blood sugar levels and bring things back to normal or to restore homeostasis.
- 1.24** Within the complexity of physiology, we won't always see three distinct components. We won't always see sensor, control center, and effector as separate. Very often, they're going to have a couple lumped together, or in the case of the beta cell, essentially all of it lumped together. The beta cell senses the change in glucose and compares it to what it should have been, and then releases the insulin in order to correct it or to effect the change at bringing things back.
- 1.25** This is just one example of negative feedback. You are going to see many examples of negative feedback as we go through some of the systems, particularly in the endocrine system, which is what I've been talking about even now, talking about insulin and the regulation of blood sugar. But nevertheless, as we go through the systems, you will see hints and echoes of negative feedback. Now, however, having described negative feedback, this being such a thorough course, I want you to be familiar with a different type of feedback that can happen within biological systems, which is positive feedback.
- 1.26** Whereas negative feedback is trying to constantly turn itself off by correcting whatever the departure from normal has been—if things got too high, negative feedback wants to bring it back low to normal; if things got low, negative feedback wants to bring it back up—all of that is negative feedback. So a system that's trying to turn itself off. Positive feedback, in contrast, is a system or a signal that's trying to ramp itself up, constantly getting more and more and more of whatever the signal is. You can see that this becomes potentially problematic.
- 1.27** We don't want something to be running out of control, but that's kind of what positive feedback is doing. This is very uncommon, and it's always temporary, whenever we have positive feedback, when the system just keeps turning itself on and on and on. If you think about positive feedback, this would be like when you have a microphone and you're bringing it too close to the speaker. It starts feeding into each other and the signal just keeps ramping up. That is positive feedback. You can start to see how that would be a problem within the body. If it's going to happen, it has to happen for a very precise reason, and it has to be able to turn itself off.
- 1.28** Now, one of the most remarkable examples of this is childbirth. During childbirth, we have one of the very few instances of positive feedback. In fact, there are other instances of positive feedback, and most of them happen to be in the female body, just suggesting how complex women really are. Within the male body, we still have some instances, but it's less common. This one being the most famous of all instances or examples of positive feedback, childbirth.

- 1.29** As the baby has been growing and is now reaching the point of maximum dimensions, baby starts to push against the cervix. We're not going to get into the anatomy, so the cervix being the bottom part of the uterus that the baby must pass through as the baby is going to be born. As the cervix starts to stretch, that sends a signal up to a top part of the brain called the hypothalamus. We're going to revisit the hypothalamus when we talk about the endocrine system later. But the hypothalamus is one of the central parts of the brain and this higher function of the brain.
- 1.30** The cervix is stretching as the baby is pressing down at the bottom of the uterus. That sends a signal to the hypothalamus. Then the hypothalamus starts releasing a hormone called oxytocin. You've probably heard of this before. There are synthetic versions of oxytocin that can be injected into a woman to induce the effect of oxytocin or to mimic its effect. That main effect is going to be to tell the uterus to start contracting. The uterus is this big, thickly muscled organ that's holding the baby. So oxytocin, which itself was a result of this getting all stretched, is now pressing on the uterus to tell it to start contracting, all in an effort, of course, to start pushing baby out.
- 1.31** But of course, as baby starts getting pushed out, what happens to the cervix? It, of course, starts to get stretched even more, which tells the signal to turn on even higher. And so this becomes this cycle. I don't want to say vicious because it's a glorious cycle. It becomes this positive feedback cycle where the signal is turning itself on to continue to ramp up. Now, here, we actually are able to see all three distinct parts of the system that I described earlier. Because when we talked about the negative feedback system and its three components, they also apply to positive feedback.
- 1.32** For example, where's the sensor? Well, the sensor is the cervix. As the cervix is getting stretched, it's what's sensing the change. Then it's sending a signal up to the control center, which is the hypothalamus. The hypothalamus is getting this signal and saying, "Hey, this signal is deviating from normal. I'm sensing this through nerves that something has changed." So the hypothalamus is the control center. Then the effector is this combination of the release of oxytocin from the hypothalamus, which is actually a big part of the endocrine system, releasing oxytocin, the hormone, going down to the uterus, telling the uterus to contract.
- 1.33** Here within this instance of positive feedback, we end up seeing all three components of the system—sensor, control center, and effector. All of this still being a reflection of homeostasis, which is the central idea that I wanted to convey to you in this subtopic within the broader topic of cell physiology. Homeostasis being the desire of the system to constantly be snapping back to a normal range. Even in positive feedback, the end result is going to be that things get back to normal, which is the baby is born, the cervix stops stretching. Even in positive feedback, it's still trying to serve a particular purpose, albeit an exceptional and extraordinary one.
- 1.34** All right, with homeostasis in mind, and again, this is a concept you're going to see reflections of in the future lectures, let's move on to membrane transport. A cell is a

well-enclosed entity in its own right. It has, in fact, this double wall of protection called a lipid bilayer. And lipid means fat. And in fact, the cell membrane is loaded with different types of fats. One reason why we shouldn't fear dietary fats, this is an essential component of literally every single cell of the body. But it is such a well-enveloped structure that it is regulated. Whatever is coming in or out is very well and finely tuned and controlled.

- 1.35** That's what brings us to the membrane transport. In order to appreciate the system's physiology that we'll get to, we need to appreciate the ability of a cell to control what's coming in or out. The learning objectives for membrane transport, I want you to appreciate and understand the difference between active and passive transport. Those are two methods that the cell is moving molecules in and out. I want you to appreciate the difference between primary and secondary active transport. When we talk about active transport, there's two separate components to that.
- 1.36** Then lastly, I want you to appreciate one of the most important cell membrane entities, which is this thing called the sodium-potassium pump. We're going to start getting really, really particular very, very quickly. This is just because the membrane transport is such a fundamentally important concept to understand the cell and then the systems that are built on top of that idea. With membrane transport, a first and primary concept is to appreciate that there are different levels of molecules inside and outside of the cell, that there's never a balance between them. In fact, balance is death.
- 1.37** There must be these differences that help the cell maintain all of its dynamic functions. I want to highlight two molecules in particular that are commonly referred to as solutes. If I use the word solute, all it means is that it's something that's been dissolved in a solution. Speaking of dissolved, we're talking about sodium and potassium. These are also known as electrolytes for reasons that will become very obvious very, very quickly. You've heard the term electrolytes. These are simply these positively or negatively charged molecules that end up creating a literal electrical signal across a cell. We'll revisit that in just a moment within this topic of cell physiology.
- 1.38** Now, with potassium and sodium in mind, there are very different levels inside and outside the cell. Again, that is by necessity. They must be very different in order to maintain function, and we'll get into that. Outside of the cell, there is a lot of sodium. In fact, there is 10 times more sodium outside the cell than there is inside the cell. If we look at this first electrolyte, sodium, and I'm picking sodium and potassium on purpose because they are so essential and they always are coordinating each other, they have a bit of a love-hate relationship.
- 1.39** There's 10 times more sodium outside the cell, what's called the extracellular fluid or the ECF, than there is within the cell, also known as the intracellular fluid, ICF. So ECF, extra- meaning outside of, 10 times higher than ICF, intra-, inside the cell. Now, that is in stark and even more dramatically, contrast to what we see with potassium, where potassium has 30 times more level, a greater level of concentration in the cell, in the

ICF, than it does in the ECF. So a substantially different ratio here. Again, not only in opposite of sodium, but even more dramatic, 30 times more.

- 1.40** And so the cell is constantly trying to balance, but to keep this—I say balance, but it's really trying to keep the imbalance. Because remember, if it ever got to a point where it was static, the cell stops to function. It must have this conflict, if you will, of these molecules warring against each other. An essential idea to understand with membrane transport is the molecular driving forces. Essentially, where we have molecules that are higher concentration in one space than another, it wants to roll downhill.
- 1.41** So if we have much more sodium outside the cell than we have inside the cell, which direction do you think the sodium wants to move? It wants to roll downhill. These electrolytes are like so many things. We're going to revisit this idea in more detail in future topics. They want to go down their concentration gradient. So wherever we have more sodium, in this case outside the cell than inside, the sodium is constantly wanting to push itself in. In contrast, of course, and even more dramatically because of the ratio, we have so much more potassium inside the cell, remember, 30 times more, that it's constantly trying to leak out of the cell.
- 1.42** To varying modest degrees, it is leaking a little. We have to have these pumps, which we'll revisit in a moment, that are constantly working to move things back into the balance that I've just described, because that is the homeostatic level. But as you're thinking about these, in this case, they're both positively charged ions; both potassium and sodium are plus or positive charged, but they do have this slight charge. And as they are moving across the membrane, along with many, many other molecules, it starts to create, not only do we have the concentration gradient, this difference in concentration from inside to outside the cell, but that begins—as they're moving, it creates an electrical gradient.
- 1.43** That is why these are referred to as electrolytes. As they are moving across the membrane, they are literally giving the membrane a charge. We'll revisit that when we talk about electrophysiology in just a few minutes within this broader cell physiology topic. But there is, in fact, an electrical charge because of the movement of these chemicals, these molecules. Collectively, we refer to this entire process as an electrical chemical gradient. This is the electrochemical gradient, an electrical gradient that is derived from the movement of these positively charged molecules, sometimes referred to as ions, but we can just call them electrolytes.
- 1.44** They have to be moving. We're constantly moving. The cell must be pulling things in and dumping things out or releasing things. Then with this idea in mind, the molecular driving force, you remembering that the ions always want to be moving down their concentration gradient, from a high concentration to low, we can then move into the idea of transport. Remember, there are two different methods of transport across the cell membrane. There's passive, which is the lazy version, and then there's active, which is when the cell is working to get it done.

- 1.45 But all transport of molecules requires some degree of energy. At some point, whether the energy had to have—the cell had to use the energy earlier or whether it's using it in the moment, there must have been some form of cellular energy in order to actually make this movement occur, this transportation of the molecules. So the passive form of transport, so the passive transport is when the cell is not using energy in the moment. It's generally taking advantage of a concentration gradient.
- 1.46 As I mentioned earlier, where you have a higher level of a molecule, say outside the cell than in, it's generally going to be just leaking across at some modest amount. That is passive. There's no energy required. Now, you might have needed energy, and I'll talk more about this in a moment, to create that imbalance. But once you've created the imbalance, everything thereafter is just a stone rolling down the hill. You needed energy to push the stone up or to concentrate one part of the body outside the cell versus in. But passive transport is when it's just now, it can do it on its own. It doesn't need any energy. It is like the stone rolling down the hill.
- 1.47 Active transport is the exact opposite. You must be using energy. Then there are going to be particular pumps that are getting powered in order to be pushing the molecules around outside or inside the cell. Let's talk about each of these two systems, each of these two processes in more detail, passive and active transport.
- 1.48 Within passive transport, there's two different methods that are used: simple diffusion and facilitated diffusion. Even the word diffusion, I'm hoping, evokes this idea of a passive, lazy process. It's just leaking out. It's just happening. Simple diffusion and facilitated diffusion. So with simple diffusion, there's an additional two concepts in mind, which is the just movement, the leaking across the lipid bilayer, the two little lipid layers that constitute the collective envelope of the cell or the membrane. Then there is channels, specific little pores that are embedded across the bilayers. Imagine that I had two balloons and now we're poking something through, not poking to pop, hopefully, but in this case, just a little pore that is penetrating through both layers of the cell.
- 1.49 Now, the first idea being the passive diffusion through the lipid bilayer, that two-level envelope around the cell, we have various variables that will play into it, particularly the magnitude of the force. In other words, what's the concentration that's trying to push it in? Also, the surface area of the cell, and then the membrane permeability. Firstly, the magnitude of the driving force. If we have a cell and there are two different molecules that are both higher in concentration outside the cell than in, but molecule A is 10 times higher than molecule B, but both of which are higher outside the cell than in, you could see that one of them, A, is going to want to push itself in much, much more than B because it just has so much more.
- 1.50 It's basically going back to the metaphor of the rock or the stone at the top of the mountain. It's, one has a very, very steep slope and one has a much more gradual or shallow slope. This is what I mean by the magnitude of the driving force. What is the

slope of the hill that the rock is going to roll down? Or in this case, what is the concentration difference? What is the delta, the disparity? If we have much, much more of chemical A or electrolyte A than we do of B, they're both going to want to be pushing into the cell, but A is going to do it much faster because its driving force is just that much higher. The concentration ends up creating this driving force, which will influence the simple diffusion.

- 1.51** Now, the next one is the surface area. If we have a broader or bigger cell, and the total amount of surface area of the cell is much, much higher because of the growth of the cell, it stands to reason that it is capable of bringing more in more quickly. The more of a surface area we have of a cell, the faster the simple diffusion of the molecule can happen. This is one of the reasons why, if someone has bigger muscles because they work out, and we'll talk about the muscle system later, then it actually becomes easier for the body to regulate blood sugar levels, back to an earlier example.
- 1.52** Bigger muscles are capable of pulling in glucose more rapidly. If for no other reason, then there's just much more of the membrane. So the glucose has more entry points than it would in a much smaller muscle. And indeed, we do see that. One of the best ways to help regulate blood sugar is to increase muscle mass. OK, so the surface area of the cell will also impact the simple diffusion and the rate at which the cell can pull molecules in.
- 1.53** Then the last one is this concept of the membrane being somewhat porous. Membranes are not this absolute impenetrable envelope. They are penetrable. They have a permeability to them that allows some molecules to get in. Indeed, with this concept in mind, the permeability and the simple diffusion being an aspect of passive movement, if you have a molecule that has a greater permeability within the cell, it will naturally move faster into the cell than a molecule that has a lower degree of permeability.
- 1.54** Now, at the end of it all, within the simple diffusion concept, that was the passive, and now we have the channel, or through the lipid bilayer, and then we have through the channels. The channel, as I described it, it's like a little pore, a little pillar or column, that is penetrating both of the layers of the cell. Usually, it's an open structure, allowing things to move in or out. Sometimes it has to be put there through a hormone signal that's preceded it. But regardless, if there is a channel, it just is allowing molecules to move in or out. That's just a simple concept. It's a passageway that allows things to move.
- 1.55** Many, many ions will use channels. Water will use a channel to move in and out of the cell. A lot of processes, a lot of molecules, are going to be taking advantage of a channel, and it is itself still a concept of simple diffusion within the broader aspect of passive transport. Now, let's leave simple diffusion behind and go into the second and final part of the passive movement of molecules, which is facilitated diffusion.

Facilitated means there's going to be something on the cell membrane that is precisely regulating the movement of whatever it is we're trying to move.

- 1.56** Usually, this will take the form of this particular protein complex that acts like a weird version of a Pac-Man molecule, where it can take a molecule from outside the cell but not be open, not like the channel was that I just described in the simple diffusion. It will be closed at its bottom, and then the molecule will come in, and as it receives the molecule, the very placement and the intake of that molecule will induce a change in its structure.
- 1.57** Now it will switch to being oppositely faced, where it was open to the outside, and then it got filled with whatever the molecule was that it wanted to move. Then it shifts in order to now be open to the inside, to allow whatever the precise amount of molecule it bound or took in, it closes and opens. Now, only that much will come back out into the cell. Then once that has happened, it shifts back, and then it will continue to do so as long as it's embedded within the membrane of the cell.
- 1.58** This is a carrier. That is the carrier-mediated process. That's another word for it. When we talk about facilitated diffusion, we have a carrier that has been designed and embedded within the cell membrane to precisely regulate the movement of molecules. It will take something from the outside, and only that much will then be allowed to move into the inside.
- 1.59** Now, one final note, when we talk about transport, is the movement of water. Water also wants to go with the flow of ions. If we have a cell that is taking in lots of electrolytes, water wants to go with those electrolytes into the cell, thereby having the cell swell, potentially. But if more molecules are leaving the cell, water will follow it through channels. Water will chase the electrolytes. In fact, as an example, not that we're going to be talking about the kidney system within this broader lecture of human physiology, but the kidneys represent an interesting case, where if people have too much body water, their blood pressure will be too high.
- 1.60** One common intervention among clinicians is to give someone a drug that will push electrolytes out into their urine because then the water will follow those electrolytes. The water is going to chase the electrolytes or the ions or the molecules. So wherever the molecules are going, water will go with it. And generally, it's happening through those pores that I've mentioned earlier, those channels. OK, all of this over the last few minutes has been us describing passive transport. Remember, passive is the lazy, slovenly version of moving molecules around. It's just something rolling downhill, if you will.
- 1.61** Downhill by being, it's going down a concentration gradient. Well, active transport was the act of pushing the stone up to the hill, to use that metaphor again. So this requires energy. We're moving something against where it wants to be going. Let's come back to the cell. Imagine, if you will, in your mind, a cell. Outside the cell, we have a

relatively high amount of stuff, whatever the ion or the molecule is, relatively higher outside the cell than inside. Based on what I described earlier with concentration gradients, where does this stuff want to go? If we have more outside the cell than in, it wants to go in. It wants to go downhill, down its concentration gradient.

1.62 However, what if we want to push it out? What if we want to take this at the lower bottom of the hill, but we want to push it to the top? Or in other words, take an area of low concentration gradient to higher concentration gradient? This is the active transport, and it requires energy. This is absolutely essential to understanding the conductance of nerves. The signal down a nerve, me animating my body now as I'm giving this lecture—all of this requires active transport. You do not have these imbalances that I described earlier, which are essential to normal cell physiology, without active transport. You have to be doing something to make one area more concentrated than the other.

1.63 Now, what is energy? A brief little tangent as we step outside of normal cell physiology for a moment and talking about cell transport, which I'm going to do sometimes. Sometimes we'll have some interesting little tangents. What is cellular energy? I have a unique appreciation for cellular energy. My PhD was in an area called bioenergetics, which is very uncommon, but it's essentially the study of energy within an organism. We're not talking about coal-burning plants or nuclear power or the energy that is lighting up the lights in this room or allowing people to tune in and watch this later. We're not talking about that energy.

1.64 The cell doesn't use energy like that. But a cell will create molecules that allow it to get work done. That is the essential understanding of cellular energy. It's molecules that allow the cell to work. So, within the cell, we have the powerhouse of the cell, namely the mitochondria, which anyone can recall from high school biology. Within the vast majority of the cells of the body, we have these little powerhouses that are capable of taking in energy from the form of fats and glucose, for example, and then breaking them down to create the molecule of energy.

1.65 It's commonly referred to as the energy currency of the cell, and that is a molecule called the ATP. ATP stands for adenosine triphosphate. Imagine if you will, we have an adenosine molecule, then we have triphosphate, so three phosphate molecules linked in a little bit of a tail. And these bonds between the phosphate molecules are called high-energy bonds, because when they break, the cell, it actually will release a sufficient energy to allow the cell to do something that the cell wouldn't have done otherwise. That's what I meant when I said that energy within the cell is basically us buying work.

1.66 So ATP is commonly referred to as the currency of the cell. It is the medium of exchange where the cell will say, "I need to get something done, and here's how I'm paying for it, by breaking ATP." And that's a process called hydrolysis, and we're going to talk a lot about it when we discuss the contraction of muscles within the muscle

system later. So ATP is the energy molecule. So anytime I'm talking about energy being required, I want you to be thinking ATP.

- 1.67** Where did that ATP come from? It came from the cell burning, if you will, fats and glucose, as the primary energy sources. So we take in those calorie nutrients into the cell, the cell will break them down, it will metabolize them, and end up getting ATP as one of the results, one of the outcomes. And indeed, a necessary outcome because as I said, it is the ATP that allows the cell to do the work, that whatever we need the cell to do. In the case of active transport, that work is moving molecules against its concentration gradient.
- 1.68** So if we have more sodium in the cell than we do outside the cell, 30 times more, we are working hard to move that sodium in. And ATP is what's allowing us to pump that water against its concentration or pump the ions, pump those electrolytes, against their concentration gradient. Energy, think ATP. ATP coming from the cell burning molecules. The first form of active transport, and there are two, there's primary active and secondary active. The first one, primary active, is just the burning or the breaking of ATP, powering some channels.
- 1.69** Now, earlier, I was acting out this shape of a transporter. In this case, we actually have the most famous of all transporters, the sodium-potassium pump, sometimes referred to as the sodium-potassium ATPase. Within the realm of biochemistry, if you ever hear that -ase at the end of a word, that suffix, that suggests an enzyme. An enzyme is a protein that's getting some work done. It's doing something. It's converting some molecule into something else. That's an enzyme.
- 1.70** So ATPase means, this is an enzyme that is capable of pulling off one of those phosphates from that little triphosphate tail of ATP, and thereby getting some work done. We have this sodium-potassium ATPase that is capable of taking sodium from inside the cell. So remember, we have much more sodium outside the cell than in. But it wants to keep that. It wants to work against the concentration gradient. So, this sodium-potassium ATPase will be faced inwards, and it will bind some sodium molecules. Then it will use ATP to open up and release those sodium back out into the cell where there's already really high levels of sodium.
- 1.71** It's using the ATP to work against its concentration gradient. But at the same time, the cell is so brilliant that it takes advantage of that work. Then we'll also pull in some potassium molecules and then restore its shape, thereby allowing the potassium to come in. In each case, moving sodium out where it's already 10 times higher, and then moving potassium in where it's already 30 times higher, and in each case, it's the same single molecule.
- 1.72** As it's using ATP, as it's using energy, this primary active version of active transport is taking the sodium, using ATP, boom, opening up, moving these molecules against their concentration gradient, effectively pushing the stone up the hill. All right, now, the

second type is secondary active transport. The interesting thing about secondary active transport is that it has to have—we often will say that it is using energy indirectly because you needed—this is the instance where it's taking advantage of a stone that's already been pushed uphill.

- 1.73** There was at some point some other work that was done, but now we're going to take advantage of it in clever ways, not directly, like with primary active transport, but in two sneaky ways through what's called a symport and an antiport. In the case of the symport, imagine the cell, and then there is a transporter, some molecule, a channel, that can allow the movement of these ions. We have ion A and we have ion B. Ion A, or electrolyte A, has much, much higher levels outside the cell than in. But ion B has lower levels outside the cell than in.
- 1.74** So A is going to easily come down its concentration gradient. This is a very steep hill. It wants to roll downhill. So we have much more of A outside the cell than we have inside the cell, and so it is moving down this transporter readily. But the sneaky thing in this case of secondary active transport is that ion B, which has a much lower concentration outside the cell than in, it will sneak in with ion A. So as ion A is moving, ion B is essentially hooking it at the arm or hitching a ride and moving into the cell. So that is symport.
- 1.75** Symport is one molecule going against its concentration gradient by piggybacking, if you will, on a molecule that is going down its concentration gradient. That's symport, a form of secondary active transport. Then the next form is antiport. I bet you can already assume what's happening here. In the case of antiport, we have a molecule that's going down its concentration gradient, and as it has opened that pore or that channel, the other one sneaks in and goes in the opposite direction while it's open. It takes advantage of it.
- 1.76** So one molecule opens the pore to go down its concentration gradient, the easy way, then the other molecule sneaks out the opposite direction. So far from being a piggyback, it's a much more hostile approach, but never resulting in a collision; they both pass each other on the way, one going in the direction it should based on physics and the chemical gradient, one going against its concentration gradient because of the nature of the movement of the other. All right, so that is the conclusion for membrane transport.
- 1.77** At this point in the cell physiology lecture, we have discussed essential topics such as homeostasis, and you're now familiar with negative feedback, and you're familiar with positive feedback in a fascinating, glorious example like childbirth. Within each of those systems, there were the three components that we discussed. Now we've been talking about membrane transport. As I have alluded to and mentioned directly, you need to appreciate membrane transport in appreciating the movement of these electrolytes to then move on to the third and final topic within cell physiology, which is electrophysiology.

- 1.78** This is when we start to discuss the animating forces of a cell and how a cell is then conveying signals, particularly through the nervous system, which will be the next big topic of when we discuss the peripheral nervous system in Lecture 2. So, let's move on to electrophysiology. Within electrophysiology, I have two subtopics that I want to discuss, action potentials, and then synaptic transmission. With action potentials, this is the idea of appreciating the movement of the electrical signal across a membrane of a cell, moving a signal from one point of a cell to a distant point of a cell, which has, of course, more relevance if it's a long cell like a neuron, and we'll discuss that explicitly.
- 1.79** To understand the action potential, we have to understand the conductance or the wave, the electrical wave. This happens through a process of polarization, moving the signal out of normal, and then repolarizing the signal. We're going to talk about this aspect of polarizing a cell membrane, all of that being part of the action potential. Then the synaptic transmission. When a signal has moved to the end of, say, a neuron, how does it then send that signal to the next structure, whether it's another neuron or some other form of tissue? That's going to be the synapse, the convergence, and the communication between two distinct cell types.
- 1.80** All right, so let's start with action potential. As I mentioned earlier, the concentration of electrolytes, those solutes like sodium and potassium, and their movement, ends up creating an electrical signal across a cell membrane. This is something that can actually be measured in voltage, in fact, in particular, millivolts. But that is the electrical potential or the membrane potential of the cell, the electrical capacity within the cell. Again, something that can explicitly be measured in electrical terms like in millivolts.
- 1.81** That is entirely a result of the movement of these ions in and out, these charged molecules. As they're moving in and out of a cell, they're creating this electrical force. In general, the general rule is, wherever you have a concentration gradient going one direction, out of a cell or into a cell, you have an electrical gradient that is directly opposite that direction. If we have a concentration gradient that is moving something out of the cell, then the electrical gradient is moving towards the cell.
- 1.82** In contrast, if we have a concentration gradient that is moving into the cell, then the electrical gradient is running downhill out of the cell. Now, what does that all mean? Basically, that will affect the sign of the charge, whether it is a negative charge or a positive charge. So anytime the electrical signal is coming from outside the cell in, it's going to manifest as a negative charge. This will become relevant in a moment when we talk about polarization and the signal, the conductance or the sending of a signal, along a cell like a neuron.
- 1.83** So if the electrical signal or electrical gradient is moving into the cell, it's going to manifest as a negative sign, with regards to positive and negative. In contrast, if the electrical gradient is moving out of the cell, that's going to manifest as a positive electrical potential or what's sometimes referred to as an equilibrium potential. This

matters because, again, it's, as we start to push the signal more positive or more negative, that allows the cell membrane to send its signal.

- 1.84** These are some of the general ideas—the movement of the concentration gradient of ions, creating an electrical gradient. But then let's move back into the ions or the solutes that you're familiar with, sodium and potassium. Now, you'll recall that there is much more potassium inside the cell than out. So, potassium wants to run out of the cell. Because its concentration gradient is trying to push the potassium out, which direction is the electrical gradient then? Always being opposite. It's coming in.
- 1.85** You'll recall that an electrical signal that comes in is negative. When potassium is moving out, it's resulting in a negative signal, a negative electrical signal. In contrast, sodium is 10 higher outside the cell than in. Sodium is constantly trying to move into the cell, thus creating an electrical gradient or signal that's going out of the cell. Remember, if the electrical gradient is going out, that's a positive sign. Here we have the movement of potassium, creating a negative millivolt signal, then we have the movement of sodium, creating a positive millivolt signal.
- 1.86** All of this comes to fruition when we talk about the membrane potential, which again is going to be part of the broader action potential, sending a signal across a cell to get the cell to do something, sending a signal throughout the body. In a rested state, in a cell that is just doing nothing, the movement of these electrolytes is such that it's typically resting, called a resting potential, at -70 millivolts. That is the normal resting potential. It's always in a slightly negative state.
- 1.87** Then polarization is the word we use to describe any time we're not at zero, which we're never really at zero. We'll pass it in a moment, as I'll discuss. But so we would say that because the cell has a resting membrane potential, its electrical signal is minus or -70 millivolts, it is polarized. It is not neutral. We have pushed it into a negative range and we're keeping it there because of the nature of how we're moving, particularly potassium and sodium. So it's always a little more negative. You'll recall, remember, which ion, the movement of which ion, is creating more of a negative signal? It's potassium. There's some insight there.
- 1.88** There's always a little more potassium moving across the membrane than there is sodium, which is why we're always keeping it in a rested state, or when the cell is rested, we're keeping it in a slightly negative domain. All right, now, that is the cell that isn't really working. That's the resting potential. Especially when we start talking about neurons and the movement of signals, me feeling and animating my fingers, I must have a long signal coming from my central nervous system all the way down. We're going to talk about the peripheral nervous system, the sending of that signal, in the next topic.
- 1.89** The first thing that must happen is we take that resting membrane potential, which is sitting at around -70 millivolts, and we have to depolarize it. We need to start pushing it

up. Depolarization is the first step in this membrane potential and getting to what we call the action potential, the sending of a massive signal, which we'll get to in just a moment. We have the depolarization. Remember, we're polarized, -70. We start depolarizing it, moving it up.

- 1.90** Now, it can't stay there. It can't stay there. But we pass zero, we get into the positive range, and we'll revisit the mechanisms in just a moment. Then once we've done that, there is this snap response of repolarizing. Repolarizing will start to bring things back. But then, in fact, in the process, what can happen is we have a hyperpolarization. We can actually move things further down, further from actually getting a signal, which has to be this signal going up into the positive range; we can hyperpolarize as well. That's also a possibility. But the most fundamental part of this is the depolarization and then the repolarization.
- 1.91** Now, the depolarization was us going from -70 all the way up to about positive 30. We're going to talk much more about this in just a few minutes. We get much, much higher up and we're getting more and more positive. Which solute, which electrolyte, when it moves, creates a positive signal? Do you remember? It's the sodium. When we talk about depolarization, it is this opening of sodium channels that allows the sodium to come rushing into the cell, creating a much more and rapid, within milliseconds, signal creating a much more positive charge.
- 1.92** Now, interestingly, at the same time, and we're going to talk much more about all of this, we then, as we get more and more positively charged, we open up these other charged potassium channels, and then potassium starts rushing out of the cell, allowing the cell to repolarizing it back down to its -70 resting membrane potential. But then there are other instances when we want to really depress a signal. We want to really turn it off. We don't want there to be any chance of it reaching some activity. And so that's when we hyperpolarize.
- 1.93** We take the -70 and we push it down even further. -80, -90. Which ion, do you remember which electrolyte makes things more negative? It's the potassium. If we have even more potassium running in, that's when we can hyperpolarize, pushing something even further from activity. This has very real consequences. If you begin to hyperpolarize, you can have things like the heart stop beating because the signal isn't going to the heart anymore. You can have neurons that are no longer conducting signals, and so you can lose sensation.
- 1.94** So hyperpolarization is a very real phenomenon, and it has everything to do, in this case, with the excessive movement of potassium. But in any case, we have the depolarization from that negative resting membrane potential, then the repolarization, resetting it. Then in some instances, we can go a little too far and have a hyperpolarization. Now, when we talk about the membrane potential and the movement of the electrical signal across the cell membrane, there are two types of potentials, a graded potential and then the action potential.

- 1.95** The graded potential is when there is a small change within the membrane, and it usually is local. You're only feeling it in one, or the cell is only acting in one very particular part. Whereas the action potential is something that is then capable of sending long signals. You can already start to see the relevance as you think about the signal coming from the brain to, say, my limbs as I'm animated and talking and moving.
- 1.96** The first one, the graded potential, this is the small change. If you think about that little -70, that low -70 millivolt signal across the membrane, we're getting little hiccups in the signal. It's never going all the way up like I described earlier with the full-on depolarization. That's the action potential, which we'll get to. But there is a smaller, lighter signal, and that's the graded potential.
- 1.97** Typically, these are activated by things called gated ion channels. Gated, meaning there must be a signal that opens it. It's like a gate when you're trying to get into a property. You have to go and punch in a code or push a button, then you can move. The same thing is happening here. There must be some sort of signal that allows some movement of the electrolytes across these channels. There are three in particular that I want to mention— chemically gated, voltage-gated, and then pressure-gated. Thankfully, the name really reveals the idea here.
- 1.98** The first one is chemically gated, and we're going to talk about some chemically gated ion channels when we start talking about the synapse in just a few minutes. When you start moving, you want to take this electrical signal from this nerve or neuron, and then pass it to the subsequent neuron across this synapse. I'm getting a little ahead of myself, but that's going to be the chemically gated. As the signal comes to the end of one neuron, it will release a chemical. One that we're going to talk a lot about is acetylcholine, but there are more than that.
- 1.99** Then acetylcholine will move across this junction and then stimulate the next one to keep the signal moving. That's a chemically gated signal. Then the next one is voltage-gated. We're going to talk about some examples of this, particularly when we talk about skeletal muscle and heart muscle. The skeletal muscle system and the cardiovascular system, both are heavily relying on voltage-gated. In the case of voltage-gated, it's when you have an electrical signal going across the membrane, and then when it gets to this channel, it opens the channel.
- 1.100** Now, that's not the same as the chemically gated, when there's a chemical that's telling a channel to open; this is the electrical signal. As it's moving across a membrane, as it is spreading and it reaches one of these channels, it tells the channel to open. That's a voltage-gated. Again, particularly relevant in muscle, skeletal muscle, and heart, which we'll get to.
- 1.101** Then lastly, we have the pressure-gated. You can see the obvious relevance of the pressure-gated is when you actually have a physical distortion. It's this mechanical sensation that is inducing a signal, and it literally is touch. So these sensations of touch

and pressure and vibration, those are all using pressure-gated systems. It's obvious, or a pressure-gated channel to conduct a signal. It's obvious when I'm touching my arm here. But one less obvious is within the visceral space.

- 1.102** When you think about an intestine that's been distended, say, because of gas—if there's any gas that's inflating the intestine, it will activate these pressure gated channels, and then you'll feel that. We have one obvious case where it's the pressure gated of my finger poking my arm. Then there is the less obvious, but still pressure gated channels, of us being able to sense a signal from somewhere within the visceral space because, say, the intestines are more distended because of gas formation.
- 1.103** Now, let's move once, final steps before we move from graded potential to the action potential. If you can imagine again, in your mind's eye, you have this negative millivolt, about -70 millivolt of the resting potential. The graded potential are these little bumps. If you're looking at the signal over time, it's these little bumps of a signal. Those are the graded potentials. Always just it's a little hiccup and then it stops. It's a little movement and it stops. It's enough to sense there's some action happening.
- 1.104** But in order to get to the true action potential, which we'll talk about now, it has to reach this threshold, a threshold of -55. You go from -70. If you just move up to -55, that is a point where you then initiate a process that can't be stopped within that few milliseconds, so it's not as dramatic as it sounds. But once you've reached it, if the graded potentials are strong enough or happening rapidly enough together, it will push up to that -55, and then it doesn't stop. At that point, you've reached this all-or-nothing response, and now you move from a graded potential, which is usually local, up to the full-on action potential.
- 1.105** If we go back to my finger tapping tapping on my arm, there would be a way, especially if it's a spot where I don't have any arm hair, there would be a possibility that you can actually contact the skin and it not be sensed. That would be the graded potential. There is something that is physically contacted, there would be some modest amount of pressure, but I'm not sensing it. That would be the graded potential. But once the signal got high enough, it has reached that threshold, because in this case, just the intensity of the signal, and it would then elicit the full-on response that I'm sensing as I'm poking my hand.
- 1.106** That's the difference between a graded potential, these small little bumps of a signal, versus a full-on action potential, which is the signal that has reached that point of -55, and then it just goes on and up. Let's move on to the action potential rather than, and leave the graded potential behind because we got more extreme. We have to keep in mind that same idea of the -70 millivolts and then a signal pushing it to the -55 threshold, and then it just keeps on going.
- 1.107** This is especially relevant when we start talking about the conductance or sending the signal across distances or just sensing anything, me animating the body, me sensing and

receiving input from various parts of my body. These are action potentials that are conveying these big signals. This is a rapid and dramatic change in the charge or the potential, the electrical potential, across the cell membrane.

- 1.108** There's four distinct steps that I want you to keep in mind. The first step is the graded potential that depolarizes the membrane from -70 reaching that threshold. You'll recall the threshold is about -55. There's some differences across various cell types of the body, but usually it's resting at around -70, and then the signal goes up to -55. The moment it reaches -55, the cell has experienced a sufficient change in voltage that now we have a voltage-gated channel opening.
- 1.109** Remember, we're moving more and more positive. Which one do you think is opening, sodium or potassium? Remember, it's the movement of sodium that makes things more positive, and indeed, that's the channel that gets flicked on or open. We have the signal that has reached the threshold, and the moment we reach that threshold, we then activate some voltage-gated sodium channels. The voltage-gated sodium channel now just throws open the doors. They're wide open, and now that sodium is just moving like gangbusters, resulting in this massive push up.
- 1.110** We're going from -70 all the way up to about positive 30, all because of just this rapid movement of the sodium. Now, in a classic negative feedback system, as we are really moving the sodium and the charge is getting more and more positive now, that ends up closing the sodium channel. Now we have a voltage-gated closing, but at the same time, we throw open voltage-gated potassium channels, to say that more precisely. So as we get more and more positive, we then, we close the voltage-gated sodium channels, and we open the voltage-gated potassium channels.
- 1.111** And as potassium starts to move, it is moving out of the cell, and it starts to create this negative signal. So we begin to rapidly repolarize. So we reach threshold, the sodium channel is open, that's depolarizing. Then we get to that maximum point at around positive 30 millivolts. And there's such a change in voltage that now the sodium channels close, the potassium channels open, and we repolarize, swinging the signal down in milliseconds right back to where we were.
- 1.112** Then the fourth and final step is, as we get back to about -70, we have restored that membrane potential. Now, what can happen in that instance is that we overshoot. It's pretty common to have a brief period of hyperpolarization. The movement of sodium, rather, of potassium, the movement of potassium was so substantial and robust that we went a little too far. But the way we correct that very temporary hyperpolarization is through that good old-fashioned sodium-potassium ATPase, or the sodium-potassium ATP pump that we talked about earlier when we discussed the active transport.
- 1.113** This was the really elegant system of moving a signal across the cell. The low -70 millivolt resting potential, dramatically increasing because of a sodium change.

Remember, a voltage-gated sodium channel. Then itself rapidly correcting by closing the sodium and opening the potassium, restoring the cell, then ultimately getting back, restoring the signal, the membrane potential, ultimately getting back to the resting membrane potential, thanks to the sodium-potassium pump.

- 1.114** Now, one interesting tangent with this whole action potential in mind, in sending signals around the body, is that you're all probably, unfortunately, familiar with lidocaine, probably because you've been to the dentist. You get an injection, the lidocaine has numbed your mouth, and it actually acts by inhibiting those sodium, the voltage-gated sodium channels. That is what lidocaine does. It takes that signal that's trying to depolarize and stops it. Because if you can't open up the sodium gated, the voltage-gated sodium channels, you don't make a signal.
- 1.115** There's no opportunity in those nerves, right where the lidocaine has gone in, there's no opportunity to depolarize. Thus you have no action potential; thus you have no sensitivity. That's how lidocaine acts. Now, with the action potential in mind, we can actually move into, and as a bit of a prelude to the peripheral nervous system lecture that will follow this one, we can talk about the movement of a signal along a neuron. When we talk about a neuron, neurons are long. They can be several feet long. These are the longest, biggest cells of the body.
- 1.116** They have what's called the axon, which is a long thread-like component of the neuron. There's going to be a cell body, and then there's going to be an axon, a long thread of a cell. We need to be able to send a signal along that neuron in order to animate the body and have it be functioning. I say animate, I don't mean just the skeletal muscle and us acting and moving around physically. I even mean all of the other processes that are happening without us even knowing— the beating of the heart, the breathing, the movement of the food through the intestines.
- 1.117** All of this is stuff that is happening because of a signal that's moving along the neuron. The brain sensing what's going on and then sending the signal back out, constantly sending the signal across these messaging cells, the neurons. Again, the most relevant part of it in this case being the axon, the long thread that will conduct the signal. As we have an action potential that's moving along a membrane, we can think of it moving along, literally moving along this thread-like axon of a neuron.
- 1.118** This is called propagation. When we're moving a signal across the length of a neuron or indeed across the length of any cell membrane, in this case, the neuron, we are propagating the signal. Then the neuron in particular has the ability to send the signal in two different ways. One called contiguous propagation, then the other one called saltatory propagation. The difference between them is actually the wrapping around of the axon of these little things called myelin sheaths.
- 1.119** Myelin is this cholesterol and fat-heavy molecule that wraps around the axon. We'll get to that one in just a moment. Let's start this discussion by talking about the contiguous

propagation. The term "contiguous" actually suggests the function, where you have this uninterrupted signal that's moving down the length of the neuron, the axon. It happens very, very quickly. It's a very fast signal, but it's not quite fast enough as the action potential is moving. It's not quite fast enough to do what we need to in many instances.

- 1.120** But some neurons of the body can rely on this method and do. Contiguous propagation —this happens when the axon does not have these myelin sheaths, these little fatty complexes that are wrapping and folding around the string. Imagine if we have a string of pearls. In this case, it's just a thread. There's no pearls. And so the signal is running down the entire length of that. Just like you'd imagine, if we were holding a rope and we jerk up one part of it, we see that flow all the way down to the other end of the rope.
- 1.121** Now, that is not the same way that it works with a myelinated axon. The myelin sheath are those fatty little envelopes that are now, if we take that thread and now we're putting a pearl on this at every half centimeter, you can see that the actual little segments of the thread itself are pretty sparse because we have these pearls that are blocking up a certain amount of space along this string. The pearls being the myelin sheath, and then the open areas being called nodes. And a node is the only part of the axon that will depolarize.
- 1.122** In other words, wherever we have the myelin sheath, that's kind of a blank space that the signal does not pass through. It's the pearl basically blocking off the function of the string or the appearance of the string. In this case, if we start a signal on one end of the axon, sending the signal along the length of the neuron, what ends up happening is what's called this saltatory propagation, where it is skipping over the myelin sheath or skipping over the pearl, and it's only touching on the node or the exposed part of this string or the exposed part of the axon.
- 1.123** This means that the signal is basically just skipping right past the myelin, and thus it is capable of a much more rapid signal. If we have two axons of identical length, indeed, the signal difference in velocity is such that if we have those myelin sheaths covering up parts of the axon and we have just the nodes being barely exposed, the signal will go along the length of the axon 50 times faster. This is already something happening in the order of milliseconds. Now, when we wrap it in myelin, these fatty little pearls, if you will, then the signal is skipping it. We just turn that up by 50 times.
- 1.124** So a substantially faster signal, thanks to the presence of the myelin and the nature of the saltatory propagation along the length of the neuron. Then when we've gotten to the end of the neuron, now what? That then moves us into the synaptic transmission. When we get to the end of the neuron, we have that blank space and the neuron says, now what? It wants to finish sending the signal. That is the synapse. All right, so at the end of the neuron, now we have the synapse. That is the last part of this cell physiology.

- 1.125** It's one thing to send the signal, but now the message itself needs to be conveyed to the actual effect that we want, the organ or the signal continues, and that is the synapse. We have the junction between two neurons, or indeed, the junction between neuron and other tissue type, which we'll talk about when we get into the peripheral nervous system next. We have the presynaptic neuron, as I'm pantomiming this out, and then we have the postsynaptic neuron, this one sending the signal, this one receiving the signal, and then continuing it.
- 1.126** At the presynaptic neuron, we have in the end of it this bud across this. We need to send some signal across the cleft. This is the synaptic cleft, the gap between these two neurons in this case. That signal, as we talked about earlier, I mentioned that we were going to revisit the chemical signal, the chemically gated channels. This is when the chemical comes in. Because at the end of this neuron, we have these little bubbles or vesicles that are filled with a neurotransmitter. The neurotransmitter is going to be the molecule. Again, it's in these little bubbles at the end of my fist as I'm acting this out.
- 1.127** The signal comes down, then it will tell these little bubbles with neurotransmitter in it to come to the end and actually open up, to kind of burp out its chemicals. Then those chemicals will pass across the cleft, across this junction, and then be received. As we describe the anatomy of it all, we have the presynaptic neuron and then vesicles, like my little freckles, containing neurotransmitters.
- 1.128** Now, at the same time, on this same presynaptic neuron, we have some other components that you have to appreciate in order to understand the sending of the signal. We have both a reuptake transporter, a way to take the neurotransmitter and pull it back in to be reused. Then we also have voltage-gated calcium channels. This is very important. We're going to talk about these calcium channels a lot. You're going to be tired of hearing about them as we go across muscle system and the heart, the cardiovascular system, and the way the heart uses the calcium as well.
- 1.129** We have the voltage-gated calcium channels also at the end on this presynaptic neuron. Now, on the other end, on this receiving end, we have the receptor for the neurotransmitter. As this neuron is going to be releasing neurotransmitters across the cleft, this one needs to be able to sense it, and thus it's no surprise that we have receptors designed specifically for that neurotransmitter.
- 1.130** Now, let's bring them all together through a series of events. The first event is that we have an action potential running down the presynaptic neuron. The signal is sensing and running down, ultimately getting to the end. As it gets to the end, the electrical or voltage signal interacts with the voltage-gated calcium channels, opening the calcium channels. Now, calcium actually starts to run in. Because there's higher levels of calcium outside the neuron than in, calcium starts to come in to that presynaptic neuron.

- 1.131** In so doing, induces the movement of the vesicle, the little bubble that has the neurotransmitter, pulling the neurotransmitter bubbles to the edge of the neuron at the synaptic cleft. The signal comes down, it opens the calcium channel, calcium comes in, then the calcium starts pulling the vesicle to the end of the neuron, thereby allowing it to merge with the membrane and basically just belching out its neurotransmitters. Then that is the next step.
- 1.132** We have the neurotransmitter, itself being released from the presynaptic neuron across the cleft, the synaptic cleft, bumping into the receptors on this receiving end, the postsynaptic neuron, which will then induce some effect like we want. It will tell it to do something else. Whatever the signal is going to be, it will tell it to sense pain, or it will tell the muscle to contract. It'll tell the neuron to just continue to send the signal. That would depend on whatever this signal is or whatever this wants to have happening. But there will be this signal that continues, all because this one neurotransmitter moved across and then opened up these chemically gated channels.
- 1.133** Now, we also have a way to get rid of this molecule. As we have the neurotransmitter moving across, we can't let it stay there indefinitely. We have ways to clear it out. Not only we have the reuptake channel on this side to pull the signal back in, the neurotransmitter back in, but we also have enzymes that can be on this side that will grab it and break it apart. We have to be able to clear the signal again in order to turn things off. We can clear that signal by pulling it in. We can degrade it, or it will simply just diffuse away from that cleft. Those are the three ways to turn the signal off.
- 1.134** That is the end. We have gone through cell physiology, talking about homeostasis and the desire of the cell, and by a larger extent, the body, to be snapping back to a normal range. We talked about the movement of molecules across the membrane, that double-layered envelope of the cell membrane, moving molecules in or dumping molecules out. These are all going to be concepts that are revisited as we go through the systems later.
- 1.135** Then lastly, we talked about the electrophysiology and the movement of the ions or electrolytes creating an actual electrical potential across a cell. Then we really emphasized the relevance of this when we discussed the neuron and moving a signal down the length or down the axon of a neuron. Thank you.

Lecture 2: Nervous System Physiology

At a glance — Nervous system physiology.

- 2.1 With cell physiology now in our pocket, firmly understanding it and appreciating it, we're prepared to go beyond the cell and start looking at systems. The first system that I wanted to move into was the peripheral nervous system. Now, peripheral is an essential adjective to understand this because the nervous system has actually myriad components. But I wanted to distinguish this part of what we're talking about from the brain, which is the central nervous system.
- 2.2 We're going immediately outside the brain to explore the functioning of the nervous system as it is interacting with the rest of the body. The objectives that I want us to cover in the time that we have is to know the divisions of the nervous system—what are the inputs and what are the outputs? Also, to appreciate the inhibition of a signal at a neuron. In the cell physiology lecture, we spoke about the conductance of the signal from one neuron to another. I want to emphasize that and highlight and complicate it a little bit.
- 2.3 Then we're going to discuss, on the output of the nervous system, the differences as it splits into various systems, specifically the sympathetic nervous system and the parasympathetic nervous system, and then spend the rest of the time talking about that, including different neurotransmitters and different signals that are getting spread through these nervous systems. All right, with all of this as the learning objectives and the layout put out there, let's now bring it back to a whole-body level of central nervous system being the brain in the spinal column, and then the peripheral nervous system.
- 2.4 The peripheral nervous system is the nervous system that is sending a signal up to the brain, and then the brain is sending the signal back out. There is a sensory receptor which is coming from any stimulus. Whatever the stimulus may be, there's a signal that's being sensed, and it's moving up to the brain through what's called afferent, letter A, afferent neurons. There are afferent signals carrying a signal to the brain. Then the brain will determine the nature of the signal and tell the body to respond in a particular way.
- 2.5 That response will go through what's called efferent neurons. So afferent is carrying it to the brain, efferent is carrying it away from the brain to the rest of the body, and then telling the body to do something. This will be manifest as some signal to an effector organ, like the muscle, me telling my muscles to move. This foot's getting uncomfortable, so I'm stepping off that foot to the other one. That's going to be the efferent signal to an effector organ, in other words, a response. So stimulus coming up, response going out.

- 2.6** Now, if we even expand that further, especially on the response going out, we start to have this efferent division of the nervous system, where when the brain starts to send the signal back out to the rest of the body, we have the efferent division, which is getting split into two larger, and then we're going to split it again in just a moment, nervous systems. We have the somatic nervous system, and then we have autonomic nervous system.
- 2.7** And very, very briefly, because we're not going to talk much more about it, we have the somatic nervous system being the part of the nervous system that we control. So the somatic nervous system goes to motor neurons, and then a motor neuron we're going to talk about when we get to the skeletal muscle system. But the motor neuron is the neurons coming from the brain that allow us to dictate what the muscles are doing. The somatic nervous system goes to skeletal muscle, the muscle that is under our control, and in particular, moving the body around, moving the skeleton around.
- 2.8** However, that was one division, the somatic nervous system. But it can also split in this efferent division into the autonomic nervous system. Then the autonomic nervous system is in charge of everything else, everything else. But it also splits, and in particular, we're going to spend time talking about the two most important splits being the sympathetic nervous system and the parasympathetic nervous system, because it's those processes that are regulating everything else.
- 2.9** I have control over my muscles, but then everything else is going to be falling under the autonomic nervous system—sympathetic in some instances or parasympathetic. All of my heart beating, my lungs breathing, my kidneys filtering, the digestive system functioning, all of this is falling under the autonomic nervous system. Again, it splits into sympathetic and parasympathetic, each of which regulate various aspects of all the other tissues of the body. We'll talk about those.
- 2.10** Now, before we get into too deeply into the efferent division, I want to just spend a moment talking about the afferent division, the part of the nervous system that's sending the signal up to the brain in the first place. That itself is split into two systems, the somatosensory system and the visceral system. The somatosensory system is the part that deals with touch. Not only me feeling the sensation of something on my skin or in my mouth, but also me feeling the sensation of the movement of my joints.
- 2.11** For example, I can close my eyes and know where my arms are, know where my legs are. I'm not looking at them, but I can sense the movement of the joint and the contraction and movement of the muscles. That's all going to be part of the somatosensory or the sensing of the body. Then the visceral system is going to be what's happening deeper within the visceral space. This is what's happening with my guts or my kidneys or anything else. That is sending a signal up to the brain as well. It's getting sensed. It's giving stimuli that are going to the brain.

- 2.12** Now, that's the afferent, sending the signal up. Now, before we leave that behind completely, there are sensory receptors that are part of this afferent system, sending the signal to the brain. There are several different types of receptors that constitute this system. We have things like photoreceptors that sense light. If you had to guess, you'd probably assume that those are found within the eye, and you'd be right. We sense light through the eye, through photoreceptors.
- 2.13** We also have mechanoreceptors, which can sense the stretching and the bending, not only stretching of skin, but stretching of muscle and contraction and movement. We have thermoreceptors, most especially in the skin, which can tell us when something is hot or cold. We have sensors or receptors that are sensitive to chemical changes called chemoreceptors. We're going to talk about those when we get to the cardiopulmonary system later. But essentially, the body's way of sampling the blood and knowing how different levels of molecules or chemicals are changing in the blood.
- 2.14** Then lastly, the last one I want to mention are the nociceptors. Nociceptors are receptors that sense pain and send a signal up. That's such an interesting and unique part of this nervous system that I wanted to spend a few minutes talking about it. First of all, we have the receptor called the nociceptor, which senses pain. It responds to a particular neurotransmitter, which has the very clever name called substance P, P for pain. Not too clever after all, but very obvious. The nociceptor or the pain receptor is specifically designed to sense substance P, which is a neurotransmitter conducting a pain signal back to the brain, which we need.
- 2.15** The brain needs to be able to know, "Hey, this is hurting the system, so let's stop it. Let's turn this or let's move. Let's make some adjustment in order to help relieve or alleviate the pain, to remove the pain." Now, if we actually look and zoom in to that synapse, to that neuron, we have a nociceptor, so a neuron that is capable of getting the signal across to the pain receptors. Earlier in the cell physiology lecture, we discussed this transmission and the physiology of the synaptic cleft or the junction. You'll recall that we had an electrical signal that went down the length of the neuron. It opened voltage gated calcium channels.
- 2.16** The same thing is happening here with the nociceptors, the pain sensitivity. We have the voltage gated calcium channels opening up, allowing the calcium to come in, resulting in the movement of vesicles. In this case, the vesicles, these little bubbles, are holding the substance P neurotransmitter. Then that will move across the synaptic cleft, telling the next neuron to signal pain. It hurts or it's something that's going to make us want to change to help alleviate the pain. This is the pain process. Signal coming down, then substance P moving across, activating these nociceptors, telling the signal, being translated to the brain as discomfort or pain.
- 2.17** Now, one interesting thing before we leave pain behind is this complementary effect that can happen from molecules called prostaglandins. That's a fairly common word, and you might have heard of it before, but a prostaglandin, these are molecules that

are released from damaged tissues. It's when the immune system starts to come into play. If we have tissue damage, we have cells that will come to this area of tissue damage called macrophages. Macrophages are like the poster child of the immune system. Wherever you have damage and you need recovery or immunity, the macrophages are going to be there. They're always a part of this.

- 2.18** A part of this process is to release molecules called prostaglandins. Prostaglandins actually act by enhancing the sensitivity of the cell to the substance P. So whatever signal substance P was eliciting before, whatever degree of pain might have been sensed from a given amount of substance P, when the prostaglandins come, they turn it from a 5 up to an 11. They basically enhance that signal, making whatever the pain is be much more acute. We feel it much more strongly.
- 2.19** Interestingly, it is actually the inhibition of this process that we get with some degree of pain-relieving medications, things like aspirin or ibuprofen. Part of the mechanism of action is to reduce the production of prostaglandins so we have a more modest response to substance P, for better or for worse, right? Sometimes you need to feel pain in order to get the body to do what you need it to, but sometimes you want to try to alleviate some of it. Some of these more famous anti-inflammatory medications or pain-relieving medications act by inhibiting prostaglandins.
- 2.20** Now, just like there is an opportunity within the body to amplify the signal of the pain through prostaglandins, there's actually an ability to depress the pain signal through endorphins. So endorphins are an endogenous, what's called an endogenous analgesic. So we have this analgesia, or this reduction of pain, because of these naturally produced molecules called endorphins. In fact, there's more than just endorphins. There's endorphins, there's enkephalins, and dynorphin. All of these are naturally produced molecules that will come in and take the signal that's been moving down the neuron and block the calcium channel.
- 2.21** You remember that we talked about the necessity of the voltage gated calcium channel, that the signal is coming down the neuron, it opens up the voltage gated calcium channel, which in this case then allows the substance P to cross the synaptic cleft to sense pain. But here, we have these endorphins that block that calcium channel, keeping the signal stuck or reducing it, thereby mitigating some of the pain. So endorphins are these natural depressing signals, but not depressing in a bad way, or rather a suppressive signal to help turn down pain sensitivity.
- 2.22** This could be why things like exercise or cold immersion, which increase endorphins, can help with the sensitivity to pain in some people. But even beyond the endorphins, that itself is actually reflective of a natural phenomenon that exists within the peripheral nervous system, namely the ability of a signal to get inhibited. This allows me to bring in one final concept within this subtopic, which is this idea of lateral inhibition. You have a signal going from one neuron, presynaptic to postsynaptic, or any other organ. But you have an opportunity from the side to come in, which is where the

lateral idea comes into play. You have the signal and you have something abutted next to it.

- 2.23** That is called the lateral inhibition, where you have one other neuron. I just mentioned a moment ago neurons that were creating endorphins. That is an example of this natural ability of a signal to get blocked. So endorphins are acting as a lateral inhibitor, physically coming in and trying to disrupt this signal at the synaptic cleft. Now, endorphins are relevant for the pain signal, for the nociceptors in substance P, and any other signal starts to get inhibited not by the endorphins, but by a molecule called GABA, G-A-B-A, gamma-aminobutyric acid, GABA, G-A-B-A.
- 2.24** In all other neurons that aren't related to pain, GABA ends up being the blocker of the signal. For better or for worse, it's going to once again block that calcium channel. Just like the endorphins were doing for a substance P to help mitigate pain, GABA can do for a host of other signals. We have this natural phenomenon within the peripheral nervous system of whatever the signal is trying to be, of ways to come in and block it. Sometimes we want it, sometimes we don't, but even still, GABA is going to be one of the more common molecules that will come in from the side, if you will, to a synaptic cleft and start to block the signal.
- 2.25** All right, now all of this is primarily getting sensed. It was the sensory receptor part, the afferent neuron sending signals to the brain. Let's put that to the side and get back to the efferent system, the signal that is coming from the brain. You'll recall that that split into multiple divisions. We had the somatic nervous system that is us controlling our limbs through skeletal muscle regulation. But then it was complemented or matched, paralleled, by the autonomic nervous system, which can split into the sympathetic and the parasympathetic nervous systems, which essentially operate the function of the rest of the body.
- 2.26** Everything except skeletal muscle is going to fall under the autonomic nervous system. Let's spend several minutes talking about that. So both branches of the autonomic nervous system, specifically sympathetic and parasympathetic, will innervate the same organs. They're just going to have opposite effects. The sympathetic nervous system is famous for being called the fight-or-flight system. That is the body responding in a way that everything gets amped up. It stands in stark contrast to the parasympathetic nervous system, which is commonly referred to as the rest-and-digest. It's when we are calm and we are maybe digesting our food because we've just eaten.
- 2.27** But the sympathetic nervous system wants to amp the body up and get it much more excitable. The parasympathetic nervous system wants to calm it down. This is reflected across virtually every organ. There are times when you want the heart to be beating very hard and fast, then there are times when you don't want the heart to be beating hard and fast. That's an example of this dual innervation to the heart. We have neurons coming to the heart that are either going to turn it on or slow it down. In other words, sympathetic neurons or parasympathetic neurons.

- 2.28** But neither is good, neither is bad. We need both of them, always constantly seeking homeostasis. The two will always be working in balance. Rather than a combative tug-of-war, it's really more of just a concerted effort. It's two different instruments within the same band playing together. All right, now, there is, however, some difference in anatomy across these two different types of signals, sympathetic and parasympathetic. Among a handful of other differences is the nature of the ganglion.
- 2.29** As we start to look at the signal coming from the central nervous system, from the spinal column out through the body, there is what's called an autonomic ganglion, which is essentially a relay station. If you are sending a signal, it gets checked and then it goes on. The signal starts within the central nervous system. It moves out, has a brief checkpoint at the ganglion, and then it moves on. The ganglion is just simply the convergence of the two neurons. It's the presynaptic neuron waiting, and then we have the cleft, and then it bumps into the postsynaptic.
- 2.30** It can amplify. There can be one presynaptic neuron combined to multiple other postsynaptic neurons. But all of that would be happening within this little relay station called the autonomic ganglion. We have the the pre-ganglionic fiber or the pre-ganglionic neuron, and then the post-ganglionic fiber or post-ganglionic neuron. But that would just be like the presynaptic neuron and the post-synaptic with this ganglion in the middle. Now, there's a difference in where they're located as this signal is being conveyed. If it is a sympathetic neuron, then the signal is at—the way station, the ganglion, is actually right next to the spine.
- 2.31** We have the first part, the pre-ganglionic fiber, coming out of the spinal column, and then right along, right adjacent or parallel to the spinal column, will be that ganglion, that relay station. Then it will immediately butt into the post-ganglion fiber, sending the signal down as we want it to send. That is not the same as the parasympathetic. Parasympathetic normally has a long pre-ganglionic fiber coming from the central nervous system, right essentially to the tissue itself. So they're much longer pre-ganglionic fibers going all the way to the organ, and then it's a shorter post-ganglionic fiber.
- 2.32** Then we have the target organ, like the heart that we talked about earlier. If we go back to the heart, and we'll talk more about this in the cardiopulmonary system lecture later, but if we had a sympathetic neuron going to the heart, it's going to be short little neuron, right from the spine, from the central nervous system to the ganglion, and then the longer one going to the heart. In contrast, the parasympathetic is going to have a longer pre-ganglionic nerve, and then there's going to be the ganglion right closer to the heart, and then we have the post-ganglion fiber.
- 2.33** There's just a difference in the anatomy of it, an interesting difference in the actual placement of the ganglion, whether it's part of the sympathetic nervous system being closer to the spinal column, or whether it is the parasympathetic where the ganglion is closer to the actual target organ. Some differences in the anatomy of it. Now, beyond

the anatomical difference in the placement of the ganglion is the actual nature of the receptors themselves. There are different signals and different receptors that are coming into play for both the sympathetic and the parasympathetic nervous systems.

- 2.34** We can basically break them down into two different families of receptor, a cholinergic receptor and then an adrenergic receptor. These are placed somewhat differently. But both nervous systems, at the ganglion, at the autonomic ganglion, both of them have cholinergic receptors. The cholinergic receptor is the receptor that is both at the ganglion of the sympathetic nervous system and the parasympathetic nervous system. They both use cholinergic receptors at that convergence, where the pre-ganglion neuron is sending the signal to the post-ganglion.
- 2.35** At that relay station, what they have in common is cholinergic receptors. But then the commonality ends. Because if we actually then look, if we follow this next post-ganglion neuron down to the effector organ—let's just stick with the heart—as this second neuron is now signaling to the heart or whatever the organ is of interest, now it's different. Where we have still, if it's parasympathetic, we have once again a cholinergic receptor. However, at the sympathetic side, at the sympathetic nervous system, we now have an adrenergic receptor, and they send different signals.
- 2.36** At the ganglion, they both use cholinergic. At the end of it all, right to the effector organ itself, whatever the organ is that we're talking about and is getting innervated and affected by these signals, the parasympathetic will once again use cholinergic. It's cholinergic all the way from the spine to the organ itself. But in the case of the sympathetic, it has switched and is now using an adrenergic receptor. There's some interesting reasons for doing so, because it allows that to respond to a handful of other signals.
- 2.37** Let's just talk about these two receptor families in a little more detail. Cholinergic receptors are responding to a neurotransmitter called acetylcholine. Acetylcholine is the main neurotransmitter that is the predominant one throughout the body. Most receptors are responding to acetylcholine just because they're so much more abundant. But then acetylcholine receptors, these cholinergic receptors, actually have two different classifications themselves, based on the original chemicals that were found to activate them.
- 2.38** The cholinergic receptors are one of two types, either nicotinic or muscarinic. You can hear some of what the original molecule is that activated them. When these receptors were first discovered, it was found that some would respond to nicotine as well. It was given the family name of nicotinic. These are always excitatory, and they open up the sodium and potassium channels like we were talking about earlier. So earlier in cell physiology, we were talking about the movement of the sodium and potassium. Those are nicotinic cholinergic receptors. They also respond to nicotine.

- 2.39** Now, muscarinic receptors are receptors that were originally found to respond to a chemical extracted from mushrooms called muscarine. And muscarine will activate these muscarinic cholinergic receptors, and they can either be excitatory, turning on a cell signal, or inhibitory, depressing it. But it will have its own post-cell effect—again, depending on the cell type, either eliciting a strong response or turning down responses. Whereas nicotinic is always excitatory, which seems appropriate for something like nicotine; the receptors that are more sensitive to this mushroom-extracted molecule, they can either be turning on a signal or turning off.
- 2.40** But remember, what they both have in common is that they're both receptors that are sensitive to acetylcholine, and thus they fall into the cholinergic receptor family, which is not the same as the adrenergic. The adrenergic receptors are the receptors that most especially respond to norepinephrine, which is a signal that's coming from the end of a neuron. Now, it also falls into a different hormone family, which we'll discuss in the endocrine systems. It falls into a family of hormones called the catecholamines.
- 2.41** Amine, if you hear that last part of that word, amine refers to particular amino acids that these are built on, and in particular, it's tyrosine. The amino acid tyrosine is used as the building block for both norepinephrine and epinephrine. If those words don't sound familiar, you may know them from another term, which is adrenaline. Epinephrine is synonymous with adrenaline. Norepinephrine is synonymous with a derivative or a similar version called noradrenaline.
- 2.42** But that is the nature of the adrenergic receptor, where it can either respond to the norepinephrine that's coming as a neurotransmitter at a synaptic cleft, or it can respond to the epinephrine that's being produced from our adrenal glands, which is a fundamental part of the endocrine system. The adrenal glands will flood the system with epinephrine or adrenaline, and activate some of these same receptors. The adrenergic receptors respond to adrenaline or the catecholamines, these hormones that are built on a tyrosine amino acid.
- 2.43** They fall into two major classes, alpha and beta, with some subclasses. There's alpha one and two; there's beta one, two, and three. They have subclasses, but we will just keep it at that superficial level of alpha and beta. They can be either excitatory or inhibitory. It's never just one or the other. The adrenergic can fall into both camps, turning something on or turning something off.
- 2.44** Now, when we start to then bring these two together is where things get very interesting. Because as I said, you will have key organs that are innervated by both the sympathetic nervous system, getting a norepinephrine signal, and the parasympathetic nervous system, getting an acetylcholine signal. They will generally do the exact opposite. I want to just highlight some of these because the examples are just so fascinating. The heart, for example, as I've mentioned multiple times now, it has both innervation signals coming to it.

- 2.45** If the sympathetic nervous system is releasing norepinephrine to the heart, it will stimulate contractility. It will make the heart beat faster and harder, which makes sense if you think about the sympathetic nervous system being reflective of the fight-or-flight system. We're running or we're panicking, we're going to feel that at our heart. That increased heart rate is entirely a function of the sympathetic nervous system innervating it and releasing the norepinephrine.
- 2.46** In contrast, in the evening, we want to be winding down, we want things to be calming down. That's going to be primarily more parasympathetic, flooding the heart with acetylcholine, resulting in a delayed, in a slower heart rate, in a softer contraction. Now, the blood vessels. Blood vessels have contrasting effects in response to the sympathetic versus parasympathetic. It's a fascinating response. If you think about the fight-or-flight system or the sympathetic nervous system, I'm trying to get away from something, for example, as an obvious example. It's no surprise that I would want blood to be moving faster through my body in that case.
- 2.47** I want to more rapidly move blood, and I can do that by narrowing the blood vessels. So sympathetic nervous innervation of blood vessels, the muscles around the blood vessel, it will contract to narrow. It doesn't occlude it, it narrows it, which seems like a bad thing. It's called vasoconstriction, but it's not if we want to move things very, very quickly. If I really rapidly want to move energy to my cells and oxygen to my muscle to let it work and rapidly clear away any of the waste products to allow the muscle to continue to function well, I want the blood to be moving quickly.
- 2.48** In contrast, if the parasympathetic nervous system is activated, then those neurons, innervating the muscles around the blood vessels, will relax and induce this vasodilation. As the blood vessels are expanding, blood flow will slow down and blood pressure will come down. We have this contrasting effect, sympathetic inducing vasoconstriction, resulting in a higher blood pressure and more rapidly moving blood. Then we have the parasympathetic inducing vasodilation, slowing down heart rate and lowering blood pressure.
- 2.49** Now, the GI tract is another phenomenon or another system that's influenced, where the sympathetic nervous system doesn't want to waste time digesting food. If the sympathetic nervous system is turned on, typically the GI tract will slow down. It will slow down a process called peristalsis, which we'll talk all about when we get to the GI system later. But peristalsis is the name for the slow and steady movement of food and waste through the intestines.
- 2.50** But if we're busy running away from something, we do not want to have to be worrying about digesting our food or stopping for a bathroom break. We want to be able to get away. It's no surprise that if the sympathetic nervous system is turned on, it slows down the GI system. In contrast, if parasympathetic is turned on, that's basically the body's way of saying, "Hey, we've got time. We can digest. We can eliminate waste. We

have the time to do that. We're in a safe enough space," if you will. We have increased activity or increased movement of the GI system.

- 2.51** Then lastly, just to end on another fascinating organ, is the lungs, which we will get into. We'll talk more about the lungs and the cardiopulmonary system later. But once again, if you imagine, what is the overall purpose of these two systems? Sympathetic, wanting the body to respond in an emergency. Parasympathetic, wanting the body to relax or to rest and digest. In the case of the sympathetic nervous system, we open up the airway. We have this phenomenon referred to as bronchodilation. We open the airway to help air move better. There's less resistance.
- 2.52** We're able to oxygenate the blood and not only put more oxygen into the blood, but clear CO₂, the waste product, carbon dioxide, more easily. With sympathetic activation, we have bronchodilation, opening the airway. With parasympathetic, we're moving air so much more slowly. It's not as much an emergency, so it's no surprise that we get a little bronchoconstriction. So, things can narrow. Mind you, that's the exact opposite of what we had with the blood vessels. With the sympathetic nervous system, blood vessels constricted and the bronchioles or the airway dilates.
- 2.53** With parasympathetic, the blood vessels will constrict, and then the airway, or the blood vessels will dilate rather, and then the airway will constrict, because we just don't need to move the air quite as rapidly as we might if we were hyperventilating to run away from something. Those are just some examples that indicate the intricate balancing of these two nervous systems, both of which are totally essential for normal homeostasis.
- 2.54** Now, one of the tissues that I like to study the most as a scientist that focuses on obesity and diabetes and generally metabolic issues, I wanted to highlight just the fat tissue, because fat tissue is uniquely responsive to the sympathetic nervous system and the adrenergic signaling. I'd mentioned to you at the outset when we talked about the adrenergic receptors, we have different classes. We have alpha receptors and beta receptors. Our adipose tissue, what we can pinch and jiggle, is very responsive to the norepinephrine that's coming from the sympathetic nervous system.
- 2.55** When norepinephrine, when that post-ganglion fiber, is butting up to the fat cells, it starts moving across, the norepinephrine from the sympathetic signal, and that norepinephrine will bind to a beta-3 adrenergic receptor and activate this process called lipolysis. Lipolysis is the technical term for the breakdown of stored fat. All of that fat that the fat cells have been holding on to, when the sympathetic signal is coming through the adrenergic receptors, it starts to break down its stored fat, taking the stored form of fat being triglycerides, and I know this is not a biochemistry class, but it starts releasing that stored fat as free fatty acids.
- 2.56** Anything that turns on the sympathetic nervous system, like exercise or cold therapy, will induce the activation of this process, resulting in the breakdown of more fat, just as

an interesting aside. The sympathetic nervous system has very real effects when it comes to overall metabolic health and even things as precise as fat-burning.

2.57 With all that having been said, you are now much more familiar with the peripheral nervous system, the afferent signals going from the body to the brain, including the pain signals and the lateral inhibition of those signals, and then the efferent signals coming from the brain to the rest of the body, and the ganglion being the relay station between the pre-ganglion, right from the spine, to the post-ganglion signal, and the differences between these two nervous systems of the autonomic nervous system, being sympathetic and parasympathetic. They're different signals and they're different receptors, all of which converge in a beautiful balance, always seeking to maintain homeostasis.

Lecture 3: Skeletal Muscle Physiology

At a glance — Skeletal muscle physiology.

- 3.1 As we continue our journey through the systems of the body, we get to the biggest, meatiest one now, namely skeletal muscle. I say biggest on purpose because by mass, most people are more muscle than they are anything else. Hopefully, that's somewhat comforting. Most of what makes us our physical mass is our muscle. Muscle is a very unique organ with a very high dynamic metabolic rate, and of course, utterly essential to moving the body around and controlling other processes. As we move through the system, we come to skeletal muscle.
- 3.2 Skeletal muscle, as the broader topic, we're going to speak about some of the unique structures within the muscle. We're going to then zoom in to the actual length of a muscle cell. No surprise, muscle cells are long and stringy things, and talk about the smallest functional zones and units of the muscle cell, and then actually talk about the process involved in making a muscle cell contract. It is really unique in that most cells are just sitting there doing what they do without physically changing. Muscle cells are capable of dynamic changes very rapidly in size or what we would call a contraction.
- 3.3 We're going to talk about the necessity of energy and ATP in that process through what's called the cross-bridge cycle, the process of contracting the muscle and allowing it to relax, and then ultimately end by discussing the different types of muscle fibers we have within the body, all of which are designed to suit all of the variety of needs and demands that may be placed on skeletal muscle and moving the body around.
- 3.4 As we start to zoom in to the muscle itself, we have the skeletal muscle, which is one big meaty tissue. If we break that down into its next level of organization, we have what's called fascicles. A fascicle is a small bundle of muscle cells, and there are multiple bundles which come together to make the muscle itself. But if we go back to the fascicle, we can go down another layer and look at what's called the muscle fibers. Each muscle fiber is a muscle cell.
- 3.5 Then if we leave the cell behind or the layer of the cell and go one step deeper, we actually have what's called the myofibrils, which we'll discuss. The myofibrils represent these intracellular or within-the-muscle cell protein structures. Then if we zoom back out, myofibrils constitute the muscle cell or the muscle fiber, also called. Then the muscle fibers will be lumped together in what's called the fascicle. Then fascicles will be lumped together into the muscle proper, giving us the big-bellied muscle, the big functional muscle.
- 3.6 Now, one other point is that if we zoom back down to the layer of the muscle fiber or the muscle cell, each one muscle fiber or cell has one neuron that's innervating it. There'll be one neuron coming down like the fingers on my hand. Boy, I'm getting a lot of action out of my arms and hands here. If here's a muscle fiber, you have one neuron

with its various little touch points, making what's called the neuromuscular junction. A neuromuscular junction is one nerve with one muscle fiber.

- 3.7** You can see that, within one whole muscle, as we go down to fascicles and muscle fibers, because each muscle fiber has its own neuron, you don't have to contract the entire muscle every time. You can contract a portion of it, getting a graded response. If you need to lift something light, you don't need the entirety of the muscle being activated. Each neuromuscular junction allows this very fine-tuning of how much of the muscle we're using.
- 3.8** Now, let's not leave the neuromuscular junction behind quite yet. Having discussed both electrophysiology and action potentials in the cell physiology lecture, and then talking about the synaptic cleft of the peripheral nervous system, you are now well suited or well positioned as students to appreciate the neuromuscular junction because there's nothing new. If we zoom back into this layer of my fingers touching on my arm, being the neuron touching the individual muscle fiber at various points, each one of these touch points creates this junction.
- 3.9** That is called the motor neuron touching it, and we call this an end plate. A motor end plate is the muscle part of it. The nerve ending is coming to the muscle. We call it a motor end plate. But otherwise, despite some differences in names, the entire process is the exact same, and by now somewhat familiar. We have an action potential running down the presynaptic neuron. That action potential, that electrical signal, will open at the neuromuscular junction, the voltage gated calcium channels, opening these pores, allowing the calcium to run in to the neuron.
- 3.10** In so doing, it will start pulling the vesicles, the little bubbles of neurotransmitters. The particular neurotransmitter is acetylcholine. Acetylcholine will come to the edge of the presynaptic neuron and get released. Then it will move across this synaptic cleft to the end plate or the muscle fiber itself. There, it will bind on to the nicotinic cholinergic receptors. The nicotinic receptors were a class of the cholinergic receptors that we discussed in the peripheral nervous system.
- 3.11** That elicits or initiates a chemically responsive signal, like we talked about in the cell physiology signaling. We have the chemical gated ion channels. The acetylcholine comes across, and now the muscle will respond and start sending an electrical signal. That is a depolarizing signal. We talked about the action potential and the depolarization. That is the response of this acetylcholine coming across the neuromuscular junction, and now the muscle is going to send this signal.
- 3.12** Now, before we go any further and indeed, deeper into the muscle, I wanted to just pause there because that very process is something that's taken advantage of, in other words, the sending of a signal telling a muscle to contract, when it comes to something like Botox. We've all heard of Botox, of course. Botox is a toxin, a botulinum toxin, which blocks the vesicle from coming out, so it prevents the acetylcholine release. So, if

anywhere you were even injecting Botox into these skeletal muscles of, say, the face or forehead, you have a neuron that's coming, the signal is coming down, calcium is coming in, but then it stops.

- 3.13** You can't actually move the acetylcholine out. So you have no neurotransmitter and you have a muscle that won't contract. In other words, you just paralyze the muscles, all in an effort to try to get rid of some wrinkles, which would be glorious, I suppose, if that was something you cared about. Let's leave that behind now and get back to the muscle.
- 3.14** Last we discussed, we have the motor neuron coming down, innervating the muscle cell, and then sending a signal along the muscle. But the muscle is really unique in that it does not want the signal to just float across the membrane of the muscle. It actually wants it to penetrate deep into the muscle. The muscle has some unique structures, one in particular called T-tubules. As we have a signal, and you can imagine a letter T that is just inserted down into a muscle cell.
- 3.15** We have the muscle fiber, and then we have the top of this letter T, the T-tubule, and the signal that originated from that neuron is coming across the surface of the muscle. But then it gets to this junction in the T. While some of the signal continues onward, now the signal also penetrates deep into the muscle tissue. We're sending this signal deeper into the muscle fibers. As the signal is coming down the T-tubule, deeper into the muscle now, it also then starts to interact with another unique structure to these muscles, which is called the sarcoplasmic reticulum.
- 3.16** The sarcoplasmic reticulum, if this is the penetrating part of the T-tubule, the sarcoplasmic reticulum is like my fingers that are running adjacent. They're not physically touching, but they're right next to it, running right adjacent. It's this whole, what's called the reticular network, or this kind of meshy network of all of this sarcoplasmic reticulum, just abundant and penetrating through the muscle, but always in close proximity to this invagination of the T-tubule. The T tubule, of course, taking the signal of the electrical signal, the action potential, and driving it down into the muscle. As it comes down into the muscle, it now starts to interact with the sarcoplasmic reticulum in a process we'll get into now.
- 3.17** We have, again, the signal moving across the T-tubule and down the T-tubule. As the signal comes down the T-tubule, it activates the DHP receptor, dihydropyridine receptor. Dihydropyridine is a receptor that is just off the T-tubule at the end of a sarcoplasmic reticulum. Remember, the sarcoplasmic reticulum are these little structures that are coming just right adjacent to the T-tubule. But it's kind of like a cork. When the DHP receptor is activated, as the T-tubule signal comes down into the muscle, it's like we're pulling out a cork.
- 3.18** As we pull out this cork, which gets popped out as the electrical signal comes down, we allow the ryanodine receptors, which are these closed little receptors, to open up.

Again, this is all at the sarcoplasmic reticulum. The signal has come across the muscle, down the T-tubule, opened up the DHP receptor, like a cork being popped out, which then allows the ryanodine receptors to open. Now we have the most important part of all of this, which is sarcoplasmic reticulum belching out all of its calcium into the main part of the muscle cell.

- 3.19** Calcium is being held within the sarcoplasmic reticulum, my fingers, and then the signal comes down, and by activating receptors, DHP receptors and ryanodine receptors, we uncork these little receptors and allow the calcium to come gushing out into the muscle cell. It's a lot of calcium that gushes out very, very quickly, all from the sarcoplasmic reticulum, my little fingers that are running adjacent to the T-tubule. Remember, the T-tubule is a way of taking this electrical signal from the top of the muscle cell and working it down into the body of the muscle cell.
- 3.20** Now, once the calcium has come out, we're not done. That was just the first and initiating step of actually getting to what we want to have happen, which the muscle is famous for, namely contraction, the shortening and the generating of force as the muscle cells go from this long to this long. That brings us to the next subtopic, namely the sarcomere. The muscle, as we zoom in to the individual muscle fibers, it has these various striations, a striation being a mark.
- 3.21** Then if you were to look at a muscle cell, zoom in as much as you can using an electron microscope, you would see these distinct little bands or markings, what's called striations, just moving in these vertical formations. The striation is a fundamental and very unique part of the muscle. In fact, if we are once again looking at my hands as an example, it's almost like you can see my fingers as a part of this, what's called the dark band, and I'll get into that in more detail in a moment. Then we have these blank spaces called the light bands.
- 3.22** Essentially, what begins happening, and we'll go into the details, is that this distance gets pushed together. It gets pushed together. Just like my hands are coming closer together, the muscle cell, the fiber, gets shorter. But at the same time, you can see as my fingers start to overlap, we start to get a thickening of the fiber. That's exactly what is happening. As my fingers are narrow in distance, they start to come together, getting shorter, the distance between them getting shorter and thicker. That is the muscle contraction.
- 3.23** Calcium is essential to this process, through the cycle we'll talk about in just a moment. But before we leave the sarcomere behind, I want you to appreciate some of the components of the sarcomere, because the sarcomere itself is the myofibrillar proteins, the proteins that are constituting the fiber itself. So these are all the proteins that are worked within the muscle, allowing the muscle to be able to move and contract.
- 3.24** We have two distinct sections of the sarcomere. When we're looking at this picture through an electron microscope, we have the A band, what's called the A band, and we

have what's called the I band. They just happen in sequence. A band, I band, A band, I band. But each of them represents, or is really constituted, based on the two different filaments, what we call filaments of the muscle fiber. We have the thick element, which is in the A band, and then we have the thin element, which is in the I band.

3.25 It actually looks a little differently. The thick band has a bit of a darker—the A band, rather, with the thick element, has a bit of a darker appearance than the I band, which has the thin filament. Now, there's another name to each of these that you need to be familiar with. The thick filament is also known as myosin. Anchoring myosin is this thing called an M line. An M line is going right down the middle of the A band. It's this line that the thick filaments or the myosin are connected to. That would be like my knuckles holding on to my fingers. We have this M line anchoring the thick filament, aka anchoring the myosin.

3.26 Then down the middle of the I band, which has the thin filament, also known as actin, and we're going to talk all about these when we start talking about contraction, we have what's called the Z line. The Z line is the anchor for the actin or the thin filament. Then altogether, from Z line to Z line, which is the middle part of the I band, we have what's called the sarcomere, the individual functional unit, which repeats again and again and again along the length of the muscle cell.

3.27 Again, we have the A-band, which is primarily made up of the myosin or the thick filament, butted adjacent by I-bands, which is the thin filament or actin. Then it is the interaction of myosin and actin, or the interaction of the thick filament and the thin filament, respectively, that creates the cross-bridge cycle that ultimately comes together to induce contraction. Let's just talk about each of these two structures, the thick filament, myosin, and the thin filament, actin, in a little more detail.

3.28 Myosin, as I'm going to start acting this out with my arms and hands, myosin is going to be this arm, which is almost like a golf club. Imagine in your mind a driver, a wood, a golf club, and it has this narrow shape, and then it has this bent little portion. Now, unlike a golf club, this is very static. Just like my fist is moving in and out, that's what ends up happening with the thick filament through a process that we'll get to in a moment. But basically, imagine if you will, a bunch of these thick filaments wound together, where they have from one common, wrapped-around thick filament, there are multiple heads coming off of it all around.

3.29 It's not just happening in one plane, but there's a head, then other heads, other heads, other heads circling around. In fact, that thick filament, the myosin, is surrounded by the thin filament or the actin. But before we get to the actin, on the head or my fist of the thick filament is an ATPase. Now, do you remember ATP? ATP was that main molecule of energy. That's what allows a cell to buy some work. If it wants to get some work done, ATP is going to allow the cell to get that work done. In the case of the muscle, this is where it's happening. This is where ATP is necessary for the muscle to

work, by cocking and recocking the head of the myosin or the head of the thick filament.

- 3.30** We're going to talk about that process as we start getting into the cross-bridge cycle or the process of binding and contracting and binding and contracting and narrowing the muscle. But it is an essential part of it. On that head is an ATPase that will allow the breaking of ATP in order to have it start cocking and literally moving around to enable contraction. The most important part then, of the thick filament or the myosin, is the ATPase on the head of the myosin. This would be like the head of the golf club. It is the binding and the breaking of ATP, this high-energy molecule, that allows it to physically move.
- 3.31** As we go through the process later, you'll see that's the process whereby we're basically ratcheting the muscle to get a little shorter and shorter and shorter, or in other words, to contract. Now, contraction cannot happen unless the myosin has something to bind. As I mentioned earlier, around the myosin wound-together molecule with all of its heads facing in different directions, we have the thin filaments surrounding it. That's the actin. Just like there's a long thread of myosin, there's long threads of actin surrounding the myosin.
- 3.32** And actin has three distinct parts. I'm calling it actin or the thin filament, but actin is only one part. Mind you, it's the majority of it. But then on the actin, which is this strand of pearls woven together, we have two other molecules. We have the actin, which is like the strand of pearls, like twisted strand of pearls, then we have on top of it a long little threadlike protein. Imagine a little string, and that's called tropomyosin. The tropomyosin is this long threadlike structure that's actually, in its resting state, it's covering up the sites that myosin wants to bind.
- 3.33** We have, on actin, these little places on each little pearl, we have a little site that is designed for myosin to come and bind it. But that binding site is covered by tropomyosin. Tropomyosin is the string that's covering the binding sites. Then on the tropomyosin, one last layer, I promise, is the troponin. So troponin is the molecule that is keeping the tropomyosin in place. So the troponin is on top of the tropomyosin, keeping the tropomyosin in place. The tropomyosin is covering the myosin binding sites on the troponin. So we've gone down all the layers of the thin filament.
- 3.34** And remember, it's those myosin binding sites where this arm wants to connect, in order to actually make the muscle contract. It was troponin on the top, covering the tropomyosin, which itself is covering the myosin binding sites on the actin, getting back to the little strand of pearls. Now, how then does this all interact to take a muscle that's this long and turn it into a muscle this long while expanding its volume? In other words, contracting a muscle and making it big and bulky. It is through what's called the cross-bridge cycle.

- 3.35** It's just a little summary before we get into each step. It is this process of this arm being the actin, or the thin filament, with actin, tropomyosin, and troponin. Then this arm being the myosin or the thick filament. It is this ongoing process of the arm, the myosin coming and binding, and then cranking and then coming off, resetting, binding, and cranking. Of course, that's happening in just milliseconds. It's happening to the degree of thousands or millions of times within every single muscle during every contraction. As cumbersome and laborious as it seems like I'm being, imagine this just happening in the order of milliseconds.
- 3.36** But that is the cross-bridge cycle. The myosin comes across, binds the actin forming a bridge. It cranks, comes off, resets, binds again. Cranks, comes off, resets. That resetting process, as I mentioned, involves ATP. Let's talk about the first part of this, imagining, if you will, a muscle cell at rest. So nothing's happening yet. At rest, we have tropomyosin, which is the long thread on the actin that is covering the myosin binding sites. OK, so that's at rest. We have the thread covering the myosin binding sites on the actin, not allowing the myosin to come bind.
- 3.37** At the same time, the myosin head is cocked. It's ready to go. It's open and ready to crank. On that head, what's very important is that the ATP has already been spent. In this rested state, as much as we still have the tropomyosin covering the myosin binding sites on the actin, on the head of the myosin, it's cocked and ready to crank, we have a split ATP molecule. That's important to keep in mind. The split ATP molecule is split into two parts.
- 3.38** ADP, taking the adenosine triphosphate, turning it into adenosine diphosphate, so there's only two phosphates there now, and the phosphate itself, but disconnected. We have a broken ATP on the head, which allows the head of the myosin to be cocked and ready. It's a spent ATP. Again, it can't bind it anyway. It can't bind the actin anyway because the tropomyosin is blocking it in the rested state. Now, all of a sudden, go back to what we discussed earlier, where you can imagine the electrical signal coming across the muscle, going down the T-tubule.
- 3.39** What did we want to get? Remember, we wanted that signal to open the cork and allow the calcium to come out of the sarcoplasmic reticulum. When that calcium comes out, now we're cooking with fire here or cooking with gas. Now something starts happening. Namely, the calcium will come and bind to the troponin. Remember, we had the actin, which is what the myosin is trying to bind. Those myosin binding sites are covered with tropomyosin, and the tropomyosin is held in place through troponin.
- 3.40** But the moment calcium comes and flushes through the system, it actually binds the troponin and causes the troponin to shift its shape, allowing the tropomyosin to move. Now what can happen? Now the myosin binding sites are exposed, all because the calcium flooded into the cell. I'm just going to say that one more time, but in reverse. We need to move the tropomyosin out of the way in order to free up the myosin binding site, although that hasn't happened yet.

- 3.41** But now the myosin binding site is exposed, all because the tropomyosin was pushed out of the way, because calcium came and bound the troponin, which was what was anchoring the tropomyosin in the first place. Hopefully that makes sense. Calcium comes into the system and basically pulls open the thin filament, allowing it to get ready to bind the thick filament. But the thick filament still has that spent ATP on it. Now that that is exposed, it comes and locks on, in its open, cocked and ready state. But it still has its spent ATP, so it's not going to crank yet.
- 3.42** At this moment, it comes and locks in, and the free phosphate gets kicked off. Remember, earlier I'd said that it was ADP and the phosphate. Well, it binds and then the phosphate itself gets kicked off. Now we just have the ADP there. Actually, as the free phosphate gets kicked off, now we have the famous power stroke. Because with the free phosphate kicked off, ADP does not want to stay. Now the ADP will get kicked off. As ADP leaves the head of the myosin, it induces the structural change, which power strokes or pulls the head of the myosin, giving us our very first moment of narrowing the muscle.
- 3.43** As we have on both ends of this muscle, myosin heads which are pulling apart or pulling together, it cranks from different directions, pulling the muscle together. Again, all of that happened, the power stroke happened, because the phosphate got kicked off first, the moment it bound, and then the moment the phosphate left, the ADP ejected, giving us this power stroke. So ADP ejected at the moment now then, there's nothing there. It has cranked, and it doesn't have any ADP or ATP, neither fresh ATP nor spent ATP. But that's the next step.
- 3.44** Basically, the ATP now will come in, sensing a blank spot. As ATP comes and binds to the head of the myosin, it kicks it off. But it's still in this contracted state. Remember, the myosin bound and then cranked. Then the ATP comes and binds the head of this myosin and pulls it apart. But it's still cranked. It's still power-stroked in this closed or contracted form. In order to reset, it needs to split the ATP, and we're right back to where we started.
- 3.45** Again, the ATP comes and binds, and then it's still cranked, and then the ATP splits through an ATPase, which you'll recall is the enzyme that will pull off the last of the triphosphates or the three phosphates. Then it resets that myosin head. It cocks it back open, ready to power cycle again or power stroke again. But it only did so by splitting up the ATP into its two component parts, ADP, the diphosphate, and the individual free, or what's commonly called inorganic phosphate. We've reset the system, ready to go again.
- 3.46** Indeed, as long as calcium is present, calcium will keep the troponin pulled up, which will pull the tropomyosin up, which will keep those myosin binding sites active. As long as calcium is there and the myosin binding sites are exposed, it will continue to happen, pulling the actin together, together, making the muscle contract. But again, the essential part of this cross-bridge cycle, to just summarize it all over again, was not only

the influx of calcium, exposing the myosin binding sites on the thin filament, on the actin, but then it was also the dynamic changes in ATP metabolism.

- 3.47** So ATP gets busted, and that allows the cocking of the myosin head. Then the phosphate gets kicked off, which allows it to bind. Then the ADP gets kicked off, which allows it to power stroke. Then the ATP comes again, which allows it to disassociate. Then the ATP splits, allowing it to reset. It's a lot, I know. But that is the cross-bridge cycle. It is that dynamic process, again, built totally on the necessity of calcium, having this dynamic flooding of the muscle cell, and then the splitting and the recovery of ATP on the thick filament or the myosin.
- 3.48** So, calcium influencing the thin filament, then ATP influencing the thick filament. Together, all coming, combining into what's called the cross-bridge cycle, which is happening in countless, countless reactions to narrow the muscle and even to relax it. Now, as I've mentioned, if calcium stays in the muscle cell like that, it will allow the contraction to continue. In fact, it will force the contraction to continue. We have to have a way of getting calcium out of the cytosol or the middle part of the muscle cell.
- 3.49** We do this through, first of all, closing up that cork. You'll recall that we had the DHP and the ryanodine receptors that gets uncorked when we have the muscle signal go down the T-tubule. We want to put the cork back in, and that happens if the neuron stops firing. That's a first signal. We stop flooding the cell with calcium by just removing the cell signal. But at the same time, we can activate these pumps called SERCA pumps, which stands for sarco-endoplasmic reticulum calcium ATPase, S-E-R-C-A, the SERCA pumps.
- 3.50** These are basically pumps along the endoplasmic or the sarcoplasmic reticulum that will start pulling the calcium back in. Now, you'll notice that when I describe that pump, it's a calcium ATPase. What do you think that means? Based on our cell physiology, do you think that's a passive process or an active process? Remember, if it uses ATP, it is active. You are pulling the calcium into a higher concentration area because the sarcoplasmic reticulum is loaded with calcium. It's because we're actively pushing the calcium in by using ATP to do so.
- 3.51** As we clear the cell from all of its calcium, locking it back into the sarcoplasmic reticulum, then the myosin binding sites get covered, which stops the myosin from being able to bind onto the actin, thereby ending a contraction. Then once the calcium has gotten pulled in, actually, within the sarcoplasmic reticulum, there's just another interesting aspect to this, which is the presence of a molecule called calsequestrin. Calsequestrin is this big molecule that wants to just hold on to calcium.
- 3.52** As we are pumping calcium back in to the sarcoplasmic reticulum, calsequestrin is going to hold on to it until the signal is once again sent to release it. Calsequestrin is the babysitter of the calcium within the sarcoplasmic reticulum. Now, throughout this process, just to highlight the essential aspect of energy, we needed ATP for only two

processes. As much as you think, in something as big and complicated as muscle contraction, we must need ATP for all of it, we only needed it for two parts.

- 3.53** The first part being the dynamic power cycle and the power stroke of the cross-bridge cycle. We needed to split the ATP to reset, or to bind ATP to reset and then split it in order to cock or to reset. That's the first part of where ATP was necessary. Then the second part was the one I just mentioned, namely the recovery of all the calcium, basically pushing the calcium away. We needed ATP, paradoxically, for both the contraction itself, and we need ATP for the relaxation, clearing out the calcium, not allowing the muscle to contract again.
- 3.54** Now, let's, just for fun, wrap all of these three distinct processes up again. Let's just go through them briefly in order. We had, first of all, the signal, which was the action potential coming down the neuron, the motor fiber, the motor neuron, the action potential coming down, opening the voltage gated calcium channels, resulting in the release of the acetylcholine from the vesicles, the acetylcholine binding to the nicotinic, muscarinic, or the nicotinic cholinergic receptors, inducing the signal to open up other voltage gated channels.
- 3.55** Then remember, the muscle cell being unique, that as the signal is moving along the muscle cell, it doesn't just stay on the surface. It penetrates down into the muscle through the T-tubule, which then interacts with all of the adjacent sarcoplasmic reticulum, ultimately resulting in the release of calcium. Then we move on to the next part, which is the contraction. When the calcium has been released, it then goes and binds to the troponin. Then as it binds the troponin, it pushes the tropomyosin out of the way, exposing the myosin binding sites on the actin, which was the strand of pearls.
- 3.56** Now, with that exposed, we have the myosin head, which can bind. Then we have the dynamic changes of ATP that ultimately allow it to not only contract, as we kick off the ADP and the phosphate, but then the fresh ATP comes. Then as the fresh ATP comes, we pull the myosin off of the actin. But at the same time, as the ATP comes in, it also pulls the calcium out by allowing the calcium, the ATPase, those little channels that want to pull the calcium back in, it pulls them out of the muscle cell, the fluid is the middle of the muscle, and pulls it back in to be stored within the sarcoplasmic reticulum.
- 3.57** As much as we say that ATP is the molecule of energy, one of the ironies when it comes to the muscle cell is that the ATP is actually more important to the muscle relaxing than it is to contracting. But of course, you have to relax before you can contract again. All right, so those were the three particular processes, the signal coming through the electrophysiology and the action potential, then the contraction itself being a function of the movement of calcium, and then the binding of the myosin to the actin, and then ultimately the inclusion of ATP, allowing them to separate, perhaps to contract again, which it can do, but also potentially relaxing the muscle by clearing out the calcium.

- 3.58** All right, now, before we finish talking about the skeletal muscle system, I wanted to highlight the fact that different muscles have a different capacity to contract. We have three distinct types of muscle fibers within the human body. We have what's called the type I muscle fiber. We have a type II muscle fiber, or IIA, and then a type IIX, sometimes referred to as type IIB. So two different types of type II muscle fibers and one type of type I.
- 3.59** The difference between them with that classification in mind, type I versus type II, is the nature of the metabolism. Type I fibers rely primarily on fat for fuel, and they have a general—it's like a diesel engine. It can run for a very, very long time, but it's not going to be fast and flashy. That's like a big diesel engine in a big truck. That's like a type I muscle fiber. It's sometimes referred to as a slow oxidative, oxidative meaning that it's generally burning fat for fuel.
- 3.60** What the type II fibers have in common, type IIA and type IIX, is that they are referred to as fast glycolytic or just fast. Whatever they're burning, they're burning it much faster than the type I fiber. It's not the diesel engine in a big rig anymore. Now it's an engine perhaps in a sports car. It's going to burn a lot more fuel and it's going to react much more quickly. But this is reflected at different layers in different functions, as we go from type I to type IIA to type IIX, if we look at something like the twitch speed.
- 3.61** If you look at the degree to which the muscle can contract, how quickly it can contract, type IIA has a slow contraction rate, whereas the type II fibers are very— sorry, type I is a slow contraction, type IIA and IIX are very fast. They have a high fast twitch speed. That is a function of how active the ATPase is. Now, remember the ATPase on the head of the thick filament, that thick myosin filament with its golf club head that's capable of cocking and resetting, how rapidly you can split apart ATP is generally going to determine how rapidly this myosin can form a crossbridge and contract to pull the actin together.
- 3.62** With that in mind, it wouldn't surprise you to learn that the type I fiber has a relatively slower ATPase. The type I fibers, which are like the diesel engine, it breaks down ATP more slowly. It's going to be a contraction that's going to happen a little longer. Whereas the type II fibers have a very fast ATPase. It can put on ATP and split it and kick it off and split it and kick it off and split it very, very rapidly, allowing that head of the myosin to twitch and be contracting and pulling an actin molecule very, very quickly.
- 3.63** At the end of it all, then, it's no surprise that also the rate at which these muscle fibers can handle calcium is also very different. That's the third and final thing I want to note about this, is that just as much as they have a difference in the rate at which they can break apart ATP and contract, they also have a much slower, in the case of type I, or faster, in the case of the type II fibers, ability to clear calcium. It can clear the calcium and dump it back out, and clear it and dump it back out. All of these combine to create muscle type, muscle fibers, that have very different functions.

- 3.64** Type I muscle fibers being very, very slow, or slower twitch, those are primarily going to be expressed in muscles that are involved in posture and endurance. For example, where we have some of the most of our type I fibers is in the back muscles, the muscles on us that keep us erect, standing erect. Those are muscles that are used to being contracted literally all day. We'll have those working. They are very much a type I, a slow, highly oxidative, it's the diesel engine-type of muscle fiber.
- 3.65** In contrast, the type IIA and IIX, one example with type IIA is something like jogging. It's much higher intensity. We're going to fatigue eventually, but we're able to get a lot more power and work than just standing. Jogging is going to be activating my type IIA fibers, say, within my legs. But then if I go from a jog or even a walk to a sprint, then I go from type IIA to type IIX. The type IIX fibers are the very high-power, rapid velocity, but also highly fatiguable muscle fibers. They're going to tire very, very quickly. Whereas my type IIA will be a little slower, but I also get less work out of them, less power.
- 3.66** Then at the other end, we have type I, generally very low power, low velocity, but can go all day, literally can go all day. Now, before we wrap up, I just want to have one tangent before a final concept of muscle fatigue. Most people are interested to varying degrees in growing their muscles. We want bigger, better muscles. You're actually now well positioned to appreciate what goes in to a muscle tissue getting bigger and thicker. That is, in fact, I already alluded to what actually happens.
- 3.67** When a muscle is getting bigger through chronic exercise and challenging the muscle, it gets bigger through a process called hypertrophy. There's some confusion. Some people will think, "I have more muscle cells." No, the number of muscle cells is the same. But each muscle cell is bigger than it was before, because we get more and more of these proteins. We get more and more of the thick and thin filaments within the protein itself, within the muscle itself. Each muscle cell is getting bigger and bigger, as we get more and more actin and myosin that start to just populate the muscle cell.
- 3.68** The muscle cell itself starts to swell, or in other words, undergo a process called hypertrophy. It's not that we have more cells. We have the same number of muscle cells. It's just each cell has gotten much, much bigger. That is what goes into hypertrophy. Then the third and final, or sorry, the final point I want to make is fatigue. What is it when the muscle starts to burn and we can't get any more effort out of it? We feel that discomfort that we call burning. What is it? What goes into that? And what are the consequences of it? So the origins and consequences.
- 3.69** But muscle fatigue, in a broader sense, is when the muscle cannot continue to put out the same degree of force that it was before. The muscle is wearing out. Usually, this is a result of, of course, high frequency and high stimulation; we're working out the muscle to the point of failure. It will fail for one of two reasons. They're not the same, but they can occur together. One is a shortage of fuel, so the muscle is just running out of energy, which is not so common, but it can at very high or very long activities like a

marathon; you can start to run a little low. But then second, and more relevant in most instances, is the accumulation of metabolites.

- 3.70** Now, what do I mean by that? A metabolite, in this case, I'm talking about the waste products. As the muscle is working and working, there's a lot of metabolism going on to fuel that. You need to create all that ATP, for example. Well, that ATP has a cost. There is an exhaust, or waste products, that are getting generated. One of them is simply the movement of potassium. If you'll recall, we're moving potassium around a lot. As potassium starts to accumulate, it can actually start to compromise the muscle cell's ability to move calcium.
- 3.71** Too much potassium starts messing with calcium. As we mess with calcium, we mess with the ability of this muscle cell to contract and go through the cross-bridge cycle. At the same time, when we split ATP apart, that technical name for that process is ATP hydrolysis. We take those three phosphates, the little tail of three phosphates, strung together, and we pull off the last one. You'll recall that was necessary for the cross-bridge cycle, and that is ATP hydrolysis.
- 3.72** One of the consequences of splitting the ATP molecule is that we start to get a lot of hydrogen. Hydrogen has a unique effect because, at the muscle, it will start to make the muscle more acidic. It starts to change the pH, which is a measure of acid-base status. As the muscle starts to get more acidic, it actually activates the nociceptors. Now, you're struggling to remember, where did I hear about the nociceptors? The nociceptors are the pain receptors, and so when you start to feel a burning in the muscle, most of it is actually a result of the hydrogen being produced from you splitting apart so much ATP in order to continue to contract and relax your muscles.
- 3.73** The hydrogen activates the nociceptors, which is being sensed as a burning pain. It's not burning, it's no temperature change. It's actually just a pH change. Now, my final point is probably me anticipating what you're wondering, which is, you're thinking to yourself, "Professor Bikman, what about lactic acid?" Some of you were definitely thinking this. Well, interestingly, humans don't make lactic acid. You don't have a single speck of lactic acid in your bodies right now. What you do have is lactate. Animals don't make lactic acid. Animals make lactate.
- 3.74** And not only does lactate not affect pH at all—it doesn't, it has no effect on pH—but it actually is used as a fuel. So the muscle is making lactate, and the liver can pull in that lactate and actually convert it back to glucose, or the brain can pull in that lactate and just directly metabolize it as energy. Lactate is just a full-on viable fuel that cells can burn for energy, and it has no effect on the pH change.
- 3.75** Where that myth came from is, if you were measuring the pH of a working muscle, you were determining, if you had a little pH probe, an acid-base probe inserted into the muscle, you would see that if that muscle keeps working, it's getting more and more acidic. At the same time, if you're sampling the blood, you would see lactate levels

getting higher and higher in the blood. Thus, you may conclude the lactate is causing the pH change, but they're both just two coincidental events, one not directly contributing to the other. They're just both a reflection of a cell that is very, very active.

- 3.76** Lactate being a waste product or an energetic product of a cell that's burning a lot of glucose. Then the ATP or the pain and the acidity, the pH change, is a function of the ATP splitting apart. There's no such thing as lactic acid in the human body. It's lactate, and it in no way contributes to muscle pain. It just happens to be the same thing that's causing muscle pain, because of the more rapid metabolism is going to be the same thing that's making lactate, which the body will use as an energy source very freely.
- 3.77** That is the skeletal muscle system in all of its complexity and wonder, allowing the body to actually be animated and do what we want it to do at any given moment. OK, thank you.
- 3.78** I had a question that I know the brain absolutely loves glucose as its main form of energy from carbohydrates, but if it can't use glucose, the only other acceptable one is ketone bodies. I know in this day and age, a lot of diets are popular, like the keto diet where you reduce carbohydrates. Would you assume that it's very therapeutic to encourage a lot of exercise if you produce carbohydrates? Because you mentioned lactate being able to be energetically favorable for glucose, or would that not necessarily be as beneficial?
- 3.79** Yeah, that's a great question. As part of my answer, I don't mean to sound like I'm plugging my old course, but in my previous Peterson Academy lecture of Why We Get Sick, we talked about the role of energy in the brain and how, when the brain starts to go hungry, it starts to manifest as problems like Alzheimer's disease. You noted, what you said is not wrong, but I also want to add some dimension to it.
- 3.80** The brain has two primary fuels, glucose and ketones, as you noted. If both are available in an equal level, the truth of the matter is the brain will actually use more ketones. If glucose and ketones were both at, say, 5 millimolar, which is normal for glucose, but pretty high for ketones, the brain is actually getting about 70% of all of its energy from the ketone, even though the glucose is just as available.
- 3.81** That's because, that's an adaptation of the brain thinking, "If my ketones are that high, I must not be eating any carbs, and so I want to not use too much glucose that the liver is making." If you're not eating glucose in the form of carbohydrates, the liver is making all of the glucose that's in your body. The brain will basically start to say, "Hey, I'm going to leave that for other tissues because I can use ketones really, really well." If the brain actually has any preference, it appears to be that it's the ketones because, again, if you match their concentration, it will shift over to ketones.
- 3.82** Now, you mentioned some of the advantage of exercise. Exercise can be an advantage for a number of ways, but also potentially problematic, depending on the dietary state of the person. One advantage could be just the production of lactate. The brain will use

lactate as a fuel. In fact, use it very, very well. One of the therapies for someone who's experienced traumatic brain injury is to give the person an IV infusion of a low lactate solution because the brain will start burning the lactate and that will help recovery much faster.

3.83 Lactate is a very viable fuel for the brain. But we published a paper finding that if you fast, but you exercise at the beginning of your fast, you will get into ketosis much more quickly. Speaking of neurological disorders, one of the themes of almost every neurological disorder, whether it is migraines or Alzheimer's disease or Parkinson's disease or bipolar disease, bipolar disorder, is that there is a reduced ability of the brain to use glucose as a fuel, which is why, probably, there's evidence that in every one of those pathologies that appear to have nothing in common, ketones end up being therapeutic.

3.84 That you take someone with Alzheimer's disease, bump up their ketones, they immediately have a clearer cognition. Not that you've reversed the disease. You take someone who's in the middle of a migraine headache and bump up their ketones, the migraine will stop almost instantly. There is this real energetic underpinning of a lot of these neurological disorders and exercise to get back to your question, is a way to amplify and accelerate the onset of ketosis because it helps bring down glucose, it brings down insulin, accelerates fat-burning. When you're burning a lot of fat, you're making a lot of ketones.

Lecture 4: Cardiopulmonary Physiology

At a glance — Cardiopulmonary physiology.

- 4.1 As we continue our journey through the various physiological systems, we now come to one of the most famous and well known, which is the cardiopulmonary system. Now, cardiopulmonary has within it, of course, two different functions or two different tissues. You can hear both of them in just how I'm describing it. It might seem, at first glance, strange that we are lumping them together, but you'll see in short order how really, fully understanding the function of the heart requires you to appreciate the involvement of the lungs and vice versa.
- 4.2 After all, the lungs are what's responsible for moving the contents, the primary contents, that will then go into blood, which the heart is then moving around the body. So you can't really appreciate one without the other. So let's start within this context of cardiopulmonary physiology or the system. Let's start with heart, the heart system itself. Then we'll move to the pulmonary system or lung physiology after. So let's start with the heart physiology at the center of the body.
- 4.3 As we go through the cardiovascular system, there are a few key learning objectives that I want you to take away from the time that we have together. First of all, I want you to understand the actual anatomy and structure of the heart and the movement and the direction of the blood flow through the heart chambers and through the key valves that are connecting these chambers. I want you to be familiar with the cardiac cycle, this process of relaxing and contracting.
- 4.4 And a part of this is the conductance system. How is the heart actually generating its electrical signal in order to be contracting as frequently and steadily as it always does? And as a part of that, I want you to appreciate the unique action potential that the heart has, which is not quite the same as skeletal muscle. Even though the process ends up being somewhat similar, it has very, very different origins.
- 4.5 And then I want you to appreciate the stroke volume, how much blood the heart is beating, and the variables that go into that, including one of the more famous physiological laws, which is the Frank-Starling law of the heart. And that's where we will end the conversation about heart before transitioning into the involvement of the lungs.
- 4.6 Now, if we zoom into the body and actually look at the heart, we see that the heart has four distinct chambers, two on top and two on bottom. The two on top are both called atria. They are the atrium, the right atrium and the left atrium. And then below them, each is connected, and indeed, each will move its blood into a corresponding ventricle. So two atria, right and left, and two ventricles that receive the blood from the atrium that is just above it, right ventricle, left ventricle. Those are the four chambers of the heart.

- 4.7 And each one of these sides of the heart is responsible for primarily moving blood around a different circuit. And if we then zoom out of the heart and look at the entire cardiovascular circuit, it's almost like we have a figure eight. We have what's called a systemic circuit of blood, where the heart is moving blood around to the body itself, all of the body except the other circuit, which is the pulmonary circuit. So there's one circuit of blood flow that just goes between the heart to the lungs to the heart again. Then there's an entirely other circuit that goes from heart to everywhere else and back to the heart.
- 4.8 And so we have the two circuits. Each circuit is primarily fed, if you will, or fueled, by one different side of the heart. For example, the right side of the heart is what's pushing the blood. The right ventricle is what's moving the blood up through the lungs. And we'll revisit the involvement of the lungs, of course, shortly. And then it is back from the lungs, we have the blood going to the left side. And then the left side of the heart will move that blood to the systemic circulation, which will come back to the right side of the heart. So there is this full, wrapped figure-8 circuit.
- 4.9 Now, let's be a little more precise. That was the overview, and let's really actually move into the movement of the blood. We have to pick a particular spot. This is a circuit. It's a never-ending infinity loop of blood flow. So we could start this description at any point in the circuit. Just for the sake of defining it and identifying it, let's start in the left ventricle. The left ventricle, when it pumps, what the ventricles have in common is that the ventricles are pumping right or left and moving the blood out of the heart. So ventricles are meatier and thicker. They have more muscle because they have to pump the blood further.
- 4.10 Whereas the atria, left and right atriums, will just contract and move the blood down. So they're not muscled. They're weak. They don't need to do a lot. Let's start in the left ventricle, which is the biggest, strongest chamber of the heart because it is responsible for moving the blood throughout the entire body. So it has to be very strong and it has to generate a lot of force. So we start the journey of blood flow in the left ventricle. It pumps to the body, which we call the systemic circulation. Then as systemic circulation is wrapping up and coming back to the heart, it gets received into the right atrium.
- 4.11 So the right atrium is getting the blood back from the system, and then it contracts and just moves the blood right down, right next door, down into the right ventricle. And the right ventricle doesn't have too hard of a job. It only has to beat the blood through the lungs. So now we have the pulmonary circuit or the lung circuit. And once the blood has moved through the lungs, exchanging gases, dropping off CO₂, and we'll talk more about this in pulling in oxygen, it goes back to the heart, and the atrium on the left side now receives that blood from the lungs.
- 4.12 So we make the pulmonary circuit from right ventricle through the lungs to left atrium. Then the left atrium contracts and moves it down to the left ventricle, and now we've completed the circuit going back into systemic circulation. So left ventricle to system,

then to right atrium, to right ventricle, to lungs, to left atrium, down to left ventricle. And once again, we've completed the circuit. So that is the entire involvement of the heart in it moving its blood through this infinity loop circuit.

- 4.13** However, let's zoom into the heart in just a little more detail because there's an interesting physiological effect, which is you listening to the beating of the heart. Anyone who has ever heard a description of the heart sounds will always say that the heart makes a lub-dub. That is the classic university textbook description of the heart sounds. Lub-dub. You hear thump, thump, thump, thump. Those thumps are entirely a consequence of the slamming shut of different heart valves.
- 4.14** So it's the heart valves that make these sounds. It's not the movement of the blood, but the blood moving through one chamber of the heart to the next, and then the valve shutting afterwards. That's what's making the two heart sounds, lub-dub or thump, thump. So if we start again in the left ventricle that's moving its blood out to the system, it has to pass through something called the aortic valve. And this is actually the second heart sound, the dub of the lub-dub or the thump, thump, thump one and thump two. This is what makes thump two sound.
- 4.15** It's that the left ventricle contracts. It moves the blood through that aortic valve, and then the aortic valve slams shut, giving an actual physically audible sound. And then it moves through the system, of course. And then it goes back up through the vena cava, and it goes back into the right atrium. No real valve moved through here. Then the atrium contracts, and then it moves it down through the right AV valve, standing for the atrioventricular valve, from atrium to ventricle, also known as the tricuspid because it has three parts. That moves through, and we get a soft little sound that you don't even actually measure when you listen to it on a stethoscope. It's so quiet.
- 4.16** And then when the right ventricle beats, it moves up through the pulmonary arteries back to the lungs, and then it comes to the left side, where we got to earlier. Now we have another relevant sound, which is moving from left atrium down through the left AV valve, also called the bicuspid valve. And then once it pumps its blood down from left atrium to left ventricle, that valve slams shut, and that is the first sound you hear.
- 4.17** So the right side of the heart, which is receiving blood and moving the blood through the lungs, we don't hear those sounds because it's so easy and low pressure. Those valves just open and close in a bit of a quiet way. But the left side generates much more force because it has to push the blood all through this systemic circulation, not just making an easy little pulmonary circuit, but the much more rigorous systemic circuit. We hear the first heart sound, the lub of the lub-dub, is the heart sound when we close that left AV valve. The left atrium contracts. In fact, both atria contract at the same time.
- 4.18** The blood goes down, that valve slams shut loudly. We hear the first lub. Then the ventricles contract, and the left ventricle in particular pushes its blood up through the

aortic valve, and then it slams shut, and that's the second heart sound. Those are the two heart sounds, both a consequence of the closing of the valves, and in particular, to be very precise, the closing of the valves on the left side of the heart, whether it was first the left AV valve or bicuspid valve, or the second sound being the aortic valve, as the blood is getting pumped into the entire system. And the left side of the heart just must work much, much harder, which is why we hear those sounds.

- 4.19** But those two valves, which are so important to us hearing the proper sounds of the heart, when you are listening, when a clinician is listening to the heart sound, if they detect a murmur, it's because of a problem with these two valves, one or both of them, usually one of them. And it can either be that the valve has undergone a degree of stenosis, which is when the valve won't open very well. And so if the blood is moving through it, you hear it, rather than two distinct sounds, lub-dub, it may be a luuub-dub, or a longer, more drawn-out sound. That's going to be a stenosis or a stenotic valve.
- 4.20** Or the valve can have undergone a change in shape, or now it's regurgitating. There's blood that's squirting back through that valve, even when it's not supposed to be, and you'll detect a murmur. So if you ever hear a murmur or you've heard people talk about them, it's when those lub-dub sounds have been somewhat stretched out, all because one or two of those valves, but both on the left side of the heart, the left AV valve or the aortic valve, aren't closing properly or opening properly, depending on the nature of the problem.
- 4.21** Now, in between all of this is the actual degree of what's called microcirculation. So when we first have the vessels coming from the heart and lungs, so we have oxygenated blood coming from the lungs to the heart, and then the left side of the heart is going to move that richly oxygenated blood through the rest of the body, we can't actually—it first goes into arteries. So right from the heart, we're going into arteries, big, thick, strong, muscled blood vessels. And then it shrinks. It starts to spread out. It's a big, thick vessel, and then it branches out into what's called arterioles.
- 4.22** But we're not done yet, because then the arterioles will branch out again into the actual business part of the blood vessel system, which is the capillary. So it's at the capillary where we're exchanging stuff, where blood is giving stuff up to the cells of the body or the organs of the body, and where the blood is receiving stuff back from the organs of the body. So it's at the capillary where the blood is actually realizing its purpose within the body, of exchanging stuff with cells of the body.
- 4.23** And then after we have left the tissue and cells behind, the capillaries start merging. The lanes of traffic start coming back together into what's called the venule and then a vein, the big, thicker, stronger veins, ultimately leading back to the heart itself. And so we have again, between all of this, from heart to heart, with of course a pulmonary circuit in between, we have capillaries at our systemic tissue, giving up the oxygen and other things, and taking in the carbon dioxide and other waste products from the cell.

- 4.24 And then we get to the lungs, where we have the exact opposite exchange, where now we have low oxygen blood giving off its CO_2 to the lungs to be breathed out and the pulling in fresh oxygen. So we have the exact opposite processes happening at the capillaries of the systemic tissue, for example, muscle, versus the capillaries that are going past the lungs, which we'll talk more about when we get to lungs in just a bit.
- 4.25 But just appreciate that we have these two different sets of capillaries, a capillary set for the pulmonary circuit, pulling in oxygen, giving off CO_2 . And we have a set of capillaries at the systemic circuit across each tissue, whether it's the liver or the kidneys or the muscle, anything has its own capillaries where those capillaries are giving off the oxygen, as an example, and pulling in the CO_2 , and then to do the exact opposite exchange at the lung. When we get to lung, we'll revisit this idea. I'm invoking the lungs a lot, even though I'm explicitly talking about the heart. All more to my point, though, you can't really appreciate one without the other.
- 4.26 OK, now let's go back to the heart, though. I've been talking about the heart contracting and relaxing. What's interesting about the heart is that both atria will contract at the same time, and it will contract immediately, followed by the contraction of both ventricles. So the atria contract, then the ventricles contract, immediately after and both at the same time. So right and left atria, boom. Right and left ventricles, boom. So it's one followed immediately by the other. Boom, boom, boom, boom, boom, boom. Within this boom-boom are two distinct cycles, two states that are defined by the status of the ventricles.
- 4.27 So when we start talking about the two different blood pressures, which you've heard of, we typically will give a blood pressure number as two, one number than another number. That is to be precise, the diastolic and the systolic blood pressure. Diastole is a state that is defined by the ventricles filling. They're not contracting, they're relaxing. But while they're relaxing, they're filling with blood. And then systole is the fraction of a moment where the ventricles contract, ejecting all of its blood out.
- 4.28 So the diastolic pressure is the pressure when the ventricles are filling. The systolic pressure is the pressure when the ventricles contract. And those are the two phases of the cardiac cycle. There is diastole, which is the ventricles filling, not the heart filling. We want to be precise. Diastole is when the ventricles are filling. Systole is when the ventricles are emptying because they're contracting. That is diastolic blood pressure and systolic, each corresponding to a state of the ventricles, not the heart.
- 4.29 Sometimes in higher education, these terms are described improperly. So diastole is when the ventricles are relaxing and filling, and then systole is when the ventricles contract because the reality is, at the end of diastole, this period of time when we're filling the ventricles, the atria contract. So the atria will contract technically during diastole. But within a fraction of a second later, these eject the blood down, boom, that initiates the systolic contraction. So technically, the atria contract during diastole

because, remember, diastole is defined by a state where the ventricles are relaxing and filling. And then when the ventricles contract, it's systole.

- 4.30** So it's not proper to say the heart is relaxed, diastole; the heart contracts, systole. No, we need to be more precise. It's the ventricles. When the ventricles are relaxed and filling, it's diastole. When the ventricles are contracting to move blood through the body, to the lungs and to the system, then that is systole. OK, now, I've been mentioning a few different words and appropriately placed, because let's talk a little bit about the ventricular volume or the amount of blood that the ventricles can hold and then eject, because that becomes a very important physiological outcome that we want to understand, which is what is the cardiac output?
- 4.31** How much blood is the heart actually moving when it's beating? That is, after all, its purpose. So cardiac output is a fundamentally important concept of the heart system. So two different terms I want you to be familiar with, the first one being end-systolic volume. End-systolic volume is the amount of blood that is left in the ventricle right after it contracts. But of course, before that, we had the end-diastolic volume, which is how much blood is in the ventricles right before it contracts.
- 4.32** So to rephrase all of this, end-diastolic volume is the amount of blood in the ventricles right before they contract. And then end-systolic volume is the amount of blood left in the ventricle once it contracts. And the difference between those, if you take the end-diastolic volume and subtract out the end-systolic volume, you're left with something called the stroke volume. And to be precise, the stroke volume is the amount of blood that is coming out from the ventricles every time they contract.
- 4.33** And again, we got to stroke volume by measuring the end-diastolic volume, how much did it fill, subtracting out the end-systolic, which is what was left after contraction, and that gives us the stroke volume, how much actually came out. Then when we know the stroke volume, how much is coming out for every contraction, we can times that or multiply that by the heart rate. In other words, how much is it beating? How frequently is it beating? How frequently is it beating, multiplied by how much is coming out with each beat, and then we get to the cardiac output, the most functional metric of the heart.
- 4.34** The heart's primary job is to move blood. Thus, its ability to move blood is the most important outcome, and we call that the cardiac output. A function, again, of stroke volume multiplied by heart rate. And we're going to revisit some of these concepts when we start talking about that famous Frank-Starling law of the heart. OK, now before we get that far, let's pause and talk about what makes the heart beat in the first place. In order to talk about this, I actually want to remind you of the skeletal muscle.
- 4.35** Skeletal muscle was interesting because it required an external signal. That was the motor neuron, so the nerve coming and innervating the skeletal muscle fiber. And then this nerve would tell the muscle when to fire, if you will, and contract. But the heart

does not require an external signal. The heart has its own ability to generate its own rhythmicity or contraction. And in fact, in so doing means that it can respond very, very rapidly to the autonomic nervous system that we spoke about earlier, when we spoke about the peripheral nervous system.

- 4.36** By having its own contraction, without requiring external nerves to tell it when to contract, it can respond immediately to a sympathetic nervous system, increasing its contractility, or a parasympathetic, slowing down the contractility. And it does so. The heart is able to do this because it has what's called a pacemaker. You've heard that term before. You've heard it in the context of an electrical device that is put in to tell the heart when to beat. But that term itself comes from a natural anatomical cluster of cells at the top of the heart that initiates its own signal.
- 4.37** So we have the SA node, which is populated by these special types of cells called P-cells, pacemaker cells, P for pacemaker. So in a normal healthy heart, we have this cluster of pacemaker cells at the SA node. Then the SA node will initiate its own signal that will go through the heart. And I'm actually pantomiming it out in the direction that it's going to go, but I don't want to go quite that far yet. So we have this central little crown of cells on the heart that conducts the signal down. And then we have the full movement and dynamic contraction of the heart.
- 4.38** But what's interesting about this is that, once again, in contrast to the muscle, each muscle cell on the skeletal muscle needed its own neuron. Remember, that was the, that was the motor unit. It was a neuron with one individual cell. The heart doesn't have any of that. The signal starts with that little cluster of P-cells, and then it moves through the cells themselves. So totally unlike skeletal muscle, you have this signal originating on the top of the heart, and then it sends a signal into the closest heart cells called cardiomyocytes, myo meaning muscle.
- 4.39** And so then it moves into the next cells down, and then the next cells down, and it continues to go, directly passing from cell to cell. Remember, the muscle cell was its own individual unit, where you could engage some of the muscle cells and not others, depending on how they're innervated. But with the heart, you get all of them. Once that signal starts, it passes literally from one cell to the next through these little channels. We have these gap junctions, and what we have connected between them all is what we call intercalated discs.
- 4.40** So you have one cardiomyocyte, one muscle heart cell, butted right up to another one. And they share these little channels or pores, where when one receives the electrical signal, it literally passes it onto the next and then passes it onto the next, through these little pores that are capable of pushing the signal along. Now, further with skeletal muscle, you'll recall that the signal starts up top, and then it goes down the T-tubule, and then it interacts with the sarcoplasmic reticulum.

- 4.41 Well, one of those two structures is present in the heart muscle. One of them isn't. So with the cardiac muscle, we actually do not have the T-tubule. We don't need to send a signal across the surface and then down into the depths because we're directly handing it from one cell to the next. So when one cell gets depolarized because of the P-cells turning on the signal, it will directly pass that electrical signal into the next.
- 4.42 And then at that point, we do have some stark similarities with the skeletal muscle, which is when that cell is experiencing this depolarization, the signal goes across the gap junction, that little pore that can conduct this electrical signal, and then it runs along the surface of the heart. So it's mimicking what the T-tubule was doing. And in so doing, opens up a type of calcium channel called an L-type. It's a long type. It takes a little, it's a slower signal. So we have the L-type calcium channel that will take calcium from outside the cell and pull it in.
- 4.43 Now, that's also unique compared to the muscle cell, because the muscle cell did not pull in calcium. It only used the calcium from within its own sarcoplasmic reticulum, if that sounds familiar. So we had the signal come to the next heart cell over, which opened up calcium channels to allow calcium from outside the cell to come in. And then this calcium opens up the ryanodine receptor. So the ryanodine receptor was part of the cork metaphor that I used earlier, where we now start to have the calcium squeezing out of the sarcoplasmic reticulum, dramatically increasing the calcium within the heart cell itself.
- 4.44 So before, when we had the signal just going from one cell to the next, we get a little bump in calcium, but that little bump is enough to activate the sarcoplasmic reticulum and have it uncork its calcium. Now it is belching out calcium throughout all of the heart cell. And just like the cross-bridge cycle with the skeletal muscle, which we won't revisit now, with that dramatic increase in calcium comes in exposing and shifting around of the thin filament. And then we have the myosin binding.
- 4.45 We have the exact same cross-bridge cycle and power-stroking that we had with skeletal muscle. It was just a different matter of how that calcium gets there in the first place. In the muscle, the skeletal muscle was only coming from the sarcoplasmic reticulum. In the heart muscle, it's coming from outside the cell first, and then that opens up the calcium channels on the sarcoplasmic reticulum, giving us the big bolus of calcium that we need in order for the muscle to contract in the heart in this case.
- 4.46 Now, the heart has to beat again. And you'll recall from our conversation of skeletal muscle that we have to clear out the calcium. We need the heart to contract and relax. We can't have it stay in a contracted state, otherwise the blood stops moving. We need to be dynamic. It's no surprise that the heart has multiple mechanisms to very rapidly clear the calcium. It needs to flush the cell with calcium and take it back out, and flush it with calcium and take it back out.

- 4.47 We already talked about how the calcium gets into the cell, but then when we pull it back in, we have a couple different mechanisms. The first one is once again similar to skeletal muscle. We have the first pump that's pulling the calcium back out of the heart cell, is the sarcoendoplasmic reticulum calcium, ATPase, S-E-R-C-A, the SERCA pump. That's the primary mechanism whereby the heart cell is pulling the calcium back into the endoplasmic or the sarcoplasmic reticulum. But at the same time, we have two other channels that are capable of dumping the calcium back out of the cell directly.
- 4.48 Because remember, the cardiomyocyte or the heart cell first pulled in calcium from outside the cell, and then it used its own calcium from the sarcoplasmic reticulum. So we have one mechanism similar to skeletal muscle to pull it back into the SR, the sarcoplasmic reticulum. Then we have two new ones, two new transporters, that pump the calcium back out of the cell, thereby allowing the heart cell to relax very, very quickly before it does it all over again.
- 4.49 OK, so let's go back to the anatomy of the heart. We had the P-cells all clustered together in the SA node. And then that is what—it sends its signal down the walls of the atria. And so the atria contract in this downward motion, helping push the blood down. Now, you can start to see a problem. If we continued to contract downwards, it doesn't work because the actual ejection site of the ventricles is on the top of the ventricle, albeit off to the side. So it receives blood down, but then it actually still needs to pump the blood back up and out through different vessels.
- 4.50 And so if we were only pumping the heart all the way down, down through the atria and down through the ventricles, we would have no way of getting the blood out because the exit for the ventricles is up at the sides. So the genius system within the heart is that the signal will start at the P-cells of the SA node, then contract the atria. Then the signal will go down these structures called Purkinje fibers. So it conducts the signal down to the bottom of the heart, the crown of the heart, which then enables a contraction up, pushing the blood out these upward structures that are at the top of the ventricles, again, albeit shifted off to the side.
- 4.51 What's so brilliant about this system is that, when it leaves that little SA node and then stops partway between the atria and the ventricles, it pauses for just a moment at what's called the AV node, the atrioventricular node, so the node that is distinguishing between atria and ventricles. And it pauses for just a moment. And think about how genius it is to have that brief little pause. We contract the blood down. We have a brief pause as the signal goes through the Purkinje fiber, or before it goes through the Purkinje fibers, to allow the ventricles to fill. And once the ventricles have had their moment to fill, now the signal goes through the Purkinje fiber, starting at the bottom, pushing the blood up and out through its exits.
- 4.52 Now, why does it not push the blood back up into the atria? It's because those vessels, those valves, are one-way valves. Once they have pushed the blood down, they slam shut until they push the blood down again. If there's any upward pressure, they slam

shut. They're shaped in such a way that the physics of the blood flow will close the valve itself, only allowing the blood to move out through the proper vessels, going either to the long circuit or the systemic circuit. So that is the contractility and the conductance of the heart. SA node, AV node, pause, Purkinje fibers through the ventricles, and contraction. It's a brilliantly designed system.

- 4.53** All right, now, all of this is affecting the contractility of the heart. And you'll recall that cardiac output is the amount of blood that the heart is injecting per unit time. So basically, it's how many liters coming out per minute. That is the cardiac output. And there are two important factors that influence that cardiac output—the strength of the contraction of the ventricle, so ventricular contractility, and the end-diastolic volume. In other words, how filled did the ventricles get while they were filling, while they were relaxed?
- 4.54** So again, it's the amount of volume that the ventricle is filled with, and it is the contractility. How strong can the ventricles contract? And that brings us to this famous law of the heart, named after two famous physiologists, the Frank and Starling law of the heart. And at its simplest, the Frank-Starling law is simply, the greater the filling, the greater the contraction. Or to say that another way, if the heart is filling more, it's stretching more. So the greater the filling and stretching of the ventricles, the stronger the contraction of those ventricles.
- 4.55** That becomes a pretty brilliant system because if you are increasing the velocity of blood flow because you're up and you're running around, the blood is moving faster. And so it allows the heart to fill more per unit time, but it also contracts harder, making sure that it can eject all of the blood that it was getting. That is the Frank-Starling law of the heart. So in other words, if we were to map this out, where end-diastolic volume is along one axis and we have the stroke volume along another, if we can push the end-diastolic volume out, we are getting an increase in the stroke volume. So the more the heart is filling, the more it is ejecting.
- 4.56** And remember, all of its purpose with the heart is to move the blood around. And then that brings us to the lungs, actually, as we transition now from the cardio part of the cardiopulmonary system into the lung system. The heart wants to move the blood, and it wants to move the blood largely because of what the lungs have done with that blood. So let's move on to lungs. So the learning objectives for the lungs is for you to be able to describe the basic function of the lung, which is not overly complicated, but also describe the changes in lung pressure and volume, or rather how the volume of the lungs affect the pressure and the movement of air.
- 4.57** I want you to also appreciate how the airway can experience greater or less, or diminished, resistance. And with that help you understand the difference between compliance of the lungs and elasticity. The lungs are very unique, how they have to be very dynamic. They're compliant, and they're elastic. As a part of this, we're going to

talk about a type of molecule called surfactant, and I'll do a little tangent of when that goes wrong in a clinical context.

- 4.58** And then we'll also get into something called the Bohr effect, another famous physiological phenomenon, which is basically the ability of the red blood cells to give off its oxygen, how readily it will give out oxygen, and how readily it will take oxygen onto itself, because it wouldn't be good if the red blood cells were too selfish. If blood cells, which carry oxygen—we'll talk about that more in a moment—if they got oxygen and only ever held onto it, then they're doing the body no good. You'd still have other tissues of the body basically suffocating. And then lastly, we're going to talk about the spirogram and what a clinician looks for when they're looking at the breathing patterns in an individual.
- 4.59** So briefly, the anatomy of the lungs are a pretty beautiful and elegant structure, where it's essentially like an inverse root system, where you have the root, the branches or the trunk in two big sections, splitting into the lungs, and then branching and then branching again and then branching again. So if you were to imagine a root system underground from a tree, that's what's happening with this—the trachea being the central air system, splitting into the bronchi, splitting into bronchioles, which are all these smaller little divisions and branches along this system, filling these two sacs that we call our lungs.
- 4.60** Then at the very tip tip of each one of these root systems is a series of little air sacs called alveoli. So the alveoli are these little balloons, if you will, that are constantly inflating and deflating as we breathe. As we take an inhalation, the alveoli have inflated, as air is coming in. Then as we exhale, they compress a little bit as the pressure changes, and we'll talk about that more in a moment. But that's not where it ends, because it's one thing to move air in and out, but it's another to then move the air or the gases where we want them.
- 4.61** So it's no surprise that around every alveolus, the singular, and around all the alveoli, are this extensive network of capillaries. Remember, when we talked about capillaries before, that's the business part of the circulation, of that microcirculation. And at the lungs, we need to exchange the opposite direction of gases that we might have exchanged, say, in the muscle. When the capillaries were flowing through the muscle, they were giving off oxygen and pulling on CO_2 . We need to get rid of that CO_2 , and we need to replenish the oxygen.
- 4.62** And thus, no surprise that the capillaries of the pulmonary circuit, as they're weaving around these little sacs of air, these little balloon air sacs that are inflating and deflating, they're doing the exact opposite. They are now giving off CO_2 and pulling in oxygen. And in fact, that's the key function. When we really look at the business part of the lung, it is the involvement in the interaction of the capillary. The red blood cells, which are carrying the gases, the red blood cells are primarily responsible for moving the gas around, the CO_2 and the oxygen.

- 4.63 So as the red blood cells are coming back to the lung, they're loaded with carbon dioxide. And we'll talk more about that in a moment. And then as they pass the alveolus, it gives off the CO_2 , the alveoli, the little air sacs. And then at the same time, it pulls in the oxygen to reoxygenate. And now this oxygen-rich blood starts to move throughout the body. So a fundamental part then of the purpose of the lung is to appreciate the role of airflow. How well is the lung system moving air?
- 4.64 And flow, airflow, is a function of two variables. One is a change in pressure from the atmosphere to the lung itself. Then the next one is the radius of the airway. When you think about the radius, so what is the size of the airway? Is it really big or is it narrowed? And so first of all, let's talk about the pressure change. Then we'll talk about the resistance because of the radius. And if you think about the lungs, how can we change the pressure of the air within the lungs, within the alveoli, the little air sacs? How can we change that pressure? We can change it by expanding the volume.
- 4.65 So as volume expands, so too does pressure, and in fact, in an exactly opposite way. So atmospheric air pressure is typically around 760 millimeters mercury. Those are the units for understanding air pressure. As we inhale, we actually have, surrounding the lungs, something called a pleural sac. And the pleural sac is, it's almost—imagine, if you will, my fist being a lung sac, and then I push my fist down into a balloon. And now, as I've pushed down into a balloon, my hand is encompassed in this sac, and that is the pleural sac. It's just like my hand being pressed down into a balloon.
- 4.66 The air pressure in that pleural sac is slightly less than the air pressure of the atmosphere. So the atmosphere is about 760 millimeters mercury. The pleural sac has a pressure that's about 756 millimeters mercury, so just a little below. But what that means is that it creates this slightly, relatively negative pressure based on what the air pressure is in the alveoli, because the alveoli are generally going to match the air pressure within the air sacs, whatever the environment is. And so if the alveoli have about 760, the pleural sac has slightly a few points less, that creates a negative transpulmonary pressure.
- 4.67 So transpulmonary pressure is whatever the air pressure is in the alveoli, the air sacs, minus the pressure within the intra-pleural space, the balloon, in this analogy of me pushing my fist into a balloon. So it's slightly negative. And the fact that it's slightly negative means that if the pulmonary sac is getting extended, the lungs will follow. It's basically pulling it. It's a pull pressure. And that matters because of another famous law of physiology, or just physics actually, called Boyle's law. Boyle's law is defined as pressure and volume having an inverse relationship.
- 4.68 So if I inhale, or if we have a—let's not go to the body yet, let's take a piston within a closed chamber—and we measure the air pressure within that closed chamber. And then if we start to pull that up, it's going to get hard, first of all. But as we start to pull it up and expand the volume, the actual pressure within that space starts to drop. So we have volume and pressure going in a different direction. And within the lung, you think

about, we have two different sites where we can expand the lung cavity. I can inhale through both my diaphragm, pulling my diaphragm down and opening the lungs from the bottom.

- 4.69 And I can expand my rib cage through what's called the contraction of my intercostal muscles, intercostal being the muscles weaving around the rib cage. So by inhaling, I'm expanding my rib cage. I'm pulling down my diaphragm. I actually expand the volume. So the volume is going up within my lungs as I'm pulling at them, because that slightly negative pressure pulling the lungs open as the volume is expanding. So the volume is going up, and as the volume is going up, remember, Boyle's law states that there's an inverse or opposite relationship. So I'm expanding the volume, and that means the pressure starts to come down.
- 4.70 And you may recall from the cell physiology lecture, in nature, things want to go down their concentration gradient. We talked about that in the context of electrolytes wanting to go from an area of high concentration to low concentration. Well, gases are just the same. Gases want to go from high concentration to low concentration. So when we expand the airway, as volume is going up, pressure starts to drop. It means that the airway and the air pressure within the alveolus, the alveoli, the little air sacs, it drops lower than the air pressure in the atmosphere.
- 4.71 So guess which direction the air is going to move? It's going to move in, because I've created a lower pressure in my lungs than exists around me. And then, of course, when I start to contract, my intercostals come back, the diaphragm goes back to normal. When I'm exhaling, I'm compressing the lung space, which is decreasing the volume. I compress the lung space, decreasing the volume, which increases the pressure. And now that air pressure within the alveolus goes a few points higher than the atmosphere, allowing air to come out because the gas, the air, wants to follow or go down its concentration gradient.
- 4.72 So when I want to inhale, I expand the volume, which lowers the pressure, allowing air to come in from the outside. When I exhale, I compress the lung space, which reduces the volume or the space, which increases the pressure, and now the air goes out, down its concentration gradient, even though it's physically coming up. But of course, the same would happen if I were upside down or laying on my back. So this is the aspect of the pressure difference. Remember, we're talking about the flow of air.
- 4.73 And the first variable to understand the flow was the change in pressure or the ΔP . Now you understand it. The ΔP is entirely driven by, what am I doing in my lung space to separate it, to create an offset, compared to the environmental air pressure, which will allow the air to move in or out. Now, the other component of flow was resistance. And this is entirely a function of the radius of the airway. And just like, we already actually talked about this when we discussed the peripheral nervous system.

- 4.74 So with that lecture in mind, we have this process where we discussed earlier the involvement of the autonomic nervous system, where if the sympathetic and the parasympathetic nervous systems are predominant, we will have this dynamic change in the size of the airway as we have muscle wrapping around our bronchials, contracting, contracting resulting in bronchoconstriction, or relaxing resulting in bronchodilation. And we want to breathe more readily when the sympathetic nervous system is turned on. If we have to run away from something, we want to be moving oxygen and air very, very well. So a sympathetic nervous system induces a bronchodilation, reducing the radius, thereby improving flow, air flow in and out of the lung.
- 4.75 In contrast, the parasympathetic nervous system, that's a lower state of the body. We don't need to move a lot of air, so it doesn't matter if the airway is a little narrower. So parasympathetic stimulation will result in a slight degree of bronchoconstriction. Now, that's the airway. But beyond the airway, we have the lung, which must be capable of, as I mentioned earlier, this really dynamic movement as we inhale and and exhale, and inhale and exhale. So the lung is unique because it has to have this high degree of compliance, which is a term referring to how much can you stretch something? And then it must have a high degree of elasticity, meaning how well can this something return back to its original shape?
- 4.76 And the lung is able to do this in part not only because of a high degree of a particular type of connective tissue called elastin—it is an actual springy type of connective tissue, within the lung—but also it has a high level of an oily molecule called surfactant. So if we go back to the anatomy of the alveoli, we have the lung sac and all the capillaries around it. We have a different type of cell that is predominant around this area, and it is called alveolar type II cells. The function of the type II cells is just to produce this slightly oily, watery substance called surfactant.
- 4.77 And just for fun, lest you think that surfactant is irrelevant, if you have at home a flat dish with no ridges to it, and you put a drop of water on your counter and you put that dish right on it, try to lift it up. It becomes almost impossible because the attraction of the water molecules is so substantial. So water is a little sticky. We don't think of it that way, but it has an adhesion or an attraction to itself that makes it reluctant to break apart. If we only had water surrounding the alveoli, we actually wouldn't be able to inflate them enough to get a full inhalation. The only way the alveoli themselves are able to move so much is that they have this slightly oily composition because of the surfactant.
- 4.78 Now, if we take that same little drop of water and we put that dish on top, we couldn't pull it up. If we slid it to the edge of the counter, got it off, and went to that same drop of water and put a little bit of oil in it, now we could lift and lower that plate perfectly easy. So the lipid, or the little bit of fatty substance within the surfactant, as it coats the alveoli, it makes the alveoli be very, very pliable. It really enhances the compliance and

the elasticity. So surfactant is a totally essential component to break up the surface tension of the water, because, again, water is surprisingly sticky.

- 4.79** And if we only had water around our alveoli, we wouldn't be able to generate enough force breathing in, in order to inflate them. They would stay in this deflated state. We couldn't overcome that surface tension of the stickiness of water, but the surfactant and its little bit of oil helps that happen. This is relevant in human disease because there's something called respiratory distress syndrome, or RDS. This is a potentially lethal disorder where a baby may be born without lungs that are fully formed.
- 4.80** And that's not uncommon, because if you think about a baby in mommy, the baby is not using baby's lungs. There's no breathing of the atmosphere in the lung because baby is getting all of its gas, all of its oxygen, through the placenta. Mom is doing the breathing for baby. And so the lungs are the last tissues to develop. So if a baby must be born prematurely, it's possible that the baby won't be making enough surfactant yet, because that is the last tissue to develop. Because again, in utero, the baby is not using baby's lungs.
- 4.81** So it's the last tissue to develop. So when baby is born, you may need to intervene and actually squirt a little surfactant down into the airway. And indeed, that helps the baby get a little longer time so the lungs can finish developing and baby is able to start breathing on their own. All right. Now, talking about the lungs, I've already emphasized repeatedly how essential the lungs are for gas transport. And there's two gases in particular that we've been talking about. Now we're really going to dive deep, which is oxygen and carbon dioxide. Those are the two gases that must be exchanged. Let's start with oxygen.
- 4.82** Oxygen moves through the blood in two different ways. A very small amount, like a fraction of a percent, is just freely dissolved within the plasma, which is the watery part of the blood. So we have a little bit of oxygen that is dissolved, but very, very little. The vast, vast majority of oxygen is carried on the red blood cell within the hemoglobin. So when you look at a red blood cell, it is almost entirely made up of this big complex called hemoglobin. And on the hemoglobin are these particular little iron sites, which is called heme.
- 4.83** And then on the heme is a site for the oxygen to lock in. So every red blood cell is going to be loaded with hemoglobin, and the hemoglobin's purpose is to carry oxygen. So the vast, vast majority of all of the oxygen we have in our blood at any given moment is bound to hemoglobin within a red blood cell. Now, as I mentioned earlier, and I was getting a little ahead of myself at the time, the red blood cell must know when to give off oxygen and when to hold on to it, right? It's not a rational, sentient little being that intuitively knows, "Oh, it's time to give up the oxygen. Now it's time to take it."
- 4.84** No, there's something called the oxygen-hemoglobin dissociation curve. In other words, it's this curve that is a manifestation of how readily the oxygen wants to be

bound to a hemoglobin, or rather to say that another way, how readily the hemoglobin wants to hold on to the oxygen. And that's primarily a function of the amount or the pressure or the concentration of the oxygen. Where you have a lower oxygen concentration, it wants to hold on to it a little more readily. Where you have a higher oxygen concentration, it's willing to give it up. And there are a couple of different variables that will influence the degree to which the hemoglobin wants to hold on to the oxygen.

- 4.85** Firstly is a higher level of CO_2 . If there's a higher level of CO_2 , of carbon dioxide, then the red blood cell will more readily give off its oxygen. And think about how brilliant that is, because that's exactly what's happening in systemic circulation. As the blood is flowing across, let's say, a working muscle, the muscle is going to be producing a lot of CO_2 , which is a consequence of just nutrient metabolism. If the muscle is burning a lot of glucose or burning a lot of fat, it's creating a lot of CO_2 . CO_2 is a product of metabolism or burning nutrients.
- 4.86** And so, as the blood is moving across the muscle, for example, it's giving up its oxygen more readily. At the same time, another key signal that helps the hemoglobin give up its oxygen is hydrogen ions. And we talked earlier about, when you are splitting a lot of ATP, ATP hydrolysis, you're always getting hydrogen ions. So once again, you think about a very busy muscle. It's splitting a lot of ATP to go through the cross-bridge cycle, and that further contributes to pushing this dissociation along with hemoglobin. So the hemoglobin is even more readily giving up the oxygen than it was before.
- 4.87** Now, what's interesting about both of those, higher CO_2 and higher hydrogen, is that both of them happen with exercise. So it's a brilliant system, whereby the muscle is basically demanding from the red blood cell and its hemoglobin saying, "You must give me your oxygen. I need it. And you know that I need it because of all the CO_2 I'm making and all the hydrogen ions." But there's one other instance where the body is smart enough to know it needs to give up oxygen, and that is when there's a higher level of a molecule called 2,3-diphosphoglycerate. 2,3-diphosphoglycerate. Sometimes just abbreviated as 2,3-DPG.
- 4.88** This is something that naturally occurs within red blood cells, but especially it happens at the placenta. So once again, we're talking about pregnancy, because if you think about mom breathing in oxygen, she's doing the breathing for the baby. She's not only breathing in oxygen for her own cells, she's also breathing in enough oxygen for her baby's cells. And then as baby is having waste products, she must get baby's CO_2 and her CO_2 to breathe off. So she's doing the breathing for two people.
- 4.89** But how does her body know to give up oxygen more readily at the placenta, so the baby can get a very high amount of oxygen? It's because her placenta sends the signal to the red blood cells to have a higher level of this 2,3-diphosphoglycerate, which allows the hemoglobin to give off the oxygen more readily. So that's a signal from the

placenta to mom's blood cells saying, "Hey, you are growing another little human very busily right now. It needs a lot of oxygen, so we're going to take a lot."

- 4.90** One of the many, many instances where we see the mother's body deliberately making sacrifices on behalf of the offspring. It's an elegantly designed system, again, largely driven by the blood demands and the oxygen demands of the placenta, which act as the site of exchange between mom and baby. It's the placenta that are basically acting as the lungs for the baby, pulling in oxygen that will go to baby, and then taking the CO_2 from baby and giving it to mom to breathe off. OK, now that was all oxygen. Let's talk a little bit then about carbon dioxide.
- 4.91** Carbon dioxide uses both of the same two systems that oxygen did. Namely, it will get carried on the hemoglobin, just like oxygen does. But then also it will have a slightly higher degree of being dissolved in the plasma, the watery part of the blood. So it has both of those in common. But CO_2 is unique in that it has a third, and it is by far the overwhelming majority of how the CO_2 is carried, namely through bicarbonate. The bicarbonate is a genius little reservoir of CO_2 . And the formula for bicarbonate is hydrogen and then CO_3 , HCO_3 . So within that is a CO_2 potential.
- 4.92** So the bicarbonate is this big reservoir which isn't affecting, it's not hurting the body. It's a necessary intermediate. But remarkably, it ends up being the biggest reservoir of CO_2 . So when we have CO_2 coming from very busy working muscles, for example, some of it will be converted into bicarbonate. And then as it gets to the lungs, it will split apart. It will break apart and no longer be bicarbonate, but will turn back into just water and CO_2 , giving off the CO_2 to be breathed out and simply leaving a water molecule behind. So bicarbonate is this fascinating intermediate step in taking the carbon dioxide from a busy working cell and then taking it to the lungs to be breathed off.
- 4.93** All right, now, what, however, makes us breathe? I want to spend a few minutes talking about the regulation of breathing, why we're breathing as deeply or as quickly as we are. So there's the depth of the breathing and then the rate of the breathing. It's largely driven by what's called chemoreceptors, or these chemically sensitive receptors, but it's also driven by what's called pulmonary receptors. So let's talk about both of them.
- 4.94** Firstly, the chemoreceptors are manifested in two different sites. There are central chemo receptors, which are receptors that are in the brain, sampling the blood. And then there are peripheral chemoreceptors that are right outside the heart. So firstly, let's talk about the brain and the central chemoreceptors. You have heard before something referred to as the blood-brain barrier. The blood-brain barrier is this protective wall, if you will, that will be very selective about what it allows, what chemicals, what molecules can move from the blood into the brain and vice versa.
- 4.95** CO_2 , oddly enough, is something that can move across, but CO_2 is not what is sensed by the chemoreceptor. The chemoreceptor is actually only sensing the hydrogen. It's

sensing the pH change. So we're not moving hydrogen, though. Hydrogen can't go across the blood-brain barrier, but CO₂ can. And so what happens is, the CO₂ will move across the blood-brain barrier, and then it will undergo this chemical change to actually get back to bicarbonate. But at the same time, as you're converting CO₂ to bicarbonate, you actually get a free hydrogen.

4.96 So the CO₂ will migrate across the blood-brain barrier, undergo chemical transformation, resulting in higher hydrogen production. Then it's the hydrogen that will activate these central and chemo-sensitive aspects of the brain, which will tell the brain, "We have a lot of CO₂, let's breathe a little faster," and thus, ventilation will increase. So again, this is a hydrogen sensitive chemoreceptor, but we have to get to the hydrogen by pulling in the CO₂ first. So the CO₂ crosses the blood-brain barrier, and that stimulates the breathing.

4.97 Now, beyond the brain and those central chemoreceptors behind the blood-brain barrier, we have these peripheral chemoreceptors that are right outside the heart. We have carotid bodies and aortic bodies. So once the aorta is coming up out of the heart from the left ventricle, so the left ventricle beats and it pumps the blood up the aortic valve. Remember, that was one of the lub-dubs we talked about. We have some receptors right there in the aorta and right in the blood vessels that are coming off the aorta called the carotid arteries.

4.98 These are the arteries that are going up our neck. So whenever you're putting your finger there to try to feel a pulse, those are some of the carotid vessels. So we have these chemoreceptors at these two sites, in the aorta and right above the aorta, in the carotid receptors called the aortic bodies and carotid bodies. They're sensing not only the hydrogen, but also the CO₂, and they will get stimulated to increase, send a signal to the brain saying, "Breathe more rapidly or slowly." If there's more hydrogen and more CO₂, it will say, "Breathe more rapidly."

4.99 And that'll be a primary mechanism whereby, when you start exercising and you start breathing faster, it's not like the body knows the muscles are working inherently. It's not the muscle directly stimulating a nerve. It's that as the muscle is working, we're getting changes in CO₂, we're getting changes in hydrogen, and that begins to stimulate both the central and these peripheral chemoreceptors in the brain and in the heart, respectively.

4.100 Now, as I mentioned, we also have the involvement of receptors within the lung itself. The lung has receptors that can detect irritants like smoke. So if you inhale an irritant, the lung can sense it and force you to start coughing or to restrict your breathing. It also has stretch receptors, so that if it senses you're reaching a point of maximal stretch, it'll stop you from breathing in anymore. Those are some of the pulmonary or lung-specific receptors that are not at all based on the chemistry of the blood, but rather irritants coming from the airway or the stretching itself of the lung wall.

- 4.101** Now, we can get to the final point of the cardiopulmonary system by talking about spirometry. Spirometry is generally a concept that is mostly involved with clinicians, but it's a wonderful way to understand the nature of a breathing problem. So if someone has a pulmonary problem, spirometry is going to reveal it, because spirometry gives us what's called a spirogram. And a spirogram is the dynamic movement of breathing. So it is going to be—it's almost like you think of something that's measuring the intensity of an earthquake. That's what we're getting when we're looking at the spirogram.
- 4.102** When you put on—nowadays, it's digital. Once upon a time, it was an immersed water movement of you blowing air in and out of a little cylinder, and it would move up and down. Now it's just all digital. But if you give the patient an electronic or a digital spirometer, you have them breathe, and there are certain tests you can do, and two key metrics that we'll talk about. But just to establish a common viewpoint, when the patient is inhaling, the spirogram will be reflected as an upward curve. When the patient is exhaling, it will shift and go to a downward curve.
- 4.103** Now, there are myriad, there are many different metrics that can be looked at and analyzed through the spirogram, and there are two that I want to emphasize. The first is the vital capacity. The vital capacity is when you look at the spirogram and you tell the patient to exhale as much as they can, all the way out, and then maximally inhale. That is the vital capacity. In other words, what is the maximal amount of air that the lungs can hold? Vital capacity.
- 4.104** Mind you, that's not all of the air. Even if you exhale as much as you can, you'll always have some air left over in the lungs. That's called residual volume. We're not talking about residual volume. The vital capacity is you breathe everything out you could, and then you took the deepest inhalation you could, or the patient did, or the subject did. That's vital capacity.
- 4.105** Another important metric that I want to highlight is something called the FEV1, the forced expiratory volume over one second. So in this same instance where we had the patient breathe everything out and then breathe in as much as they could, and then if we told them, "Now blow all of your air out as rapidly as you can, as forcefully as you can," that would give us the amount of air that could be moved in one second. In other words, it would tell us the FEV1.
- 4.106** Why am I highlighting those two markers in particular? Because there are about a dozen total markers that you can look at on a spirogram, because each is a reflection of a fundamental lung pathology or a disorder. Within the lungs, we can break up lung problems into two general families of problems. A restrictive problem or restrictive pulmonary disorders or diseases, or obstructive pulmonary disease, restrictive and obstructive.
- 4.107** Restrictive is typically a disease that happens through fibrosis or scarring of the lung—like, for example, asbestos. Asbestos is a chemical that can be inhaled and it can

damage the lung, creating this scarred fibrotic lung sac, creating a restrictive pulmonary disease. In the case of a restrictive disease, the lung has become leathery tough, and it does not expand as well as it used to, so the compliance has been lost. And thus we have a reflection.

4.108 If we tell the person to inhale as deeply as can, which was part of the vital capacity, we will see that in this person, if they have this fibrotic lung disease, a restrictive disease, the vital capacity will be significantly diminished. So a reduced vital capacity is a key sign of a restrictive disorder, like the person's inhaling dust particles or smoke. That can cause that damage, particularly.

4.109 Now, in contrast, an obstructive lung disorder can inhale perfectly fine. The problem isn't the wall or the alveolar sac. In this case, the problem is generally the airway. The airway won't allow air to exit very well. So the person can inhale perfectly fine, but when they try to breathe out, things get really wheezy and narrow. It's like you're trying to breathe through a straw. The lung can totally hold as much air as it could before. The complication is, you can't move the air very well. And in this case then, the FEV1 ends up being the essential component.

4.110 If a patient inhales and we tell them to exhale as quickly and forcefully as they can, what would happen immediately in a healthy lung, they breathe all of their air out in just seconds, it will take them many, many more seconds. The slope of the exhalation is much shallower and more gradual in the lungs in this person's airway, all because, and all of indeed a reflection of, the compromised air flow. So reduced FEV1 reflects an obstructive disorder, so a problem in the airway itself, and then reduced vital capacity is a reflection of a restrictive disorder, suggesting a problem of the lung sac.

4.111 And with all of that in mind, you are now well informed of cardiopulmonary physiology in the system, the balance and the intricate ballet and waltz of the heart moving around the lung's product, or in other words, the gases that have been exchanged from the lungs. Thank you.

4.112 Could you kind of describe what's going on in the body when a mother has preeclampsia?

4.113 Now that we know how the body works, one of the other classes I teach at BYU is how things work when they're not working well, including preeclampsia. So in preeclampsia, that is one of the most common pregnancy disorders that a woman can have, and it is manifest as, essentially, a mounting kidney failure. So she has this problem of the kidneys. Now, why am I invoking the kidneys at all? One of the key metrics that the clinician is going to be measuring, if the woman starts to manifest with preeclampsia, which is a really high blood pressure, they have her pee on a little stick every time she comes in to measure how much protein is coming out in her urine.

4.114 So what's happening is that, as she is developing her placenta, this is basically the lungs for the baby, right? So it's no surprise that it should be massively vascularized. You

need tons of blood vessels. The placenta is just loaded with blood vessels. In order to stimulate the growth of the blood vessels, she has to have this signal called VEGF. So VEGF is a protein that stands for vascular endothelial growth factor, V-E-G-F, VEGF. VEGF will start to get expressed from her body, from the placenta, activating the placenta, so what's called an autonomic stimulation. So it's a cell releasing a hormone that activates itself. We'll talk about that type of stimulation in the endocrine disorders tomorrow.

- 4.115** So we have this autocrine stimulation of the placenta producing VEGF, which acts on the placenta to stimulate the growth of new blood cells. Now, pause on that for just a moment. The kidneys are heavily vascularized as well, and they are very busily filtering all the time, filtering, only letting out what should come out. But it needs VEGF in order to maintain its normal filtration. That's one of the other aspects of VEGF. As much as VEGF is necessary to stimulate blood vessels, we also need it to maintain normal kidney function. And I'm actually acting out what's called the glomerulus, where we filter out stuff.
- 4.116** But if we don't have VEGF, the glomerulus starts to fall apart. And in this case, it's allowing things to leak through that shouldn't be getting through, like protein. Now, what does all of this have to do with VEGF? In some women, for some unknown reason, her placenta starts to release these things called soluble VEGF receptors. So the VEGF was a protein coming from the placenta that should go right back to the placenta and activate the placenta. But as it's releasing VEGF, it starts also, for some unknown reason, releasing these soluble receptors that basically gobble up all the VEGF.
- 4.117** And so she starts to have lower and lower levels of VEGF, which on one hand can create a problem of the placenta in the baby, but that's not what's going to kill mom. What will kill mom, and indeed, preeclampsia is one of the most common causes of mortality with pregnancy, it will be that the kidneys no longer work. So she basically starts experiencing kidney failure, which is very lethal, all because her placenta is just misbehaving. Now, it ends up being fine for baby, another reflection of just how much mom's body is sacrificing for offspring. She is suffering, but the baby actually is generally doing fine.
- 4.118** The baby is going to get—the placenta is going to be very greedy, but she may, I mean, so preeclampsia is the technical term before the kidney problem has gotten so bad that she starts to have seizures. Once she starts having seizures, it's no longer pre anything. It's eclampsia, and usually she dies shortly after. There's almost nothing you can do at that point. So it becomes this constant battle between, can we give mom more time or give baby more time to finish growing, before this preeclampsia is going to become so problematic that it kills mom?
- 4.119** So it's a constant tug-of-war with the clinical team, but always monitoring the protein, how much protein is coming out in your urine, because that's going to be a reflection of the kidney damage. How much longer do we risk letting the placenta stay in? Because

it's the placenta that's causing the problem. So the solution is immediate and obvious, which is, get the placenta out. The moment the placenta is out, within hours, the kidneys restore function, her blood pressure goes to normal, and she's perfectly healthy. But if the placenta comes out, the baby must come out with it.

4.120 And so it is this absolute balancing act of how much longer can we give baby, especially time for baby's lungs to develop? And so if a woman has sufficiently bad preeclampsia, they'll usually start injecting her with steroids. And I don't mean anabolic steroids, but I mean anti-inflammatory steroids like cortisol molecules, because one of the many effects is to stimulate the development of the lung.

4.121 So you can basically drive baby lung development early, knowing, "Hey, mom's really getting bad. We're going to have to get baby out early. Let's preempt this by helping baby start developing baby's lungs a little earlier than it normally would, so that when the baby is born preterm, baby will have a greater chance of having optimal or functional lung system." Anyway, that's a long, long, long answer to preeclampsia, but I just happen to be expert on the topic.

Lecture 5: Gastrointestinal Physiology

At a glance — Gastrointestinal physiology.

- 5.1** As we continue our journey through the physiological systems of the body, we get to one of the more misunderstood but still appreciated systems, which is the gastrointestinal system. Now, anatomically, the gastrointestinal system is enormous. It encompasses everything from top to bottom, or to be a little more crass, from mouth to anus and everything in between. That is the GI system. We're not only going to take that journey through this tube of the gastrointestinal system, as we discuss the GI system, but we are going to also discuss, we must include two accessory organs, namely the liver and the pancreas.
- 5.2** Even though they're not actually part of the tube itself, they're connected to it and there is no understanding of digestion and then absorption, which we'll get into in a moment, without appreciating the relevance of those two accessory organs. The outcomes for the lecture that I hope you get, I want you to understand the different sections of the GI system and the accessory organs. Within this process, I'm going to come back to four themes as we move down the intestinal system. We're going to talk about motility, secretion, digestion, and absorption.
- 5.3** We're going to talk about some of the key functions of the GI tract and the accessory organs, particularly the liver and the pancreas, as noted. We're going to discuss some of the influences and the role of things like bile and the digestive enzymes. Then lastly, we're going to discuss some gut hormones. That's a little bit of a teaser for the upcoming endocrine system lecture that we'll get to after. But it's even more relevant nowadays because the most common weight loss drugs are actually designed to mimic hormones that come from the intestines. You'll become expert on the nature of those drugs, how they work and why.
- 5.4** All right. Now, as I mentioned, the fundamental process and purpose of the GI system is to take the essential stuff from what we're eating and drinking and then get it into the body, get it into the blood. We have to have this process of not only moving the food through the system, but also breaking it down into its most base component parts, and then moving those parts from the intestines into the bloodstream, where they will do the good that we want them to.
- 5.5** Now, the GI tract is the tube that I already mentioned, from mouth to anus and all of the connections in between. We're going to make that journey and talk about it from top to bottom. But then, as I noted, we'll include two primary accessory organs that are not only facilitating, but in fact are totally essential to proper nutrition and digestion. Let's start in the mouth as we are starting from top to bottom.
- 5.6** The very act of chewing, of course, is grinding up the food, which is preparing the food for further digestion. We could say that's the beginning of digestion. In fact, that is

facilitated because, within the mouth, there is in fact the secretion not only of saliva, and the very liquid of the saliva will help facilitate the movement of the food stuff as it goes down, but we also have an enzyme within the mouth. We talked about enzymes in previous lectures, but the enzyme is a molecule that is facilitating some reaction. It's making something else happen.

- 5.7** The particular digestive enzyme that we have in the mouth is specific to carbohydrate digestion. It's an enzyme called amylase—salivary, or it's based in the saliva, salivary amylase. And amylase will take a long stringy carbohydrate, which is a polysaccharide. It's a connection, basically, of glucose molecules, and it will start cutting them up a little bit. We get a little bit of carbohydrate digestion or starch digestion in the mouth. But there's no absorption. We're not actually moving any nutrients from the mouth into the bloodstream yet.
- 5.8** So there's a little digestion, snipping up a bigger molecule into smaller pieces, but we're not doing anything else with it. It's staying in the mouth to move down. Indeed, let's move down. Let's go from mouth to esophagus, which is the tube connecting the mouth from the throat down to the stomach, from pharynx to stomach. The top part of the esophagus is actually skeletal muscle, meaning we can control it. That is when you deliberately close your throat or you swallow. All of that is part of the skeletal muscle part of the esophagus.
- 5.9** Then the last two-thirds of the esophagus are going to be what's called smooth muscle. That is a muscle we can't control. We can initiate the swallowing and the movement of food from the mouth down into the esophagus. But then once it's past that initial point of us pushing it down, then the rest of that action is happening through a process called peristalsis, which is something you're going to hear me mention multiple times. But peristalsis essentially refers to this steady, rhythmic, smooth muscle contraction that is moving food steadily down from top to bottom, the steady movement.
- 5.10** Now, at the top, we have what's called the upper esophageal sphincter. This is what we can close. Whenever you hear the word sphincter, it's something that can be pinched off or closed off. We control the top part, this upper esophageal sphincter, because it's a skeletal muscle. But then the lower esophageal sphincter, right between esophagus and stomach, as it actually tucks through the diaphragm, which we discussed when we talked about breathing, that is a smooth muscle, so we can't control that one. It will open and close as necessary.
- 5.11** Now, as we have swallowed the food from mouth through esophagus, and the peristalsis continuing to move it down, we get through that lower esophageal sphincter, and then we get to the stomach, one of the really first moderately complicated parts of the digestive system. Everything up until that point has been pretty simple. The stomach is not some uniform structure, though. It is broken up into distinct sections. We have at the top this curved outward portion called the fundus.

Then we get to the body of the stomach. Then we have what's called the antrum, or the distal, or last portion of the stomach.

- 5.12** There are actually different cell types even expressed across the stomach. So even down the length of the stomach itself, there are different types of cells doing different types of things. And I'll mention, too, in particular in just a moment. But the stomach is also, and this is reflected actually throughout much of the intestine, the GI system, but mostly in the stomach, it is highly dynamic. It is capable of just substantial volume changes. It can go from a very small state to a state that is easily four or five times larger volumetrically than it was before. So it is capable of substantial expansion.
- 5.13** And part of this expansion is enabled by the folds and wrinkles within the stomach. If you were to open up the stomach, you would see all of these deep, deep folds of tissue, and that allows it to expand. And then, of course, the folds would get a little pushed out as the stomach is expanding. But that's a part of the anatomy of the stomach, to enable this expansion so that it can hold the food.
- 5.14** Now, I say food, but at this point, we actually invoke a different term. As the stomach starts to contract and relax, it starts to mulch this food. Now, we start to use a term called chyme. So chyme, C-H-Y-M-E, chyme is now the term to refer to this sort of mushy, mulchy mix of the food as it's starting to churn. As it's churning with the stomach starting to compress and relax and start to mix it all up, we have the involvement of two particular cells that I alluded to a moment ago, parietal cells and chief cells. Each are very important in digestive processes.
- 5.15** The first part, the parietal cells, one of the products of the parietal cell is just hydrochloric acid, a hydrogen and a chloride coming together, creating an extraordinarily acidic environment. Now, the actual process of the stomach lining, moving that hydrochloric acid out, involves some of the same processes we discussed in the cell physiology lecture, namely, some active transport, especially. Because if you are continuing to pump hydrogen ions out into a space to make it ever more acidic, you're pumping the hydrogen ions against their concentration gradient.
- 5.16** You'll recall that's a process that requires energy. It requires ATP. That's like us pushing the stone up the hill. That's very much like the lining of the stomach, what is called the epithelium, pushing the hydrogen ions from the cells of the epithelium into the body of the stomach. Now, of course, we don't want to wreck the stomach lining, and so we have also the heavy production of mucus. The stomach is a heavily lined mucus lining of itself, because the mucus will explicitly protect its own cells from getting chewed apart because of the acidity of the stomach environment.
- 5.17** Now, the acidity is remarkable. The human stomach is about as acidic as almost any animal on the planet, which suggests some pretty interesting things about what the diet should be for humans and what we are perhaps designed to eat. But our stomach is about as acidic as a wolf's stomach, not quite as acidic as a scavenger, in other words,

animals that eat dead things; their stomachs are even a little more acidic. But the human stomach has a pH of about 2, which is exceptionally acidic within the animal kingdom.

- 5.18** It means that we are well suited to really clean foods as we eat them because that pH environment will really destroy pathogens very, very well. If there's an inclination of viruses or other pathogens that we consume, ideally, and by design, they're going to largely get broken apart because the acid will just kill them. Another thing that the acid will do is start to denature the protein. When you eat a protein, proteins are actually these heavily folded-up chains of amino acids. We have individual amino acids strung together, and then they start to fold.
- 5.19** But in order for us to digest them, we need to unfold them and open them up. The acidity of the stomach actually starts to tease apart some of these connections that are folding it. Now, it's not cutting at the amino acids, but it's starting to cut at some of the other pieces that are holding it together in a folded state. The pH, the very acidic environment, starts to open up some of these proteins, which is going to be helpful because we want to start digesting them even in the stomach.
- 5.20** Then the last thing that I'll mention, especially when I get to the next cell type, is that when you have this heavily acidic environment, it actually activates the first of what's going to be several protease enzymes. Protease. Just in the name, you can start to hear which type of food or which macronutrient it's going to be digesting. Just as a reminder, within the diet, we have three macronutrients, three main parts of our diet. We have fats, carbohydrates, and proteins.
- 5.21** The acidity of the stomach helps activate an enzyme called pepsinogen, which we'll discuss more when we get to the next cell types. But this starts to activate pepsinogen, turning it into its active form, called pepsin. Then pepsin actually starts coming in and cleaving or snipping apart some of those peptide bonds, starting the first process of actually digesting protein. So pepsin, when it's active, is a protease.
- 5.22** All right, now we're not done with the parietal cells yet. As much as I was getting a little ahead of myself and alluding to the next cell type, the parietal cells also produce one other very essential digestive factor, which is something called intrinsic factor. Intrinsic factor is essential for the absorption of an essential vitamin, vitamin B12. Vitamin B12 is essential for healthy cell function. You must have it. If you don't have enough, it actually results in a very lethal form of anemia that I won't get into. But suffice it to say, you must have vitamin B12.
- 5.23** Not only must you eat it, which is only going to come from animal-sourced foods—there's no plant-sourced food for vitamin B12, and humans must have vitamin B12—but it's one thing to eat it, but then it's another thing to actually move it from the intestines into the bloodstream. And intrinsic factor is the necessary intermediate, if

you will. It essentially hitches arms with the vitamin B12 and escorts it from the intestines into the blood, to allow all of the cells of the body to function properly.

- 5.24** All right, so just to briefly review, the parietal cells give us both a very acidic environment by creating the hydrochloric acid, which then is promoting the beginnings of the first protease or proteolytic enzyme, pepsinogen, which we'll revisit in just a moment. But at the same time, it's producing intrinsic factor, which is essential for the absorption of vitamin B12 that we're getting from the diet. It's one thing to eat it, but it's another thing to actually move it into the bloodstream.
- 5.25** Now, let's leave the parietal cells behind, as we go from one cell type of the stomach to the next, which is the chief cells. The chief cells are actively secreting the pepsinogen. You'll recall that I just mentioned, pepsinogen is the name of this early form of this proteolytic enzyme. It's in an inactive form. When the chief cells first release pepsinogen, this is a molecule, an enzyme, that has the potential to split up a protein, but it can't yet. It is still in an immature form. It needs to be converted from its immature form, pepsinogen, to its active form, pepsin.
- 5.26** The hydrochloric acid is what's doing that. It's snipping off a redundant part of the molecule, and then the pepsin can work. Then the pepsin, in turn, will take those opened or denatured proteins and start doing the first cutting, breaking them down into smaller peptides, which are smaller than—we have a full protein, a string of amino acids. Then we start cutting them apart into smaller pieces, and these are called peptides. But it's not done yet. We're going to have more protein digestion. We've not cut them into small enough pieces yet. We need the involvement of other protease enzymes that we'll get to after we leave the stomach.
- 5.27** But before we leave the stomach, let's just review. I'd mentioned at the very outset of the lecture that, as we go through this journey from mouth to anus, there are four things that I'll always touch on. Motility, so the movement, secretion, digestion, and absorption. Now, with motility, the stomach has the potential to store food, and normally it will store food on the order of a few hours, 3 to 4 or 5 hours, during which time it's churning and churning, helping produce this process of chyme, this gastric slurry of partially digested food stuff.
- 5.28** That's the motility. It will hold on to the food for a while, and then it'll slowly start leaking the food out from the stomach into the small intestine, which we'll get to. The secretion, there are some secretions. The stomach is, in fact, releasing some things into the stomach. The lining of the stomach is releasing some things to help with digestion, like pepsinogen, which gets converted to pepsin because of the secretion of hydrochloric acid. Then at the same time, we have the secretion of intrinsic factor. Those are the secreted variables when we talk about the stomach.
- 5.29** Then digestion, we don't have any further digestion other than a little bit of the protein getting digested because of pepsin. But we also still have the salivary amylase that is

continuing to be able to work, even as it moves through the stomach. We still have just those two parts of digestion. Still a little bit of carbohydrate digestion and a little bit of protein digestion now. No absorption, though. Even though we're starting to break these molecules from big into smaller, we're not actually moving anything into the bloodstream yet. That's going to happen in the next part.

- 5.30** Let's move from the stomach past the antrum or the distal or farther part of the stomach into the small intestine. This is the business part of the GI system. Everything that we really need now to happen is happening at the small intestine. Everything else has just been a prelude to what the small intestine is going to do, because it's at the small intestine where we digest everything into its most base components and where we start absorbing and ultimately absorb all of it.
- 5.31** Whatever we don't digest and absorb in the small intestine essentially will not have any other action. It will just remain in the intestines all the way out, with one brief exception that we'll talk about in the large intestine. Now, the small intestine is very, very long, and it has three distinct sections. It will move from the duodenum, which is first; that's what's called the proximal, the most proximal part or the first part. Then it goes to the jejunum, and then it goes to the ileum. Duodenum is right after the stomach, which progresses to the jejunum, which then progresses to the ileum.
- 5.32** Just as an interesting tangent, I'm going to touch on this at the very end of this lecture, but not in any great detail—you are all familiar with gastric bypass surgeries, what's now sometimes called metabolic surgeries, where you will go in, a surgeon will go in and trim a part of the stomach away. The most common form of gastric bypass is what's called the roux-en-Y bypass. What they've done is actually snip the stomach apart, taking a big round-bodied stomach and creating just a teeny little pouch. They will trim off the stomach and close that one off now. Staple it so it's shut.
- 5.33** But at the same time, they actually cut the duodenum. We have the stomach connecting to the duodenum. Then the duodenum goes down into the jejunum. They'll cut away between the duodenum and the jejunum, and basically move that stomach and duodenum to the side, and then take that new cut small intestine and connect it to the now-little pouch of this newly formed stomach.
- 5.34** Based on what I just said, where most of the digestion and absorption's happening not only in the small intestine, but primarily in the duodenum, now that we have bypassed not only a big portion of the stomach, but also the duodenum, it's easy to see how someone loses so much weight so quickly after they've had a gastric bypass procedure. You not only can eat very, very little food because the stomach is now so small, but you've also bypassed the most important part of the small intestine for reasons we'll get into in just a moment, which is the duodenum.
- 5.35** That's where most digestion and absorption is going to occur. The person not only is eating much less than they were before following this surgery, but even the little bit

they are eating, they're not digesting it very well and they're not absorbing it very well, which creates for, to put it mildly, some significant GI distress, where the food is moving through their intestines far too quickly and they will get terrible diarrhea and GI problems because of it. But they will lose a substantial amount of weight, but it comes at a substantial cost.

- 5.36** Interestingly, by cutting off the stomach, they lose the access to the chief cells, which, as you'll recall, produces—or sorry, the parietal cells—which is producing intrinsic factor. Now, even if they were eating a food with vitamin B12 in it, they cannot absorb it because they have lost access to the intrinsic factor, and thus for the rest of their life, they must go in and get vitamin B12 injections. Again, this is a substantial operation. It is messing with fundamental parts of the intestines, all to just basically force a person to lose weight. Anyway, now you're a little more familiar with that, but we will revisit the gastric bypass procedures at the end when we talk about the gut-derived hormones.
- 5.37** Just to recap where we're at now, we've left the stomach, then we've gotten to the small intestine, and it's three distinct sections, duodenum, jejunum, ileum, with the duodenum really being the primarily important part. Now, what is common across all of the duodenum is the nature of the small intestine wall, what's commonly referred to as a brush border, a brush border, because it's almost like, if you were to microscopically zoom in, it's like you're seeing the hairs of a brush, where they have these finger-like structures that are, again, microscopic.
- 5.38** But as you zoom in, you start to see these finger-like structures called villi. They are these little round extrusions up into the intestinal wall. Then along the villi, if we zoom in even further, are more finger-like structures called microvilli. We have finger-like structure on finger-like structure, microscopically, as we're zooming in. What this does is fantastically expand the surface area of the entire small intestine, meaning that we have an incredible amount of space for food to start getting absorbed or for the micro-molecules of the food to start getting absorbed.
- 5.39** Because the more surface area we have, the more potential points of contact there is, and more potential points of contact means more absorption, so greater movement of food stuff. Within all of the finger-on-finger structure, namely the microvilli, it is not only heavily vascularized, so loads of capillaries, which you'll recall, the capillary is the smallest unit of the blood vessel, but it's the most important because that's where everything is moving. We're moving things into the blood vessel, but at the same time, we have lymph vessels called lacteals.
- 5.40** We have capillaries representing the smallest functioning unit of the blood system or the blood-carrying system. Then we have lacteals, which is the analog or the corollary, where this is the smallest functioning unit of the lymph system. Now, you may be wondering, why on Earth do we have lymph there? It's actually because the lymph

system is involved with the absorption of one of the macronutrients, which we'll get to in just a moment.

- 5.41** All right, so that is the nature of the cellular structure of the epithelium or the lining of the small intestine. Fingers on microscopic fingers, or in other words, villi with microvilli on top of them. Now, let's go through those four same variables—motility, secretion, digestion, and absorption. Motility, there's still very much this peristalsis, this wave of motion, steadily moving things ever closer to just being ultimately excreted. That's the peristalsis, of course.
- 5.42** There is no direct release of digestive enzymes from the small intestine into the intestinal space. However, we are going to get into the accessory organs in just a moment, namely pancreas and liver. But there is one relevant enzyme that's being released. Even though it's not directly involved in digestion, you're going to see soon that it's necessary for proper digestion, which is an enzyme called enterokinase. The small intestine is releasing this enzyme into the lumen or into the tube of the small intestine. It's releasing enterokinase, which will become relevant for reasons we'll get into in just a moment.
- 5.43** But enterokinase itself is not a digestive enzyme, so it's not directly secreting digestive enzymes. By that, I mean the wall of the small intestine is not producing small enzymes that are involved in digestion. Now, however, the small intestine is the main site of digestion, but that's because it's right in the duodenum that we have the insertion of the pancreas and the liver. That's why most digestion is in fact happening in the small intestine, not because of the small intestine itself, but because of what's connected to it. Then it's also the main site of absorption. We'll get to absorption later after we discuss digestion.
- 5.44** Now, let's then take a step out of the lumen or the tube of the small intestine and explore these two accessory organs. The pancreas is involved in most digestion. Now, we will revisit this idea when we start talking about the endocrine system in the next lecture. But the pancreas has two functional parts. It has what is commonly referred to as the endocrine pancreas, which is the part of the pancreas that's releasing hormones into the bloodstream.
- 5.45** But then it has a very robust part of itself called the exocrine pancreas, which is the part that is involved in the production and release of digestive enzymes, that the pancreas is releasing through this duct into the intestine. The pancreas is very, very busy, not only helping with digestion in the small intestine by releasing digestive enzymes as a part of its exocrine pancreatic function.
- 5.46** But then interestingly, it's even involved on the back end, which is, once digestion has occurred and we've moved the food stuff in through absorption, now we have nutrients running through the bloodstream, and then the endocrine pancreas is relevant for telling the body what to do with those nutrients. The pancreas is very, very busy when

it comes to overall metabolism and nutrition. Again, facilitating and indeed being necessary for digestion, and then telling the body, the cells of the body, what to do with the food molecules that have been brought into the body itself.

- 5.47** Now, in addition to those enzymes, which we'll get to and we'll define the three of them, each corresponding to the three macronutrients, but before I say that, I want to just emphasize that the pancreas is also dumping bicarbonate into the small intestine. Now, we discussed bicarbonate in the context of being a reservoir for carbon dioxide in the blood. That has nothing to do with this. In this case, we're talking about bicarbonate because of its ability to buffer acid. Everyone knows from chemistry that if you have an acid and you sprinkle in some bicarbonate, it will start to resolve the acidity. It will start to turn this into a neutral solution.
- 5.48** Now, why on Earth would we want the pancreas to be squirting bicarbonate into the small intestine? It's because we have all that acid coming from the stomach. As the acidity from the stomach is moving with the chyme, from the stomach into the small intestine, we want to stop that acidity because we don't want it to start hurting the small intestine, because it does not have that thick mucus lining that the stomach has to protect itself. Indeed, it has a very, very small lining, which we need because we want to start absorbing nutrients soon.
- 5.49** We need the small intestine to have only very little mucus to facilitate absorption of the nutrients. But at the same time, that leaves it exposed to the acidity of the stomach juices as they're coming down. So the pancreas takes care of it. The pancreas begins releasing a lot of bicarbonate, negating the acid right there, because the pancreas has its insertion point right at the top of the duodenum. So the moment that food is coming from the stomach, it's bang, interacting with the products of the pancreas. A genius design.
- 5.50** So, we have negated the acid. Now we can move on to the other digestive enzymes that are coming from the exocrine pancreas. They can be lumped into three distinct families. Again, each reflective of the three distinct macronutrients: proteins, carbohydrates, and fats. The first family is the protease family, proteases or the protease enzymes. Then we have amylase, and then we have lipases. That is protease for proteins, amylase for carbohydrates, and then we have lipase for fat or lipids. Let's start with the proteases.
- 5.51** Now, earlier, you'll recall that I mentioned, so we not only have the pepsinogen being cleaved or split into its active form pepsin, and that was happening in the stomach, but now we have some other ones that, as they are released from the pancreas into the intestine, they are released in their inactive form. They're not turned on yet. There are two primary ones I want to mention, trypsinogen and procarboxypeptidase. Now, they are different, but first and foremost, they are inactive. Before I talk about each of their specific effects, they need to be cut apart.

- 5.52** Now, that is the relevance of the enterokinase. If you recall, I mentioned that the one secretion of the small intestine itself is enterokinase, this enzyme that will take these immature protease enzymes and turn them on. Enterokinase will take trypsinogen, cut it, and turn it into active trypsin. Then trypsin will act as what's called an endopeptidase. It will take a long line of peptides, as we've started some process of protein digestion, and start cutting it in the middle. So cutting it from the middle out to the edges. It's an endo-peptidase.
- 5.53** And then procarboxypeptidase will be active as enterokinase activates it, and it will turn into carboxypeptidase. So we had trypsinogen turned to trypsin, which acts as an endo-peptidase, cutting the peptide chain from the middle. And then we have enterokinase converting procarboxypeptidase into carboxypeptidase, which acts as an exo-peptidase. So cutting the peptide chain or the protein chain from the ends. So trypsin is starting in the middle, as an endo-peptidase, and then carboxypeptidase is an exo-peptidase, starting to cut from the outside. So trypsin from the middle, carboxypeptidase from the ends.
- 5.54** Ultimately, giving us these either individual amino acids or a di- or tripeptides. Nothing will be bigger than three individual amino acids connected. So it's either a tripeptide or a dipeptide or an individual amino acid. That is going to be the final product of all of these peptidases or proteases activating or acting on the protein. All right, that one was a little complicated with a lot of complicated words. Let's leave the proteases behind, now that we have a fairly good understanding of them, and go to the remaining two.
- 5.55** I've already discussed the nature of amylase because we just have another type of amylase that is continuing and indeed finishing off what the salivary amylase started. Salivary amylase started by taking this chain of glucose and cutting it into pieces. Now the amylase from the pancreas is finishing it off, cutting them down into its most basic components, most especially glucose. So glucose is generally going to be the end product of the amylases.
- 5.56** Then we also have lipase. Now, we're going to have to talk about the liver in a moment to truly appreciate the actions of lipase, which we will do. But lipase is going to take a triglyceride molecule, which has what's called a glycerol backbone. There's this little backbone molecule, which would be my knuckles right here. Then the lipase is going to snip off an individual fatty acid, snip off another individual fatty acid, and then we're going to be left with what's called a monoglyceride. We still have the one little backbone and one fatty acid. That is going to be the end result that the lipase enzymes want.
- 5.57** Again, taking a triglyceride, pulling off one fatty acid, pulling off a second fatty acid, and then leaving one last fatty acid with its little backbone. But before we can do that, before the lipase enzymes can get to that most basic unit of fat, the triglyceride, we have to take the fat globule and then start to break it apart. That brings us to the liver. Now, before I mention that aspect of the liver, I just want to highlight just how

important the liver is in overall health and its relevance within the GI system, which is only a part of its function.

- 5.58** The liver is involved in many, many processes. Indeed, there are few tissues in the body that are as involved in as many processes as the liver is. In fact, there may be none. The liver is involved in the clearance of red blood cells. When red blood cells are old and need to die, the liver takes care of it. The liver also produces most of the proteins that are in our blood, the main one being albumin, which is not a protein we've talked about, but albumin is the majority protein in the blood, carrying all kinds of molecules, serving all kinds of purposes. All of that comes from the liver.
- 5.59** It also secretes a variety of hormones, some of which we'll discuss in the endocrine system lecture coming up. But let's get back to the GI system itself and highlight two other aspects—one, which is the elimination of waste products from the blood, but also, most importantly, dealing with toxins that are coming from the intestinal tract, which we'll talk about more in a moment. Then lastly, very relevant to fat digestion, the liver is producing bile. The bile salts are necessary for the digestion of fat.
- 5.60** But before we talk about bile, I want to just pause and highlight the liver's role in clearing the blood. Now, there's an interesting feature of blood flow when we start talking the liver. Normally, as we discussed in the cardiopulmonary system lecture, you'll recall that, in a normal instance, the heart is moving blood. We have blood that has just come from the lungs, freshly oxygenated, loaded with nutrients, loaded with oxygen, and that moves to arteries, then to arterials, then to capillaries.
- 5.61** Then the capillary bed is what's found through each individual tissue. That's where the blood is giving up and taking in products from the cells. Then it will go from capillary to start merging into venules, then veins, and then get back to the vena cava, and all the way back to the heart. However, the liver has its own little set of rules. The liver does have its own blood flow, this hepatic artery that will feed the liver and give the liver all of its nutrition by exchanging molecules across the capillaries.
- 5.62** But it's also part of what's called the hepatic portal system. There are only two instances of a portal system in the body. We're going to talk about one. Then we're actually going to talk about the other when we get to the endocrine system upcoming. In this case, a portal system means you have blood vessels coming down, splitting into capillaries as they go across tissues, merging again as bigger vessels, then going through another tissue into a big split-up capillary, and then merging and going back to the heart. That's exactly what the liver is doing.
- 5.63** So for example, you have blood flow coming down and feeding all of the intestines, capillaries, capillaries through all of the intestines. Then those capillaries begin to merge together into what's called the hepatic portal vein. It merges, and then it goes into the liver and spreads out into capillaries, and then it comes back to the body. The

liver is right on the front lines of whatever is getting absorbed from the intestines. So, everything that's getting absorbed from the intestines goes to the liver first.

5.64 The liver will decide what to do with all of these things, including taking in anything that came in that shouldn't have come in. It could be taking in bacteria, harmful pathogens; viruses that come in and get absorbed from the intestines, it goes to the liver first. The liver is the detoxifier of the bloodstream. It will filter all of the stuff coming from the gut. For better and worse, the good and the bad, the liver will know just what to do with it. It is literally the front lines as stuff is getting absorbed from the intestine. So thank goodness for the function of the liver.

5.65 Now, that also means it's the first one to see a lot of the nutrients, but it also has a direct role in digestion. So let's then go back to digestion and talk about the biliary system. Within that term, you can hear the word bile. So the biliary system includes the liver, the gall bladder, and then the ducts that are connecting the liver and gall bladder to the small intestine. The liver is constantly making bile salts, and then the bile salts are getting stored in the gall bladder. Then as fat starts to come down the intestine, then the gall bladder will contract and eject all of its bile into the small intestine.

5.66 Now, bile is like a detergent. Any time you're, of course, scrubbing the dishes in the sink, you know that if you put in a detergent, a soap, it's going to break the fat into smaller pieces. That is a process called emulsification. The bile salts are acting as emulsifiers. When we eat fat or when you have fat in a watery solution, it wants to glob together. There will be these large globules of fat moving from the stomach into the small intestine.

5.67 But in order for the lipase enzymes to start pulling off individual triglycerides or the fatty acids from triglycerides, we have to break this fat globule down into smaller parts. That's the action of the bile salts. The bile salts will come into this fat globule and start carving out smaller little things called fat droplets. So fat globule, bile salts come in almost like little invaders and start carving off little pieces called fat droplets. And then the fat droplet is small enough that we can now have the lipase enzymes come in and start pulling off the individual pieces of the triglycerides.

5.68 Now, one other part about the bile salt that is very unappreciated or totally overlooked, if it's known at all, is its function in protein digestion. Now, I understand, I just got done talking about how bile is necessary for fat digestion, and it is. But it's a little-known fact that bile also activates and enhances the proteolytic enzymes, or the proteases. In other words, if someone eats a pure load of protein, they will digest and absorb a certain amount of that protein. If they eat that protein with fat, they will digest even more.

5.69 This could be why studies in humans find that if you have a human working out and give them a load of protein, they'll get a certain amount of muscle protein synthesis or muscle growth. If they eat that same amount of protein with fat, they'll get even more

muscle protein synthesis or muscle growth. It could very well have, just simply, it could be a function of the digestive capacity of the protein has been improved because of the bile salts. You will not release bile salts unless you eat fat. I think there is a lesson there.

5.70 In our modern-day environment, if you'll pardon the tangent, we have a pronounced fear of fat. And yet in nature, there is no such thing as fat that comes without protein—sorry, protein that comes without fat. In our fear of fat, we've pulled the two apart, only focusing on the protein and vilifying the fat. But in nature, there is never, zero instance of protein alone. Indeed, every single example of a protein comes with fat. There is no exception. I think that's how we should eat it.

5.71 Just again, as a tangent, there's something to be learned from nature and the structure of the human body and the digestive system to eat these proteins the way God intended, if you will, which is protein comes with fat, no exceptions. That's how we should eat it. Indeed, the digestive system suggests that's the case. Because when we include the bile salts, which are only released when we eat fat, we digest protein even better.

5.72 All right, now one other tangent is just to focus on the gall bladder a little more. As I noted, the gall bladder is holding on to bile salts that are constantly dripping in from the liver, and only contracting and injecting the bile when it senses fat coming down. Unfortunately, if we don't contract the gall bladder often enough, the bile salts start to get saturated, and indeed, what's called supersaturated, and start to crystallize. What was nicely dissolved into this slurry solution of bile starts to become gallstones. That happens again, primarily because we're not ejecting the gall bladder.

5.73 When do we eject? When do we contract and move the bile out? When we eat fat. One of the most common causes of gallstone formation, and the eventual necessity to just remove the gall bladder entirely, is because a person stops eating fat. This has long been known, we've known this for decades, that the more a person avoids dietary fat, the more likely it is that they're going to have to get their gall bladder removed because they have so much stones getting formed in the gall bladder. So a healthy gall bladder is emptying itself often, which, of course, means a healthy person is eating fat often.

5.74 All right, now, we've talked a lot about digestion, not only because of the involvement of the small intestine and the release of enterokinase enzymes to help with protein, but also the involvement of the pancreas and its proteases, lipases, and amylases, but then also the inclusion of the liver. We've digested, we've broken everything down into its smallest pieces. Now we want to actually move it from the intestine into the blood. That's why we ate it in the first place. Let's start with carbohydrates.

5.75 As I noted, the optimal outcome of carbohydrate digestion is generally going to be glucose and a little bit of fructose as well. But typically, that's what we will have gotten it down to. We will have broken everything down into glucose or fructose. Now, and of course, the majority of it being glucose. Fructose has its own transporter. Actually, it's

called the glucose transporter, but it's GLUT5. That is just an open door that allows the fructose to come in as needed. It's specific for fructose.

- 5.76** Oddly enough, despite being called the glucose transporter 5, GLUT5, glucose does not get absorbed through that transporter. Glucose gets absorbed through what's called the sodium-glucose symport. Now, you will recall the term symport. A symport was a form of transporting molecules where one molecule could hitch a ride as down the other one's concentration gradient. So glucose is hitching a ride or it's piggybacking on sodium's concentration gradient. So the sodium is getting pulled in through this sodium-glucose symporter, what's called an SGLT. So the SGLT opens for sodium and the glucose just slips in.
- 5.77** Interestingly, and then afterwards it's in the epithelium and then it can move into the bloodstream at the back end of the cell. Interestingly, that is a point of attack for a very popular anti-diabetic drug class. With diabetes, that's when blood glucose levels get too high. We'll talk about that in the endocrine lecture coming up. One intervention that a clinician may take advantage of is having the patient take what's called an SGLT2 inhibitor. This is a drug the person can take that will block those glucose transporters.
- 5.78** Even though it's wanting to pull in the sodium, it wants to pull the glucose in with it, we block that down. It will improve blood glucose levels. It does improve the diabetes. But unfortunately, by keeping the glucose in the intestine and not allowing it to get absorbed, it stays in the intestine and it pulls a lot of water with it. A common outcome of these drugs is some potentially catastrophic GI distress. There's a lot of bubbling, there's a lot of movement going on in the intestines, to the point that a person can have some socially awkward situations.
- 5.79** Or I would say, as just a very rational being, if we're trying to block the glucose from coming in, maybe just have them eat less glucose. If that's what the drug is trying to do, spare yourself all of the GI distress and the potentially socially catastrophic situation, and just eat less glucose. But even still, now you understand how it moves in. Glucose moves in through a symport, taking advantage of sodium's concentration gradient. We're done with glucose then, so let's go to proteins.
- 5.80** Once we have finished, once the peptides have been acted on, the peptide being the chain of amino acids, by all of the proteases, you'll recall that we were left with three potential products— tripeptides, three amino acids, dipeptides, or individual amino acids. They each have their own method of transportation. The smaller, like the tripeptides, they can come in through a process called transcytosis. They actually have a little bubble that will form around them and escort it across the cell and then release it into the bloodstream on the back end.
- 5.81** But then otherwise, we have symports. We have a symport that will move the smaller peptides and the individual amino acids, where it will be moving symport with hydrogen or symport with sodium. But once again, we have two symports. You recall

the symports from the cell physiology lecture and the discussion about membrane transport, moving things into or out of the cells. We have, again, this transcytosis, a bubble that forms around the bigger peptides, bringing them in. Or more commonly, we have symport, where the smaller peptides or amino acids are slipping in with either hydrogen or with sodium.

- 5.82** All right, that covers carbohydrate, absorption of the smallest base units of it, and the absorption now of protein. Let's get to the last one, which is fat absorption. Now, you'll recall that we needed the involvement of bile salts in order to carve off small little fat droplets from the larger fat globule. Then once we did so, we could have the lipase enzymes come in and start pulling off individual fatty acids, ultimately leaving us with just a monoglyceride, the backbone, a small little backbone, and then the fatty acid itself. Those are going to be the final products.
- 5.83** Once we get to those final products, in the wall, it's actually an interesting process. We break down a triglyceride into its smallest parts, the fatty acids. Once it gets into the epithelium, which it can do freely because it's a fat, fat is able to move through the cell surface on its own. Because this is a fat, the surface of the cell is a fat, the membrane is loaded with fats, and it just senses the fat and allows the fat to pass right through. It doesn't need a specific transporter.
- 5.84** The fatty acids and the monoglyceride move right into the cell of the epithelium very freely. Once they've done this, it actually undergoes a process called re-esterification. It basically takes the parts of the triglyceride and then brings them back together, forming the triglyceride again. It took the triglyceride, broke it apart, brings it into the cells, and then forms it again.
- 5.85** Then those triglycerides begin interacting with cholesterol that can be coming from the diet, which again, freely moves through because it's a fat, and proteins forming what's called a chylomicron. A chylomicron is basically this big bus that's carrying a lot of fat and some proteins. But interestingly, unlike the carbohydrate-based units and the protein-based units, so glucose, basically, and then the amino acids or peptides, those go right into the blood. The chylomicron does not.
- 5.86** If you'll recall, when we talked about the villi and the microvilli, we described how the microvilli have both capillaries and lacteals. The other molecules we just mentioned are going into the capillaries of the microvilli. But the chylomicron actually moves into the lacteal, which is like a lymph capillary. It moves into the lymph. The first movement of these fats into the body, actually into the body, because when it's in the intestines, we're not really saying it's in the body, but it's not yet. As it moves from the intestines into the body proper, it does first move into the lymph.
- 5.87** Then the lymph has its own vascular system, if you will, which will then dump the, anything moving in the lymph gets back into the blood at a few different entry points. There is this direct interaction. What goes into the lymph will only stay there shortly,

and then it will get into the bloodstream. But not all fats do this. Not all fats get lumped into this big fatty bus called the chylomicron, which then goes into the lymph.

- 5.88** As a little bit of a metabolic tangent, and I am a metabolic scientist, that's what happens for what's called long-chain fats, which, admittedly, is the overwhelming majority of different types of fat that we eat. Almost all of the fats we eat are long-chain fats. However, there are medium-chain fats, and then there are short-chain fats. Short-chain fats are very uncommon, but let's start with medium-chain fats.
- 5.89** Medium-chain fats come from things like coconut oil or goat dairy. In fact, all of the medium-chain fats are named, have a goat-like name—caprilic, capric, caproic. It's all of these terms derivative of an old, I think, the Greek word for goat. But they all come from goat dairy. You'll notice, if you've ever had goat dairy, it tastes a little more tart than milk from a cow, for example. That's because as the fats get ever shorter, they get a little more tart.
- 5.90** We go through the medium-chain fats, which do not become part of the chylomicron. I'll talk about the fate of them in just a moment. But then as we get a little shorter even still, we get to the short-chain fats, which are even more tart. In fact, tart to the point of tasting like vinegar. Vinegar is, in fact, the shortest of the short-chain fats. Then we have a couple others that are very short as well, like butyric acid, which is uniquely enriched in butter. Butter coming from the word butyric acid, one of the short-chain fats.
- 5.91** What's interesting about the short-chain fats is that they are almost only a product of fermentation. If you taste a cucumber and then you taste a pickle, they will not taste the same. One will be much more tart than the other. If you eat some stewed cabbage and then eat some kimchi, one will taste much more tart than the other. It's because the process of fermentation has produced these short-chain fats.
- 5.92** In fact, as a metabolic scientist, one of the reasons I'm so enthusiastic about fermented foods is that when you ferment something, you've allowed bacteria to, first of all, eat the sugars, which is the only thing they'll eat. Bacteria will only eat starches and sugars, so carbohydrates, but what they leave behind are short-chain fatty acids, which have a pretty meaningful metabolic benefit. They really punch well above their molecular weight when it comes to metabolic benefits, improving all kinds of metabolic perks, which I won't get into.
- 5.93** Now, back to digestion. As I noted, long-chain fatty acids will get absorbed through a chylomicron into the lymph and take a long, steady route through the body. The shorter, so the medium- and the short-chain fats get directly absorbed. They don't go into the lacteal. They not only do not become part of the chylomicron, they also do not get absorbed into the lacteal or the lymph. They go right into the blood.
- 5.94** Now, recall the movement of blood, where we have blood that goes from the intestines; it doesn't go back to the heart. Remember, as the blood flows through the

intestines, it goes into the hepatic portal system. A portal system means, as we use that term in cardiovascular system, capillaries to vessel, to capillaries again, and then back to the heart. The liver starts seeing a lot of these medium- and short-chain fats, and they cannot be stored. As the fats are shorter, only long-chain fats can be stored as fat. Short-chain fats only have one fate, which is they must be burned.

- 5.95** As the liver is forced to burn these fats, it burns them so quickly that it starts to create ketones. If you've ever heard about using something like coconut oil or, let alone, short-chain fats like apple cider vinegar to stimulate ketogenesis, now you know why. Because it has a direct line to the liver and they must be burned, they cannot be stored.
- 5.96** All right, now let's finish our journey through the GI system before we talk about the gut hormone. The final part, of course, is the large intestine. This is at the end of the small intestine, we've now entered just a phase of essential—essentially, it's just mass movement. In other words, taking all of the waste, which is anything that was not absorbed through the small intestine, and now it's what's going to become waste that's exiting the body in the form of feces. It has a slow journey through the large intestine, and then it will connect to the rectum, which will then, of course, exit through the anus.
- 5.97** The motility here through this large intestine is once again peristalsis. The steady contraction and movement of the remaining mass, at this point waste, so we won't even call it food. There's no secretion, there's no digestion, it's already done. There is a little absorption, but it's now only water. Any water that is left is coming out just to help dry or dehydrate the feces, so it's not just this big, watery mess. Otherwise, that would be diarrhea.
- 5.98** In a normal, healthy state, we're absorbing just a little water and we're dehydrating the feces on its way out. Then, indeed, we get out. As we move from the, at the end of the large intestine, at the end of the colon, we get to the rectum, then the anus. Now, the one final point to discuss is the actual function of the internal anal sphincter and the external anal sphincter. There are two sphincters stacked right on top of each other, essentially. The internal is smooth muscle. Remember, smooth muscle is muscle we cannot control. That is just subject to the autonomic nervous system.
- 5.99** If the sympathetic nervous system is turned on, that is closed. If the parasympathetic nervous system is turned on, it will open. Now, thankfully, we are not subject to the whims of the nervous system when it comes to evacuating feces because the external anal sphincter is skeletal muscle. That's one we can control, which gives the human this thankful grace period to control when they will actually defecate. But again, both of those sphincters must relax in order for the feces to actually exit.
- 5.100** All right, we have finished our journey from mouth to anus. But before we leave this topic behind completely, just because it has now become so relevant with the birth, well, indeed, the explosion of very now popular weight loss drugs, I wanted to briefly

talk about a class of hormone from the small intestine called the incretins. The incretin hormones are these known small intestine hormones that have powerful metabolic functions.

- 5.101** The discovery of the incretin hormones actually happened in a group of humans that underwent roux-en-Y gastric bypass. The same surgical procedures that I described earlier are the actual origin for the discovery of the incretin hormones. Again, this is taking the stomach and snipping it into one small, teeny little pouch, and then cutting off the duodenum and the remainder of the stomach, basically shoving it to the side, and then taking that further-down tube beyond the duodenum, and now connecting it to the smaller pouch of a stomach.
- 5.102** What they found was twofold in these people. The first thing they found was that their diabetes got better almost immediately. These were all these people, once upon a time, you would not even be eligible for gastric bypass unless you were both overweight and you had serious type 2 diabetes. They found that the diabetes improved very, very quickly. That gave birth to questions from these clinical teams. Why is it that this person is still fantastically obese, and yet now their diabetes is gone? They are as non-diabetic as a lean person. What has happened?
- 5.103** In looking at the blood of these individuals, they found that there was these huge changes in some hormones, namely the incretins. The one that I want to talk about, because this is the most famous and the one that is leveraged the most in modern weight loss drugs, is a hormone called GLP-1, or glucagon-like peptide-1. GLP-1 is the hormone that these weight loss drugs are trying to mimic. But before I get into that, let me just mention what GLP-1 is doing. On one, it has multiple effects, all of which are beneficial metabolically. It makes sense to want to take advantage of what GLP-1 does.
- 5.104** Firstly, it increases fat oxidation, so the burning of fat. GLP-1 will directly act at fat cells to start activating a process called lipolysis or the breakdown of stored fat. It will also induce or activate satiety centers in the brain, telling the brain that, "Hey, you're full, you don't need to eat." Combined with the next effect, that becomes very powerful, which is, it slows peristalsis. GLP-1, when it goes up, it slows the movement of the food through the intestines, in fact to a point that it can potentially become problematic, which I'll get to in just a moment.
- 5.105** Now, independent of those three effects—increased fat burning, activating satiety center, so diminishing hunger centrally, and slowing down the movement of food through the intestines, which also helps the person feel full longer—GLP-1 inhibits a hormone called glucagon. Now, we're going to talk a lot about glucagon in the upcoming endocrine disorders lecture. But glucagon, its primary job is to increase glucose levels in the blood. It's insulin's opposite, as we'll discuss in the next lecture. Whereas insulin wants to lower blood glucose, glucagon wants to increase it.

- 5.106** So if GLP-1 goes up and it lowers glucagon, that helps lower the glucose levels, thereby improving diabetes. That is, in fact, the main reason why they found that people who had higher levels of GLP-1, it appeared to eradicate their type 2 diabetes. That is one effect. Now, the weight loss, once these drugs started being used for weight loss, because at the earliest stage of the use of these drugs, it was just to help lower glucagon to improve diabetes.
- 5.107** Then they found, "Hey, these people also lose a little bit of weight. So let's just increase the dose and increase the dose by five times of these molecules that will act like GLP-1." So these drugs now, these weight loss drugs, fall into a family called GLP-1 agonists. And you have celebrities bragging about using these drugs on red carpets at all these glamorous events. At this point, when they have increased the dose by five times, you still get that glucagon inhibition which improves blood glucose, but you have really slowed the intestines down now. Indeed, you've slowed them down to a potentially lethal degree.
- 5.108** Now, you'll recall that we had peristalsis, which is the steady contraction and movement of the mass of food and waste from esophagus all the way out. When these drugs are activating the GLP-1 receptors so much, it almost paralyzes them. I say almost, but there are some instances when, in fact, it has. There are some people who take these drugs and it freezes their intestines completely, which can be very lethal. They can die within just a couple of days. But also, they find that food ends up staying in the stomach for potentially up to 24 hours.
- 5.109** You'll recall, I had said earlier that food should only be in the stomach for just a few hours, maybe four, maybe five hours tops. But by slowing everything down so much, the food is literally festering and sitting in the stomach. There's something—I don't want to mention the names of these drugs, and there are nowadays so many of them, but the person will get these putrid burps as the food is sitting there and just festering in their stomach because it's not able to move on. Then the gases of digestion start coming up through the esophagus and out the mouth. These are a class of drugs that definitely will result in weight loss, but they come at a substantial cost.
- 5.110** Now, we have finished our journey of the GI system from mouth to anus, discussing all of the intricacies, the motility, the secretion, the digestion and absorption, moving down from mouth through esophagus to stomach to small intestine, ultimately colon, and then the way out. We discussed also the inclusion of the accessory organs, namely liver, and especially the pancreas, namely the exocrine pancreas. That's how we get the nutrition and the fuel we need. Thank you.

Lecture 6: Endocrine Physiology

At a glance — Endocrine physiology.

- 6.1** The endocrine system is a bit of an odd fit when we are talking about the system's physiology of the body, because all of the other topics that we've discussed have distinct sections. It's one particular tissue, for example. It's the heart, or it's the lungs, or it's the nervous system and the neurons we're talking about. When we talk about the endocrine system, we're now actually branching out from one individual organ to talking about a system-wide effect.
- 6.2** Because after all, the endocrine system really is this process of moving one signal from one part of the body and to another part of the body. It's sending a signal. It's transmitting. But before I get that far, let's just go over the learning objectives for this particular topic. Firstly, I want you to understand the purpose of hormones. Why does the body have a system like this built into it? Next, I want you to know how hormones are classified and how they're created, how they move through the blood, and how the structure of the hormone will determine how it moves.
- 6.3** I also want you to be familiar with the most famous endocrine axis, which is the hypothalamic-pituitary axis. We'll talk about some of the key hormones that are a part of that. Then we'll talk about the pancreas and its role. Then we will talk about specific endocrine disorders. Just a little highlight because it's just so obviously—this topic lends itself so obviously to some clinical relevance. We'll touch on that.
- 6.4** Now, as I alluded to a moment ago, a hormone is a signal, or it's a way to send a signal from one part of the body to another part of the body. For example, how does fat tissue know that it's time to start breaking down fat when the muscles start exercising? There's not a direct neural connection from muscle to fat, but the muscle will begin releasing hormones that will tell the fat, "Hey, we need energy," and the fat will start breaking down its stored energy. Again, the purpose of a hormone is for one cell type in one tissue to send a signal to a different cell type in a different tissue. It's the telegram or the email that is moving.
- 6.5** Now, what is the nature of that email or that message? The message, or that is to say, the hormone, comes in three different forms. A hormone is going to fit into one of three families. It's going to be a peptide hormone, so protein-based. It's going to be an amine, which is built on an individual amino acid, or it's going to be a steroid hormone, which is built on cholesterol. As much as we vilify cholesterol, it is totally essential for the synthesis of steroid hormones. Those are the three categories of hormones—peptides, amines, and steroids.
- 6.6** Most hormones in the body are peptide hormones. That is the majority of hormones. Indeed, most of the hormones we're discussing and have been discussing are peptide-based hormones. All of the hormones that come from the pancreas, peptide. All the

hormones that come from the hypothalamus or the pituitary, they're also peptide-based. Even the gut hormones that we just talked about and mentioned in the previous lecture, those are all peptide-based. And peptide, I just mean it's a little string of amino acids. That is the nature of the hormone.

- 6.7** Now, amines are much, much smaller. These are usually built on a single amino acid or maybe two amino acid products put together. It's going to be tyrosine or tryptophan. You've heard of both of those amino acids before. The amine family of hormones includes some very famous ones, like thyroid hormone and the catecholamines that we've discussed previously, namely epinephrine or norepinephrine, synonymous with adrenaline or noradrenaline. Those are both built on tyrosine.
- 6.8** Then even another famous hormone called melatonin is built on tryptophan. There are a variety of hormones that are just built on an individual amino acid. Then we come to the biggest, most complicated, which are the steroid hormones. That is an enormous family of hormones. That encompasses the adrenal corticoids, which we'll talk about, like cortisol, for example. That's a steroid hormone. Even the very famous sex hormones, all estrogens, all androgens, are built on this, what's called a steroid nucleus, so a part of a cholesterol molecule.
- 6.9** All right, so those are the three classes of hormones—peptides, amines, and then steroids. The different structure actually then dictates how the hormone actually moves through the bloodstream. Some of them can just be dissolved in the blood, and they're just moving through the blood happily, no need to be regulated. That includes what's called the water-soluble hormones. The water-soluble hormones are the peptides. Any peptide hormone is water-soluble, freely just being released into the blood to move at will.
- 6.10** And the catecholamines, so adrenaline and noradrenaline or epinephrine and norepinephrine. They also are free to just move through the blood, unregulated. Now, the lipid-soluble hormones are a little more complicated. This involves steroid hormones and thyroid hormones. You can sense the division within the amine family. Some of the amine hormones, like the catecholamines, are water-soluble, and they can just flow freely. Others of the amine hormones, like thyroid hormone, is not. It is lipid-soluble and it needs, like the steroid hormones, a carrier. It must have an escort taking it through the bloodstream.
- 6.11** When we look at the lipid-soluble hormones, we're actually left with two different pools of the lipid-soluble hormones, including sex hormones and thyroid hormone. One of the two pools is going to be a free pool, but it is an exceptionally small amount; is free just to flow through the blood. It's a very, very low amount. Then the majority, and I mean like 1,000 times more, will be part of what's called a bound pool. There is a specific protein that is carrying that hormone through the blood, escorting it, actually protecting it from getting degraded.

- 6.12** The bound pool of the lipid-soluble hormones acts like a reservoir, where if you need some of the steroid hormone very, very quickly, you can have the protein unbind the steroid hormone and allow it to flow freely. But remember, there's only a very little amount, like 1,000th of the amount is free. Then 1,000 times that is bound. That, again, just acts as a reservoir because it's only the free version of the steroid hormones that can actually act and do what we want the hormone to do.
- 6.13** Then while it's bound, it can't act; it can't signal like we want, but at least it's being protected and held on to for later use. We have the free pool and the bound pool of the lipid-soluble hormones. Now, the hormones themselves can act or signal in different ways. It's one thing to be carried or moved through the bloodstream, but then what does it do? There are three different ways a hormone can signal or send its signal. I've mentioned this a little bit in a previous lecture, but there is autocrine signaling, there is paracrine signaling, and endocrine signaling.
- 6.14** Autocrine is when you have a cell that is releasing a hormone that activates itself. It comes right back to it. It seems a little odd, doesn't it? Why not just keep the signal within itself? That is what it is. So autocrine is when a cell is releasing a chemical substance, a hormone, which will then act on itself. Paracrine is when you have two adjacent cells. So these are two cells that are literally butted against each other, and one sends a signal to the other. We'll talk about that in a moment because that happens in the pancreas, between alpha cells and beta cells, producing glucagon and insulin.
- 6.15** They have a direct communication between each other. In this instance, with paracrine signaling, it's one cell type releasing a hormone that acts on a neighboring cell, but of a different type. This isn't the autocrine. It's not the cell activating itself or sending its own signal to itself. It's sending the signal to a neighbor, but the signal never actually gets into the blood. It doesn't need to go that far. It's just acting in its own little neighborhood, cell to cell.
- 6.16** Now, endocrine signaling is when you have a cell actually releasing its hormone product into the bloodstream itself to move through the blood. That is unique compared to the other two, because autocrine doesn't need to go into the blood. It just stays within its little domain. Paracrine, the same. It just stays in its own little cellular neighborhood. But endocrine, classic endocrine signaling, is when the hormone is released into the blood to move throughout the entire system, having effects everywhere.
- 6.17** Now, those same two classes of hormones, water-soluble or lipid-soluble, have varying effects at the cells throughout the body, based on how far into the cell they can get. Now, remember, the membrane of the cell is basically this fatty little envelope. It's actually two layers, but it's a fatty, both of them are fatty layers. They're loaded with fats. Lipid-soluble hormones cannot get through that membrane. They can only come to the surface and knock on a door.

- 6.18** There will be a specific receptor designed for this water-soluble hormone to come to. The water-soluble hormone can't go inside. It will knock on the door, and then there will be a series of what's called second messenger molecules. Just like if someone comes to my house and they want me, they will knock on the door. Then there's going to be a series of events, including a dog barking, my kids fighting each other to run to the door to see who can answer it first, then one of them running to come get me, and then me actually coming to the door. The person who knocked didn't come in, but the end result is a series of events, all of which are second messengers, eliciting the desired outcome, whatever it may be.
- 6.19** That's what happens with every water-soluble hormone. That's how peptide hormones will act, and that's how the catecholamines will act, because those are the classes of water-soluble hormone. That is not the same as the lipid-soluble hormones. The lipid-soluble hormones are like thieves in the night. They are able to just slip through their fellow fatty molecules of the cell membrane. This would be the individual just coming directly into my home and doing what they wanted, finding me, perhaps.
- 6.20** The lipid-soluble does not need a second messenger. It is the messenger. It comes into the cell and either binds to receptors within the cell or goes straight into the nucleus to induce and turn on the synthesis of different genes, which will become different proteins. It will elicit the effect all on its own. It doesn't need a second messenger. The nature of these hormones, whether it is peptide or amine or steroid, that not only will determine how it moves through the blood, free or bound, but it will also determine how it acts or signals to its receptor cells.
- 6.21** Does it just get to the membrane and no further? Knocking on the door and then using second messengers to send its signal? Or is it able to just go right into the cell and just do what it wants to do? Now, that is some of the basics of the endocrine system. Let's leave the basics behind and start to get a little more specific. I've mentioned in the learning objectives that the first thing we want to talk about is this famous endocrine axis, meaning an involvement of different organs, the hypothalamus to the anterior pituitary. Here we have, it is the most famous axis.
- 6.22** So the hypothalamus is going to send what's called hypothalamic hormones through a portal system. You remember when we talked about the hepatic or the liver portal system in the GI system lecture? This is an instance where the blood flow is going to the hypothalamus, branching into capillaries, coming back into a vessel, and then branching into capillaries again as it moves through the pituitary gland. The hypothalamus is the first. Then it goes through this little portal system to the pituitary, sending a hypothalamic hormone down a very, very short distance to the pituitary. Then the pituitary will send the signal out into the body.
- 6.23** Depending on the pituitary hormone, we're going to go through a few different examples. We're going to talk about growth hormone, then we're going to talk about thyroid hormone, and then we're going to talk about some adrenal hormones. All of

these are subject to the hypothalamic-pituitary axis. We basically have a CEO, being the hypothalamus, then we have a manager, being the pituitary gland; and then, depending on the nature of the pathway, we can have a grunt worker, a line worker in the factory, or whatever the case may be.

- 6.24** The first one, growth hormone, and actually, let me back up. Where we have the line worker, there is the inclusion of one last gland. The signal will start in the hypothalamus, go to the pituitary, and then in the case of growth hormone, growth hormone comes directly from the pituitary. It acts and goes out and does what it wants to do. However, when we talk about the next two that we'll discuss, thyroid and the adrenal hormones, the signal comes from the interior, the pituitary, the interior pituitary, and then goes to the thyroid, then the thyroid gland will release its hormone.
- 6.25** Or in the case of the adrenal hormones, the signal comes from the pituitary gland, goes down to the adrenal glands, and then the adrenal glands will release the end hormone. When we talk about this then, we actually can split it up into three different levels. The end gland is often referred to as the primary gland. Then in that instance, the pituitary is referred to as the secondary gland. Then the hypothalamus is referred to as the tertiary, so the CEO that's dictating everything. It is up at the top. It's not directly involved. It's two layers removed. We have tertiary to secondary to primary, depending on whatever the primary gland is, of course, based on whatever hormone of interest.
- 6.26** So let's just go through and talk about the regulation of these three families or classes —growth hormone, thyroid hormone, and then the adrenal hormones. We'll talk about multiple there. Then that will allow us to really get a good idea of the hypothalamic-pituitary axis. Then we can move into the pancreas, one of the more relevant endocrine glands, but separate from this axis. It acts on its own. It doesn't act on the directions coming from the hypothalamus or the pituitary gland. All right.
- 6.27** Now, one thing that's very important to remember in the discussion of the endocrine system is negative feedback. Indeed, one of the reasons I love the endocrine system so much is that it is so rational. If you can remember negative feedback, then you can almost always anticipate the direction these hormones are going, up or down. When it comes to growth hormone, the hypothalamus will send a signal through the portal vessel called growth-hormone-releasing hormone. Not a very tricky name for a hormone. The effect is going to be quite clear.
- 6.28** When the pituitary gland gets that signal from the hypothalamus, it releases growth hormone. Growth hormone is unique among all the pituitary hormones because it directly acts on the body. Most of the pituitary hormones will then go to another gland, like the thyroid gland or the adrenal gland or the gonads, that being the testes and the ovaries. It will have another gland that it needs to turn on. But growth hormone is unique because it acts itself. It is its own hormone.

- 6.29** Now, what does growth hormone do? Before we talk about the specifics, I want to just mention it does have a direct effect on muscle and bone, stimulating some degree of muscle growth and the growth of the long bones. But I'll get more into that in just a moment. But it also has a direct effect at the liver, telling the liver to produce more hormones. So growth hormone is unique because it can both act on its own, but also stimulate the release of other hormones.
- 6.30** We don't typically think of the liver as an endocrine gland, but when the growth hormone comes to the liver, the liver will start releasing complementary hormones in a family called somatomedins. These are the famous insulin-like growth hormones, or the IGFs, the most famous of which being IGF-1. So IGF-1 is a product from the liver that is being released in response to growth hormone. Growth hormone does have a signal that will stimulate the release of other hormones. But then with IGF-1, especially, and growth hormone, we now have the collective effort doing what we typically think of when we think of growth hormone.
- 6.31** With growth hormone and the somatomedins, or IGF-1, we have the classic stimulus of growth. The body will get bigger and thicker, but we also have some oft-unappreciated aspects or effects of growth hormone itself. As much as we only invoke growth hormone in the context of growing, it actually is intimately and exquisitely involved in nutrient metabolism, the net effect of which, growth hormone wants to mobilize energy to fuel growth.
- 6.32** So, it affects two key tissues in order to do this. First and foremost, it will stimulate lipolysis at fat tissue. It wants fat tissue to break down its stored fat. It will increase free fatty acids, which is the product of breaking down stored fat, because it wants to use that energy to fuel growth. You can't just tell a cell to grow without giving it some energy to grow with. It mobilizes fat. Then at the same time, it mobilizes the stored glucose from the liver. The liver has been pulling in and storing glucose as glycogen.
- 6.33** When growth hormone is elevated, it will come to the liver and say, "Hey, liver, you've got all that glycogen stored. The body needs it." Start releasing the glycogen, and that will increase blood glucose to fuel growth, ideally. Now, I say ideally because if you have instances where growth hormone is too high, you have that happening too much. Now, what are the instances when growth hormone is too high? There's two different names for growth hormone hypersecretion, depending on when the hypersecretion is occurring.
- 6.34** If the hypersecretion begins before the bones have stopped growing, which happens during puberty, the child is growing, the adolescent is growing, and at the end of adolescence, the bones have stopped. The long bones, arms and legs, the person has stopped growing, at the end of adolescence. So mid-teens and girls, late teens, even early twenties in boys, depending on the individual. If growth hormone has been hypersecreting before the bones have closed off, the person will have what's called

gigantism. The name is quite obvious. Those are the people that can get to 8 feet tall and beyond.

- 6.35** This unnatural level of growth because of this unnaturally high level of growth hormone, which is almost always the result of a growth hormone-secreting tumor. Growth hormone is just being pumped out like gangbusters, and it's amplifying the growth of the long bones. Now, however, if a person has this growth hormone hypersecretion after the bones have already closed their growth opportunity, so growth height has stopped, now they will have something called acromegaly. They will not be particularly tall. They may be perfectly normal height.
- 6.36** Indeed, if the bones had closed off, they will not be any taller than the average person of their same sex. Heights may be a little tall, but usually they're normal height. It just depends on when the hypersecretion started. But as the hypersecretion continues, even though they've stopped growing, they still can get some growth of certain bones, not the long bones of the arms and legs, but they can continue to get growth of the facial bones, the maxillary bones, and the hands and feet.
- 6.37** So they can start to develop ever more coarse features, as the jaw is getting a little bigger and the cheeks are getting a little more sunken. So the face can continue to grow. That's just down here, not the scalp, but the face can continue to grow, giving very coarse features. And again, as I noted, the hands and feet. Those are bones that will still respond to the growth signal, so giving a bit of a unique appearance. But even beyond that, which can become just perhaps a little odd-looking and uncomfortable, what is more substantially relevant is the growth of soft tissue, particularly the heart.
- 6.38** So as this growth signal just won't stop, the heart in particular can get bigger and bigger, and the person, and indeed, will ultimately, will often kill the person, is something called hypertrophic cardiomyopathy. That is when the heart muscle just keeps getting thicker and thicker and thicker. If you'll recall from the cardiopulmonary lecture, we need the heart to be able to fill up. It needs to be able to relax and open up and fill up. However, as the myocardium, the muscle of the heart, is getting thicker and thicker, it starts to compress the chamber. Now the heart simply cannot fill anymore. It's gotten too thick. That is hypertrophic cardiomyopathy.
- 6.39** Then adding insult to injury, while that is often what will ultimately lead to the person's demise, we have the growth hormone constantly telling the liver to release more glucose. As the glucose is chronically elevated, that results in chronic elevations in insulin, which we'll discuss at the end of this lecture. They will often have insulin resistance and type 2 diabetes, all because the growth hormone simply will not stop mobilizing energy, and they always have too much glucose in their blood, again, pushing them into this metabolic disarray of insulin resistance and type 2 diabetes.
- 6.40** All right, let's leave growth hormone behind and now talk about, arguably, a more famous pathway of hormone, namely thyroid hormone. The thyroid hormone starts in

the—the signal starts in the hypothalamus as thyroid-releasing hormone. Then thyroid-releasing hormone, TRH, will go to the pituitary and tell the pituitary to release thyroid-stimulating hormone or TSH. Then TSH will come from the pituitary to the thyroid gland itself, telling the thyroid gland, "Produce thyroid hormone."

- 6.41** Now, before I talk about the actual thyroid hormone itself, let's just take one step back and look at the signal coming from the pituitary to the thyroid in the form of TSH, thyroid stimulating hormone. TSH has the obvious effect at stimulating the production of thyroid hormone, but it has the less obvious effect of stimulating the growth of the thyroid gland. You can maybe already see where I'm going with this. One of the most common obvious features of a thyroid problem is the goiter or the substantially enlarged thyroid gland.
- 6.42** So TSH, thyroid stimulating hormone, has that direct effect of telling the thyroid gland to grow, which is not the same as telling the thyroid gland to produce more thyroid hormone. Those are two different signals. Now, they're certainly complementary, but they are different signals. But in order to follow closely an upcoming part of the thyroid discussion, namely the development of a goiter, you need to remember that TSH is having two different effects, stimulating the production of thyroid hormone and stimulating the growth of the thyroid gland.
- 6.43** Now, back to the thyroid hormone itself. When the thyroid gland is releasing thyroid hormone, it's actually producing most—most of what it's producing is a hormone called T4. T4 is an inactive form of thyroid hormone. The active form of thyroid hormone is T3. But even though the thyroid gland is releasing a little bit of T3, there's a little bit coming out, most of it's T4. Then that T4 will circulate through the body and be pulled into a cell or a tissue and be metabolized or converted into T3 and then released.
- 6.44** Now, in fact, one of the more interesting sites of conversion is the brown adipose tissue. This is something we've not talked about, and it's a whole other topic. But brown adipose tissue is a very high metabolically active fat tissue that is sprinkled through the thoracic cavity or the chest cavity of humans. But it becomes particularly active at converting T4 to T3, in fact, to such a degree that it becomes the main producer of T3. Most of the active form of thyroid hormone will be coming from brown adipose tissue when brown adipose tissue gets turned on through cold exposure.
- 6.45** So nowadays, it's very popular to have cold bath or cold immersion for numerous reasons, and I'm generally an advocate of such approaches, but it's interesting to note that in those instances, when the body starts to get cold, the brown adipose tissue becomes the main producer of the active thyroid hormone, T3. Now, remember, it didn't create it itself from scratch. The original T4 still came from the thyroid gland; it just is using the brown adipose tissue, this very high metabolic rate fat tissue that produces a lot of heat, in order to convert the inactive form, T4, into the active form, namely T3.

- 6.46** Now, what does T3 do? What does thyroid hormone do? It does a lot. It will affect every cell of the body. The thematic effect of thyroid hormone is to basically control, it's like a metabolic choke valve. It will control the idle of the engine. So whatever the cell is doing, if thyroid hormone is elevated, it's going to do it faster and more. If thyroid hormone is depressed, the cell will be depressed. It will do whatever it's doing more slowly than before. So a bit of a compromised function.
- 6.47** Now, of course, you want thyroid hormone to be in a very normal range, so the cell is just operating at a normal rate. But again, thyroid hormone is basically pressing the gas pedal, accelerating the engine or easing off the engine too much, depending on its level being high or low. And indeed, let's talk about it being high or low. Let's just discuss some of the consequences of hyperthyroidism and hypothyroidism, and then explore that in the context of some specific thyroid disorders.
- 6.48** With hyperthyroidism, as I mentioned, it's controlling this metabolic idle of the engine. If thyroid hormone is high, we're pressing the accelerator and the engine is just running faster. Typically, a person will lose weight quite rapidly. They will also just be physically hot. They will be hotter than they were before. They have a heat intolerance. They walk into a room and they just can't wait to take their jacket off. They're just so uncomfortably warm. They're thirsty because they're almost literally panting and they're always sweating a little bit. The skin will be not only warm, but even a little moist because sweat is a good way to try to cool the body off.
- 6.49** Beyond that inconvenience, the person will have a rapid pulse. Remember I said everything's happening faster. The heart is beating faster, even to the point that it can develop arrhythmias, these sporadic spasms of the heart, compromising its function. It also will activate clot formation more quickly, too quickly, so the person can start to experience blood clots. Even the clotting pathway in blood starts to be turned on too much. The neuromuscular junctions, neuron innervating muscle is turned on too much, so the person can have spasms and muscle tremors.
- 6.50** Then even the parastalsis, the motion wave moving food stuff through the intestines, is happening too fast, giving rise to potential diarrhea. All of those are a consequence that you could almost guess would happen if we just looked at each tissue system and dialed it up to an 11. Some of the less obvious is actually an effect that happens behind the eye, but it becomes a somewhat obvious feature of a person with hyperthyroidism, which is that when thyroid levels are artificially elevated into this hypersecretion range, there is a swelling of fat pads behind the eye.
- 6.51** We have the eye, then we have a cushion, and the high levels of thyroid start to expand the cushion, making the person have bulging eyes, a condition called exophthalmos. The eyes start to press physically out, and they start to bulge out. They're not popping out, thankfully, but they're bulging out, where the person will start to experience what's called a lid lag, an eyelid lag. If you're looking at a person, if I were putting this remote in front of each of your eyes at eye level, your eyelids are at a certain level, as

your eyes are at a certain level. But then as I lower it, and your eyes start to tip down, the lid will come down with it.

- 6.52** However, in someone with exophthalmos, as you lower the pencil or the remote in front of their eyes, even though their pupils are starting to look down, their eyelids are staying up. They're not making it over the curve of the eyeball very passively. The person, thankfully, can still blink, but it does, in fact, require more effort because it has to overcome the curvature of the bulging-out eye. Again, that's exophthalmos, a sign and symptom of hyperthyroidism.
- 6.53** Then hypothyroidism, everything's opposite. Whereas the person with hyperthyroidism was losing weight and they were always hot and sweaty, the person with hypothyroidism is gaining weight and they're always cold. They are always trying to bundle up. The heart was too fast with hyperthyroidism, now it's too slow. The muscles were twitchy, now they fail to respond to stimuli, and reflexes are slower. For example, if you do what's called the patellar tendon reflex, where you bend the knee and knock it with the hammer right below the knee, it will kick out immediately. In someone with hypothyroidism, you'll tap it and then it will kick out, noticeably later. So everything is delayed.
- 6.54** Neurons are delayed with their signal, muscles are delayed with their twitching, and then the intestines, rather than moving too quickly, have slowed down too much. So the person can have constipation, the opposite of diarrhea. And then lastly, one very interesting feature is the eyes, once again. Whereas in hyperthyroidism, there was a bulging of the eye, in hypothyroidism, coincidentally, it almost appears as if the eye is more sunken.
- 6.55** Now, the eye, in fact, has not delved deeper into the ocular socket, but the person starts to have edema or fluid accumulation around the eyes. And so the eyes start to look really puffy. Not the eyeball, but the skin around the eyes. So it looks like the eyes are more sunken as all of the skin is a little more swollen around the eye, so it ends up being this opposite effect. Far from being a bulging, it's more of a sunk-in feature. Those are the general features of hyperthyroidism and hypothyroidism.
- 6.56** Let's talk about some specific thyroid disorders, namely thyroiditis, Graves' disease, and then a goiter, which doesn't quite fit in with either directly. Now, thyroiditis can come in a few different forms, and the most common is Hashimoto's thyroiditis. In fact, indeed, so common that I suspect every one of you here and listening at home know someone who's had Hashimoto's or does have Hashimoto's. It is the most common version of a thyroid problem. It's the most common in the US.
- 6.57** And what happens with Hashimoto's is, the autoimmune system is attacking the thyroid gland. And so the person will actually have, at least in the early stages of the disease, this frenetic hyper and hypothyroid. So the thyroid gland will start to be attacked, and then the immune system will ease up and the thyroid bounces back, and

then it goes a little too hyperthyroid. Then it goes low again, and then it goes high again. But it's this ever-reducing level on its way down to ultimate destruction.

- 6.58** Hashimoto's thyroiditis is this inflammation. Anytime you hear the suffix -itis, it means inflamed. It can be hyper and hypo thyroid bouts, but always working its way to ultimate demise, to death, where it will be a true hypothyroid. Now, one very interesting and unfortunately probably controversial aspect of Hashimoto's thyroiditis is, in fact, something that is reflected in every single autoimmune disease. An autoimmune disease is a condition when the body is fighting itself. The body's immune system is fighting its own cells for some reason. There's no good reason for this.
- 6.59** Most—indeed, I want to say every, but I suppose there's an exception to this—most autoimmune diseases have an environmental trigger. There's something the person has been exposed to. They've inhaled or they've consumed, usually incessantly, something that's triggered their immune system to start attacking itself. Hashimoto's is no exception. The most common environmental trigger is actually the oft-maligned gluten. I know it sounds crazy and you'd think I'm making it up; I'm not. The literature on this topic is robust.
- 6.60** People who have gluten sensitivities or outright celiac disease are about 20 times more likely to develop Hashimoto's. There is a fantastically strong connection. If you are consuming something that is eliciting an immune response, what happens as the immune response continues and continues and continues, it begins to forget who it's supposed to be fighting. In other words, addressing the gluten in this case, and it starts fighting itself. Very often, it ends up being the thyroid gland. In someone with Hashimoto's, very often the environmental trigger is that they actually have an underlying celiac disease, or at least on that spectrum, and they've been unwittingly exposing themselves to a trigger.
- 6.61** Now, Graves' disease is also an autoimmune disease, but of a very different sort. In Graves' disease, the person's immune system is actually making antibodies, but not antibodies that will target a cell to destroy it, like is happening with Hashimoto's, but antibodies that actually act like TSH, thyroid-stimulating hormone. So these antibodies are mimicking the signal that's coming from the pituitary gland.
- 6.62** You'll recall we had TRH, thyroid-releasing hormone, from the hypothalamus, which will then induce the release of TSH from the anterior pituitary, which then goes to the thyroid gland, telling it to both produce more thyroid hormone and grow. Remember, those are two different things, complementary yet distinct. In Graves' disease, the person is making antibodies that are acting like TSH. It's mimicking TSH. It comes to the thyroid gland, and the thyroid gland thinks it's seeing TSH. But it's just these antibodies.
- 6.63** The body is swimming in these antibodies because the immune system is just pumping them out like gangbusters. The thyroid gland is not only very overactive, resulting in the classic hyperthyroidism and all the signs and symptoms we talked about earlier, but

they also have this growth. Again, they're hyperthyroid, and we have the thyroid gland just getting constantly told that it needs to grow bigger and bigger and bigger.

- 6.64** One of the consequences of hyperthyroidism, again, in addition to everything else we mentioned earlier— the exophthalmos, the heat intolerance, the sweating, the weight loss—all of that is still happening, but they also now start to have a goiter developing. So let's talk about a goiter. A goiter is the term to refer to an enlarged thyroid gland. It can be subtle or it can be substantial, substantial to the point that the person literally can't move their chin down. That's not often, that's not frequent in the United States, but in countries around the world, that can happen.
- 6.65** Now, people will sometimes assume that a thyroid gland represents a hyperthyroidism, but not always. In the case of Graves' disease, it is. You have both a goiter and hyperthyroid, but that's not always the case. You can have a goiter and have hypothyroidism, just as well as you could have a goiter with hyperthyroidism. Let's just distinguish between the two as we discuss this final point about thyroid regulation.
- 6.66** Let's say, in fact, the most common form of a goiter is an iodine deficiency. If a person is not eating enough iodine, iodine is an essential molecule to create thyroid hormone, and this requires you to think about negative feedback. If I don't have enough iodine in my diet, I cannot make thyroid hormone, full stop. It just cannot happen. And so the brain, the hypothalamus and the pituitary, they sense, "Hey, there's no thyroid hormone, so there's nothing to turn me off." There's no negative feedback to come back to the brain and say, "Hey, look, thyroid hormone is being produced. It's time to turn down your production."
- 6.67** Does that make sense? So there's no signal to turn it off. And so if you actually measured TSH levels, because the thyroid gland can't make thyroid hormone, what do you think TSH levels would be? Very, very high because there's nothing to turn it off. There's no negative feedback happening. The TSH is just being flooded from the pituitary gland, coming to the thyroid gland, telling the thyroid gland, "Make more thyroid hormone." But the thyroid gland can't because it doesn't have the necessary building block, namely the iodine.
- 6.68** While the thyroid gland cannot respond to the TSH by producing more thyroid hormone in the absence of that necessary building block, it still responds to the growth signal of the TSH. In this instance, in an iodine deficiency, which is the most common cause of a goiter, just due to nutritional insufficiencies, the person will have an enlarged thyroid gland, a goiter, but have all the other signs and symptoms of hypothyroidism. Because the TSH is trying to tell the thyroid gland to make the thyroid hormone, but in the absence of iodine, it cannot.
- 6.69** Then we already discussed an instance where the person has a goiter and hyperthyroidism, something like Graves' disease, or it could even be a TSH-secreting tumor on the anterior pituitary, which is one of the more common pituitary tumors. If

there's either too much TSH coming down, it will be hyperthyroid and hypergrowth, or if there's too many of these TSH-mimicking antibodies, like we have with Graves' disease, we have the exact same effect. Too much thyroid hormone production and too much thyroid hormone growth, resulting in a hyperthyroidism and a goiter.

- 6.70** When you see a goiter, we cannot assume anything. We cannot assume that we know the person's hyperthyroid or hypothyroid. We have to look a little further, dig a little deeper. All right, now let's leave the thyroid gland behind and go to the last gland we'll discuss within the hypothalamic-pituitary axis within that line, and namely the adrenal glands. Now, the adrenal glands are complicated. Complicated because they're like an orange, although an oddly shaped orange. There's a thick peel called the adrenal cortex. Then there is a fleshy middle called the adrenal medulla. They're very, very different types.
- 6.71** They may as well be totally different glands. They're just wrapped up in each other, but they have nothing to do with each other, oddly enough. Then to make it even more complicated, if we go out of the fleshy little center, the medulla of the adrenal gland, and just go back to the cortex, the peel, the peel is so substantial that it actually has three completely distinct sections—these zones, they're called. Within each of these three zones, we have the production of different steroid hormones. We're going to talk about these to some degree.
- 6.72** There is a clever little mnemonic that I use to help students remember the production down these three layers of these hormones. The adrenal cortex can release hormones in what's called the mineralocorticoid family, regulating salt. Then the next layer down, it regulates the glucocorticoids, cortisol, which will influence glucose. We'll discuss that more in a moment. Then the last layer is a layer that produces sex hormones.
- 6.73** In fact, that's a very important layer because, in menopause, as the woman's ovaries have taken a bit of a back seat in producing sex hormones, that layer— the zona reticularis, it's called—that last layer of the adrenal gland actually becomes the main site of sex hormone production in the postmenopausal woman. But as we go through these layers, the way to remember the production of hormones is salt. Pardon me, this is a little crass. "Salt, sugar, sex. The deeper you go, the sweeter it gets." It's a little irreverent, like I said, but you'll probably remember it.
- 6.74** That is the three layers of hormones producing, hormones involved in salt regulation, hormones involved in sugar or blood glucose regulation, and then in sex regulation. Salt, sugar, sex, as you go down the three layers. Then, of course, in the middle of it all is the medulla. Let's start with a discussion of hypersecretion. Just for the sake of time, we're going to talk about the hypersecretion of cortisol.
- 6.75** When you have too much cortisol, that is a problem called Cushing syndrome. Cushing syndrome is elevated plasma cortisol for whatever reason, whether there is a tumor that's creating too much of the stimulus signal from the pituitary, ACTH is what that

hormone is called, or whether it's a tumor directly of the adrenal cortex itself. Any instance of too much cortisol is Cushing syndrome. Now, that has catastrophic consequences because cortisol is a powerful signal. I want to highlight its effects by discussing five particular influences or impact, five impacts of elevated cortisol, which is its impact on sex hormones, its impact on catecholamines, so epinephrine, and then its impact on the three macronutrients, carbohydrates, fats, and proteins.

- 6.76** Firstly, cortisol doesn't result in the increased production of catecholamines like epinephrine and norepinephrine, but it enhances the body's sensitivity to it. Now, you'll recall that epinephrine and norepinephrine are the main neurotransmitters of the sympathetic nervous system, the fight-or-flight nervous system. When the body is amped up, the heart's beating harder, the body starts sweating, and the body is ready for action. Cortisol is enhancing the sensitivity to those signals. It's not increasing the signal directly, but it's enhancing the reception of the signal.
- 6.77** So whatever given amount of catecholamines, of epinephrine or adrenaline there is in the blood, if cortisol is high, it's amplifying that signal. So if a given amount of epinephrine or adrenaline is resulting in a particularly elevated heart rate, if we sprinkle in some cortisol into the mix, it turns it up a notch. So everything's happening a little more. Sex hormones are impacted in a negative way. If cortisol is elevated, cortisol is the primary or it's the poster child of stress hormones. It basically tells the gonads, being the testes in men or the ovaries in women, "Hey, we're too stressed to reproduce right now, so let's turn off everything."
- 6.78** No surprise, estrogens and androgens go down. Ovulation can stop. Sperm production can be compromised. It basically is saying, again, "Hey, we do not want to reproduce right now. We are in an unsafe environment." Now, again, this is when cortisol is too high, artificially too high. We have a hypersecretion. In reality, their environment may be perfectly fine, but the body is being tricked into thinking that it's not because the cortisol is too high.
- 6.79** Cortisol will take any aspect of fertility and try to turn it off. Not good for the species, and of course, potentially very hurtful or unfortunate for the individual, depending on where they are at in life. Now, let's get into the regulation of metabolism. Fat, protein, and carbohydrate or glucose. Cortisol has a very interesting effect when it comes to fat metabolism, and it's not a universal effect. If cortisol is elevated, like in Cushing syndrome, it will have a selective lipolysis or breakdown of fat, coupled with a selective lipogenesis or storage of fat.
- 6.80** In particular, cortisol will activate lipolysis in the limbs of a person. The peripheral fat will be mobilized. It's stimulating fat breakdown all the way up from the legs, up through the buttocks, and then it will stimulate the fat breakdown all the way through the arms. But while it is stimulating fat breakdown or mobilization in the peripheral of the body, at the same time, it's stimulating fat synthesis in the middle of the body.

- 6.81** Someone with chronically elevated cortisol will develop something called truncal obesity. Trunk, referring to, from basically groin to the head. I do emphasize head because the person will start to get a lot of fat stored around their neck and face. It's not just their belly, it's going all the way up. In fact, they're storing their fat in such a unique pattern that they'll often develop what's called a buffalo hump, which is a big pad of fat on the back at the base of the neck. Mobilizing fat from the limbs to store that fat on the trunk, just a central fat storage.
- 6.82** Now, with protein metabolism, It's a fairly substantial and unfortunate effect. Cortisol wants to strip the proteins from muscle and bone and skin. It's breaking down muscle protein. It's breaking down collagen in the skin because it wants to get those amino acids to send them to the liver, because then it will tell the liver to make glucose. If cortisol is chronically elevated, as it is in Cushing syndrome, it will start destroying lean mass, including skin, in order to make more glucose. It wants to just increase glucose, make it ever higher. It just loves glucose.
- 6.83** Cortisol will do whatever it needs to do in order to increase blood glucose levels, including breaking down lean tissue. The person will start to develop stretch marks on their skin. They will start to lose muscle mass, and they will have a weakening of the bones, all to increase glucose levels. Mind you, at the same time, they're getting a central adiposity or central fat storage. It ends up resulting in a very unique body shape, and not in a good way.
- 6.84** Then last, with cortisol effects, is the effect on blood glucose. I've already alluded to this. Cortisol wants to increase blood glucose at any cost. It will strip the amino acids from lean mass to do so. It will stimulate the liver to break down its stored glycogen. It will also make the muscle and the liver become insulin resistant in order to keep the glucose high.
- 6.85** Muscle is the main consumer of glucose, and we'll discuss that effect more in just a bit. But it makes the muscle very insulin resistant, which just contributes to the glucose levels getting ever higher. A common outcome metabolically of someone with Cushing syndrome will be insulin resistance and eventual type 2 diabetes, where they're both insulin resistant and they have elevated glucose levels.
- 6.86** All right, now, as I mentioned, the adrenal gland does more than just release cortisol. It also has the ability to release very high levels of sex hormones, even to the point where it's releasing the wrong sex hormones based on the sex. There can be instances, due to tumor formation, where, in a man, that one layer of his adrenal gland, of the adrenal cortex, is producing too much estrogens, and that will result in a process called feminization, making the man a little less manly and a little more physically feminine.
- 6.87** In women, it can have the opposite effect, a term called virilization, where her adrenal glands may be overproducing androgens, which has the effect then of making her body more masculine. There are instances of this hypersecretion of sex hormones from the

adrenal cortex. Of course, the effect is going to be more pronounced across the sexes, depending on which sex hormone is being overreleased. Of course, just by way of clarity, male bodies, just like female bodies, have both both estrogens and androgens. All sexes have androgens and estrogens. It's just the particular ratio. Females have, of course, relatively more estrogens than males do. Indeed, substantially more.

- 6.88** All right, now, before we leave the adrenal gland behind, let's just go one last layer deep into the actual medulla, the adrenal medulla itself. It is possible for someone to have a tumor of the adrenal medulla, which is hyper-secreting the catecholamines like epinephrine or adrenaline. This is a condition called pheochromocytoma. That's a mouthful. Pheochromocytoma. That is an adrenal medullary tumor secreting too much epinephrine. With too much epinephrine comes too much sympathetic nervous system.
- 6.89** The person just feels like they're having a panic attack all the time. They always feel like they're anxious. They're hot, their heart is pounding, they're breathing heavy. They have sweaty palms. They also have elevated glucose levels because the epinephrine is always stimulating the liver to release more glucose. The person will always feel like they're in a fight or flight, which we can all appreciate would be exhausting and very difficult to ever relax.
- 6.90** All right. Now we have finished the part of the endocrine system lecture where we discussed the hypothalamic, pituitary, then primary gland axis, that line of signaling, and all of the complexity of negative feedback and the consequences of the hormones being messed around. I want to step out of the hypothalamic-pituitary axis and talk about one final gland that is supremely important in endocrinology and metabolism, namely the pancreas.
- 6.91** Now, when we talked about the pancreas in the context of the GI system, we really elaborated the exocrine pancreas, or the aspect of the pancreas that's producing digestive enzymes to be released into the intestines to help with digestion. But as I alluded to at the time, there is an alternative side or aspect of the pancreas that we refer to as the endocrine pancreas. This is the part of the pancreas that is releasing hormones into the bloodstream.
- 6.92** That is best represented by these little, what's called islets, these little pockets of cells that are sprinkled through the pancreas. Within the islets, we have alpha cells, which produce glucagon, and we have beta cells, which produce insulin. They have their own little paracrine signaling. They have a bit of an odd relationship because, on one hand, you have the alpha cell producing glucagon, which will stimulate the beta cell to produce insulin. But then at the same time, as the beta cell is producing insulin, it will feed back to the alpha cell, telling the alpha cell to stop producing glucagon.
- 6.93** So whereas glucagon is stimulating insulin release, insulin in turn is inhibiting glucagon release. If there's ever a direct battle between the two hormones and what they want

to do in the bloodstream, insulin will always win because it will be directly turning off glucagon. That is relevant because they do have a tug-of-war. That tug-of-war is happening over what's going on with blood glucose levels. As I mentioned earlier, whereas insulin is famous for wanting to lower blood glucose, and indeed, a main stimulus is when blood glucose gets too high, insulin wants to lower it, glucagon is just the opposite. It is the yang to the yin, metabolically.

- 6.94** Whereas insulin is stimulated when glucose is too high, glucagon is stimulated when glucose is too low. Whereas insulin wants to push glucose down, glucagon wants to bring it up. So they have opposite effects. This discussion of insulin and glucagon is very relevant because the single most common health disorder worldwide is a disorder of insulin, and that is a condition called insulin resistance. Now, worldwide, roughly half of all adults are overweight, and more than half of all adults have insulin resistance. In fact, within the United States, it's estimated that upwards of near 80% of US adults have some aspect of insulin resistance.
- 6.95** Unless you think it's a uniquely American problem, it is not. These statistics are even worse, not only in our backyard, namely in Mexico, but also as we look across the Middle East and Asia and the Pacific Islands; it's even higher rates of these metabolic derangements than we have here. This is a global problem. Over half of all adults, well over half, have insulin resistance. It is important to understand it.
- 6.96** Now, what is insulin resistance? Let's look at the level of the cell. Let's really zoom into the level of the cell for just a moment, and then we'll zoom back out and look at the whole body. If we look at the level of the cell, insulin is flowing through the bloodstream, and it will come as a peptide hormone. It will knock on a door of the cell called the insulin receptor, eliciting a second messenger response that we discussed earlier. Or we'll just call it an action, because insulin affects literally every cell of the body. I don't use the word literally too liberally like the kids do these days. I mean, literally, every single cell of the body will respond to insulin.
- 6.97** But that means what insulin does at the brain cells is different from what it does at the bone cells. What it does at the liver cells is different from the lung cells. We'll just call it an action. Now, insulin's most famous action is opening up glucose doors in tissues like muscle or fat tissue to allow the glucose to come from the blood into the muscle and the fat. Mind you, other tissues of the body don't need insulin to open the glucose doors. The glucose can just come in. It's only tissues like muscle and fat that need the insulin signal first to come and knock and then open the door.
- 6.98** But it does so, and thereby, lowering blood glucose, insulin's most famous effect, but not its most important. Insulin does all kinds of things that are very important. It's the most famous effect. Insulin resistance is where some cells have a diminished action of insulin. What was once a really robust response to the knocking of insulin on the door becomes a very diminished response. The cell is just not listening very well. That is

insulin resistance. One consequence of that is that blood glucose levels can start to climb up, which we'll revisit in just a moment.

- 6.99** But at the same time, if one molecule of insulin politely knocking on the door is no longer sufficient to get the response we want, then we need more insulin. And what was once a polite knocking on the door becomes an angry mob, pounding on the doors of the cell, all in an effort to try to get that action back to where it was before. Within this definition of insulin resistance, there are two vital components that you must remember to understand insulin resistance. First and foremost, it's that some cells of the body aren't responding to insulin anymore. Not all cells, some cells.
- 6.100** Then second, blood insulin levels themselves are actually higher, a condition called hyperinsulinemia. Now, if we imagine the average individual who's aging through life and they're getting ever closer to type 2 diabetes, which is increasingly common, before they ever get to type 2 diabetes, they must pass through insulin resistance. Insulin resistance is so often missed, and indeed, part of the reason why it's one of the most common health problem is, we don't detect it early enough.
- 6.101** Insulin resistance is missed because we have a glucose-centric paradigm when it comes to metabolic health. We only look at the glucose and we ignore the insulin. If we were able to expand our scope to look at insulin, we would see, over the life of this person, even while glucose is staying normal, insulin is getting ever higher and higher and higher. It is the early metabolic signal or the canary in the metabolic coal mine. It is the early signal, warning that there's a problem. That is only when the body gets so resistant to insulin that it can't well control glucose anymore, then now the glucose starts to climb, even though the insulin is still elevated.
- 6.102** At that point, we detect the problem, and we call it type 2 diabetes. But that can happen 10 or 20 years later. Imagine how powerful it would be if we could detect the problem when it was just here. High insulin, but enough, working well enough to keep the glucose in check, before it was so resistant and now we can't control the glucose anymore. That is still correctable, thankfully. But imagine the power if we shift the focus to include insulin as a measurement.
- 6.103** There are three key causes to insulin resistance. One is stress, including some hormones that we've already discussed, being cortisol and epinephrine. Those are the poster-child hormones when it comes to stress. One thing they have in common, despite all of their differences, is they want to increase blood glucose, which puts them at odds with insulin who wants to reduce blood glucose. The more those stress hormones, cortisol and epinephrine, are elevated, the more insulin has to work harder and thus leading to insulin resistance. So that is the stress origin.
- 6.104** The next origin is inflammation. Inflammation is obvious when you think about something like an illness, and indeed, it's relevant. If a person has an illness, including even an autoimmune disease, they will be more insulin resistant than they were before

that illness or the inflammation from that disease. But also there are some unexpected sources of inflammation that can contribute to insulin resistance, including the inhalation of harmful molecules, like cigarette smoke or diesel exhaust. I have, in fact, published some of those reports myself from my own lab, looking at exploring the various stimuli that contribute to inflammation-induced insulin resistance.

- 6.105** Then the third and final cause of insulin resistance is too much insulin. If the body is getting too much insulin being released because of the incessant consumption of refined starches and sugars that are constantly spiking up glucose, then the body never gets a break. It is a fundamental biological principle. In fact, I would even say a fundamental principle of our existence, body and soul, that too much of something will result in a resistance to that something. Too much insulin, no surprise, results in a resistance to the insulin, all in an effort to try to maintain homeostasis.
- 6.106** Now, my final point with all of this, as we wrap up the endocrine system lecture, is to just highlight the relevance of insulin resistance ultimately contributing to type 2 diabetes, because there is no type 2 diabetes without a foundation of insulin resistance. You cannot have ever gotten to that without first having had this big substantial foundation of insulin resistance. Generally, my view is that it is a fat-first process, that the first part of the body that becomes insulin resistant, for reasons I won't get into now, is that the fat cell, and in fact, I focused on this more in my other Peterson Academy lectures of the Why We Get sick series, so I can refer you to that.
- 6.107** But also fat starts it. Then once the fat cell has fallen, if you will, it's like the first domino in a row of stacked dominoes. We tip that one over, and then it falls into the other dominoes that matter when it comes to glucose control, specifically muscle, liver, and the alpha cells of the pancreas. My concluding thoughts are these. Firstly, the muscle. Muscle is the main consumer of glucose in the body. 80% of the glucose, if we eat or drink a starchy or sugary meal, 80% of that glucose goes into the muscle, so it consumes the lion's share of glucose.
- 6.108** As I noted, muscle must have an insulin signal to come and knock on the door to open the glucose doors, and then the glucose will come flowing from the blood into the muscle. But if the muscle isn't getting that signal because it's insulin resistant, it doesn't listen very well, and it doesn't open the glucose doors, and the glucose stays in the blood, contributing to ever-higher glucose levels. This moves the person further into that realm of type 2 diabetes from just being insulin resistant.
- 6.109** Now, they go further into the type 2 diabetes realm when the liver starts to have a problem. The liver controls all nutrients, whether it is ketones or lactate or glucose or fat. The liver will always know what to do with them. But when the liver becomes insulin resistant, it's supposed to be storing the glucose as glycogen, but that signal gets broken down. Now, rather than storing the glucose as glycogen, if the liver becomes insulin resistant, it now is releasing the glucose. It's breaking down the glycogen rather

than building it up, this storage form of glucose. Rather than pulling it in and storing it, it's releasing it as glucose into the bloodstream.

- 6.110** Even though insulin is trying to work hard to bring the glucose down by telling the liver to store it, insulin is basically telling the liver, "Liver, I want you to pull in the glucose and hold on to it." But the liver is not listening. It's become a disobedient child. Rather than holding on to the glucose as stored glycogen, it's breaking down the glycogen, releasing it as glucose into the bloodstream, resulting in glucose levels climbing even higher.
- 6.111** Then the third and final aspect to this compelling story, I hope, of diabetes and its progression is going back to the islets, or the little pockets of cells within the pancreas itself. Going back to that little neighborhood of alpha cells and beta cells, producing glucagon and producing insulin, respectively. As I noted, insulin and glucagon have opposite effects on glucose. Whereas insulin wants to lower blood glucose, glucagon wants to increase blood glucose. Insulin normally wins that war by directly inhibiting the alpha cell. Insulin tells the alpha cell, "Hey, I'm in charge. I want to lower blood glucose, so I need you to lower your production of glucagon." And normally, the alpha cell will listen, and glucagon production goes down.
- 6.112** However, what happens when the alpha cell becomes insulin resistant, which it can? And so now insulin is trying to tell it to inhibit glucagon, but the alpha cell doesn't get that signal anymore. It's not listening. It's disobedient. Rather than responding politely, helping the body lower blood glucose, glucagon, the alpha cell begins pumping glucagon en masse, which is then going to the liver and telling the liver, "Hey, you don't need that glycogen. Break it down as glucose," further pushing up the glucose.
- 6.113** Once the insulin resistance has spread to the alpha cells of the pancreas, to the liver itself, and to the skeletal muscle, now you have this inflection point where the glucose starts to climb dramatically. What was once this seemingly benign state of insulin resistance now has converted and matured into full-on type 2 diabetes.
- 6.114** We've reached the end of the endocrine system lecture, and we have covered a lot. We revisited negative feedback, which is an essential aspect of understanding the intricate balance of hormones. We started with the famous hypothalamic-pituitary axis and went through some of the key hormones there, namely, growth hormone, thyroid hormone, and the adrenal hormones. Then we ended with a non-HP axis, hypothalamic-pituitary axis gland, namely, the pancreas, and highlighted two of its most important signals of the endocrine side of the pancreas, namely insulin and glucagon. Then we were able, with that topic in mind, to discuss the single most common health disorder worldwide, namely insulin resistance. Thank you.

Lecture 7: Adipose Physiology

At a glance — Adipose physiology, closing the systems tour.

- 7.1** As we come to the end of discussing the physiological systems of the body, I have left the chubbiest topic for last, namely the adipose system or the adipose tissue or fat tissue of the body. This is actually a topic that is uncommonly discussed when it comes to the physiological systems. Indeed, you would be hard-pressed to even find a chapter devoted to fat tissue within a textbook of physiology.
- 7.2** But you happen to be taking this version of physiology from a fat cell scientist, so I have made an effort to include this as a topic because the fat tissue deserves its own chapter in every physiology textbook and certainly deserves its own lecture in this series as well. As we go through fat tissue as a topic, I want you to appreciate its development. What are the variables that constitute its growth, the growth, the expansion of fat tissue? I want you to appreciate the different types of fat tissue that humans have, namely white and brown fat and their differences functionally.
- 7.3** I want you to appreciate some of the endocrinology of the fat tissue. Like so many tissues are these days, fat tissue is also a hormone-secreting tissue, and some of those hormones are very interesting. I want you to appreciate the difference in how fat can grow, namely through hypertrophy and hyperplasia, two different methods of growth. And then at the end of it all, what are the actual signals that tell the fat tissue to expand? A topic that most people are interested in because it seems we all struggle with wanting less body fat. So if we understand what are the triggers promoting fat growth, it lends itself to understanding the answer.
- 7.4** All right. Now, if we look at the growth and development of fat tissue in a human, there are different phases of life, and with these different phases of life comes different methods of fat growth. Throughout the first years of a human's life, through, this is infancy and childhood and then adolescence, as they are growing, they're making more fat cells. So the actual number of fat cells is expanding during this time, during these developmental years. That is a process that is called hyperplasia. So again, infancy, childhood, adolescence.
- 7.5** Then at the end of adolescence, hyperplasia generally stops. And then forever after, for most people, the rest of their adult life, the number of fat cells they have is going to be set. Now, that does not mean fat cells are immortal. That is not true. If you've ever heard that, it's incorrect. Fat cells are, however, very long-lived. They can live upwards of 10 years, which is long for a cell. But for every one fat cell that dies, it is simply replaced by another one for the majority of the person's adult life. But again, up until adulthood, which is the end of adolescence, it's primarily hyperplasia, and then that levels off.

- 7.6** Then any future fat growth, because we all know, much to our chagrin, a person can find it very easy to continue to get fatter in their adult years. That is not a function of hyperplasia, the multiplication of the fat cells, but now it's a function of hypertrophy, which is the expansion of the fat cells. So fat cell number is no longer changing, but that does not mean that fat mass isn't changing. It can still, in fact, continue to go up. And we'll talk about the stimuli that elicit that response later in this lecture.
- 7.7** OK, now within the body, there are three different sites where we store fat, three depots. The least amount of fat we have is the brown adipose tissue, and I'll talk more about brown fat in just a moment. But brown fat is a fat that is sprinkled or woven through the thoracic cavity of the body. So it's near our rib cage and the clavicles up at the top. So it's just these small little pockets of fat woven through the thoracic space.
- 7.8** Then we have the majority of fat on humans, which is our subcutaneous fat. That is the fat that we can pinch and jiggle. That's the majority of the fat we have. And then we also have a fairly substantial amount, depending on the person, most especially depending on the person's sex, male more than female. We also have the potential to store more fat in what's called the visceral space. And I'm trying to convey this three-dimensional movement here, where the visceral fat is being stored within the abdominal space.
- 7.9** It's not the same as having a big belly that is pinchable and jiggable. It's the kind of belly where the person is fatter, but it's big and it's distended and it's hard. That is fat that is more visceral. So three distinct depots—visceral fat, subcutaneous fat that pinches, it can be pinched and jiggled, and then we have the brown adipose, which is woven through the thoracic space. Now, beyond the difference in where we store it is very much how we look at it or how the fat looks. This is another level of distinction.
- 7.10** So I've just defined or distinguished fat based on storage site, brown adipose, subcutaneous adipose, visceral adipose. But then we have one other level of distinguishing them, which is based on their color. Brown fat, as the name suggests, is in fact a very dark reddish brown in color. If we were to take a little biopsy of brown fat from a human, it would appear very reddish brown, for reasons I'll get into in a moment.
- 7.11** In contrast, if we pull a biopsy of subcutaneous fat or visceral fat, it's very white. Not quite as white as the walls around me, but a little bit of a yellowish white, but still very, very light, light-colored. The difference in the appearance, and beyond that, one final difference is that white fat cells are about 1,000 times larger than brown fat cells. Now, I'm talking about the cells themselves. So brown fat cells, or to be technical, brown adipocytes, are very, very small by even any cell standard. In contrast, white adipocytes are 1,000 times bigger.
- 7.12** So the fat cells are capable of significant growth. There is not a single cell that can match the degree to which white fat cells can expand, going from the size of a brown

fat cell to 1,000 times bigger. And we'll talk about some of the consequences of that in just a bit. Now, why is white fat so whitish-looking and brown fat so reddish, brownish-looking? It has everything to do with what the cells are made of.

- 7.13** Whereas white fat is almost fully made up of what's called a lipid droplet, it's one big bubble of fat, brown fat has small little lipid droplets, but it also has a lot of mitochondria. And the mitochondria are the powerhouse of the cell. So it's common to see a lot of mitochondria in things like muscle or heart or these high metabolic rate tissues. You wouldn't expect to see a lot of mitochondria in fat tissue, whose primary job is to store energy, because after all, mitochondria want to burn energy. That's what the mitochondria are for. That's the powerhouse.
- 7.14** But brown adipocytes or the brown fat cells are very different in that they have a high amount of mitochondria, and the mitochondria are like an engine that's just revving, revving, revving. Because brown fat exists primarily to make heat, whereas white fat cells, subcutaneous and visceral, exist primarily to store energy. Rather than burn it for the sake of heat, which is what the brown fat cells are doing, the white fat cells want to miserly hoard the energy and store it for later use, in times of, say, fasting or any time calories may be insufficient.
- 7.15** Now, it begs the question, why have a fat storage that only exists not only in small little pockets, but also just to produce heat? In order to really appreciate the role of brown fat in humans, let's actually go back in time to our earliest years as a baby. In fact, earliest months, to be a little more precise. When a baby is freshly born, what the baby lacks in muscle tissue it makes up for with cherubic, angelic, and adorable chubby fat tissue. So the baby is very fat, and indeed, humans are the only mammal on land that is born obese. There's no other animal in the animal kingdom that is born obese, except humans.
- 7.16** When we are newborn, we are unique in that we are very, very chubby and we primarily lack muscle. And that's a problem if we need to generate heat. Because if I were to take any adults or even kids listening to this and put them in a cold room or cold water, the primary mechanism of warming your body would be shivering. And the shivering is the spasming of the muscle. And as the muscle is spasming, it's creating heat.
- 7.17** However, for a newborn baby, which does not have a lot of muscle, indeed not enough, and not enough innervation to the muscle to spasm with shivering, fat tissue becomes the primary mechanism of warming the body. So as newborns, we actually have a surprisingly high amount of brown fat. Then as we get older and our muscles get bigger, the body decides, well, we don't need that wasteful heat-producing fat. And we get, as muscle expands, brown fat goes away. And so then we're left with only a little bit of fat just up in our chest area. We don't need it as much anymore because the muscles have taken care of it.

- 7.18** However, that doesn't mean brown fat becomes irrelevant. In humans that either have more brown fat or have more active brown fat, we generally find that they are more resistant to obesity and diabetes and metabolic problems. In other words, they have an easier time staying lean and metabolically healthy. So even though we don't have quite as much relative brown fat as we did when we were newborns, we still want it as a metabolic advantage, to help us be lean and again, metabolically sound.
- 7.19** With that topic, with that idea in mind, my lab has actually published a series of manuscripts that I just can't help but mention because it's so salient to the topic, where we found that ketones actually elicit this conversion, a slight conversion of white fat, which is generally metabolically inert, very low metabolic rate, and makes the white fat start behaving a little more like brown fat. So the metabolic rate in humans, as we measured it in our studies, went up by three times. So metabolic rate increased by three times in fat tissue when people were in ketosis or had elevated ketones.
- 7.20** In contrast, and we will revisit this idea later when we highlight other people's research, insulin will suppress the metabolic rate even in brown fat cells. So insulin wants to turn the metabolic rate down, whereas ketones want to turn the metabolic rate up. So there are these contrasting signals. Now, let's leave brown adipose tissue behind and start talking about the majority of our fat, namely the white fat, as it is divided between our subcutaneous fat depot, the pinchable, jiggable fat, versus the visceral adipose, the fat tissue that's woven through our internal organs.
- 7.21** Now, within the topic of energy storage, I wanted to address this two different ways, which is not only how much fat the body is being told to store, but also where the body is being told to store fat. Much of the debate, and indeed, we will wade into these waters when it comes to fat tissue is how much fat the person is storing. But before we do that, I wanted to discuss where the body is storing fat, because it's an interesting topic.
- 7.22** And to really put a fine point in it, insulin determines how much fat the body stores, and we will talk more about this in a few minutes. But another enzyme, another signal, is what tells the body where to store fat. Those are not the same thing. So insulin is dictating how much fat the body is storing. But then another signal called lipoprotein lipase is determining where the body stores fat. Now, you'll recall from an earlier lecture that I discussed that, whenever you hear that suffix -ase, A-S-E, we're talking about an enzyme, a molecule that is mediating some effect. It's turning some process on.
- 7.23** Lipoprotein lipase is expressed at fat cells throughout the body. And wherever we have more lipoprotein lipase, we will have more fat storage. And a primary stimulus for where we express lipoprotein lipase is sex hormones. This is why men and women look so differently as they go through puberty and end up as adults, as opposed to when they're just children. If you look at a little boy and a little girl, you generally couldn't tell

the difference between them if their genitals were covered. They have the same overall body type.

- 7.24** However, when they start to go through puberty, estrogens begin to dictate the expression of lipoprotein lipase at prototypical female fat storage sites, specifically breasts, buttocks, hips. So little girl starts to look woman. In contrast, the androgens, which are substantially higher in the boy, are beginning to express more lipoprotein lipase centrally. That's it. So generally, as his body is gaining fat mass, he's gaining it only around his belly. In contrast, as her body is gaining fat mass, she's generally gaining it on the butt and hips because lipoprotein lipase is determining where the fat is being stored.
- 7.25** Now, in the absence of any sex signal, of a sex hormone signal, and there are some pathologies where a person isn't making sex hormones, the body will actually default to a generic female pattern. It wouldn't necessarily mean the man is storing a lot of fat in his chest as breast tissue, but it will typically mean there's more fat being stored on the buttocks and hips. So there's a kind of a default generic fat pattern there.
- 7.26** Now, regardless of where the person is storing their white fat tissue because that's the type of fat we're talking about, as there is more and more fat being stored, there is ever more of a particular hormone being released, namely leptin. So let's transition into just briefly talking about the endocrinology of the adipose tissue. I know I just got done talking about the endocrine system, but humor me. What is so fascinating about the fat tissue is that fat actually is a highly active endocrine organ. It secretes dozens and dozens—more than 50 hormones have been identified that come from fat tissue. So this is a fantastically active endocrine organ.
- 7.27** And one of the most famous is leptin. And as I mentioned, leptin will go in the same direction as body fat. So as we have more and more body fat, we have more and more leptin. Now, you've heard of leptin before, and you've probably heard that it is a satiety hormone. I just wanted to briefly share with you part of the problem with relying too much on leptin when it comes to understanding body fat levels.
- 7.28** So leptin was first discovered in a group of mice within this massive lab. They had all these colonies of mice. One colony had an animal that developed a spontaneous mutation in a gene, indeed, the gene that encoded for leptin. At the time, leptin was not discovered. What they found in this line of animals is that they would become fantastically obese, in fact, adorably fat. Normally, mice are very, very lean, and it's hard to make them fat. This little strain or this colony of mice became fantastically fat and just insatiable appetites. They were constantly eating and constantly getting fatter.
- 7.29** Now, of course, this lab wanted to understand what had happened with these animals. They found that there was some protein in their blood that was missing, that other animals had plenty of. And they named it after the Greek word for thin, "leptos", they

named this signal leptin, because they found if they took that protein and injected it into these animals, they got lean.

- 7.30** And this exploded into this massive, tens-of-millions-of-dollars drug deal, thinking we have found the cure for obesity, because in these animals, if we give them leptin, we correct the obesity. And indeed, if you give leptin to a human who does not have leptin, which is a very, very rare genetic disorder, they will get very lean very quickly. Now, however, the problem is, when they started giving the leptin to normal obese adults, it did absolutely nothing.
- 7.31** The whole thing was a fantastic waste of money, a huge loss for this drug company, and there's no longer any research in that area at all, because the reality is, most people who struggle with obesity or overweight have plenty of leptin. Leptin just isn't working particularly well. Remember what I said about insulin resistance, how one of the key causes of insulin resistance is too much insulin? The same thing happens with leptin. Too much leptin creates a leptin resistance. And so even though they were giving the leptin to the obese humans, it did nothing because they already had plenty of leptin.
- 7.32** Now, again, to be clear, if you give leptin to someone who has a mutation in their leptin gene, which can happen, albeit very rarely, it will work perfectly, and indeed, they need it. It is almost a life-saving therapy. Now, leptin often is only discussed in the context of turning the brain hunger centers off. Or in other words, we only consider leptin as being a satiety signal. But leptin has myriad effects throughout the body, stimulating fat breakdown, stimulating muscle glucose uptake, so a lot of very good things, and stimulating the beta cells, or rather, to be a little clearer, inhibiting the beta cells.
- 7.33** So when leptin is coming from a fat cell, it wants to turn the beta cell off, to have it inhibit insulin. And this is important because in a leptin mutation state, human or animal, if you don't have leptin, you have a much, much harder time turning off insulin. So even those first scientists who first discovered leptin, one of their quotes at the time was that insulin levels "go through the roof." So these animals constantly have elevated insulin. And that will matter when we get to this next topic now, which is how does fat grow?
- 7.34** Now, before I talk about the relevance of insulin in fat growth, I want to just revisit the two methods in which the body is gaining fat mass, which was, if you recall, hyperplasia and hypertrophy. And we have a normal state of hyperplasia in the early years of our life, and then hyperplasia turns off, and any future fat gain is occurring through hypertrophy. So we have two different methods of getting fat— hyperplasia, or gaining fat mass, hyperplasia, which is when the fat cells are all pretty small but abundant, versus hypertrophy, which is when the number of fat cells is static, but the size of the fat cells is growing.
- 7.35** Unfortunately, it is the latter that explains how most people get fat. Now, some people have a genetic ability to make fat cells, but that is uncommon. That is only about 10%

of people who are obese. Most people who continue to gain fat through adulthood have the hyperplasia turned off, and they're getting fat through hypertrophy. Again, that is unfortunate because that creates metabolic problems for two reasons. So first and foremost, you'll recall that I said that the fat cell is capable of substantial growth. It can expand 1,000 times over its original size.

- 7.36** And in fact, it begins to reach a point of such maximum dimension that it cannot grow anymore. The membrane will start to literally break apart, if it continues to grow. It's like a balloon that we filled to the maximum dimensions with water. Any more effort to try to put any more fluid in there is going to pop it. The fat cell follows a similar pattern. Insulin has been promoting this growth and growth and growth, not allowing it to be breaking down its fat. But the fat cell gets to the point where it basically runs out of room. It has no more vacancy. It can handle no more energy.
- 7.37** And so even though insulin as a signal is telling the fat cell to grow, grow, grow, it says, "Insulin, I can no longer listen to you because if I continue to grow, I'm going to pop. And that will be very bad for everyone. That's not a neat, tidy cellular death. It'll be messy. I have to stop listening to you and stop growing." And so it does stop growing. Having reached this point of maximum dimension, it becomes insulin resistant to stop any future growth. So it's doing so, it's becoming insulin resistant to ensure its own survival, but that begins to result in a problem, being that it's starting to leak free fatty acids.
- 7.38** It's breaking down fats, even though insulin is still forcing more in and insulin levels are high. So it becomes insulin resistant to stop future growth. At the same time, as the fat cells are expanding to 1,000 times their normal size, they begin to cause some suffocation of fat cells themselves. So normally, a cell must be within what's called a few microns or micrometers of a capillary. You'll recall the capillaries from earlier discussions, where that's the business unit of the cardiovascular system, where we are exchanging oxygen and taking in CO₂ from the blood, between the blood and the cells. Again, a cell must be in very close proximity to a capillary.
- 7.39** However, as the cells begin to expand 1,000 times, they start to push each other further and further away from capillaries. In so doing, the cells essentially begin to suffocate, or to use the technical cellular term, they experience hypoxia. Now, they don't want to die from hypoxia. And so as an adaptation, the hypertrophic, now-suffocating fat cell begins to leak out a host of pro-inflammatory proteins called cytokines, because some of these cytokines will signal the growth of new blood vessels. So new capillaries will start to bud off and grow over to where these other fat cells are.
- 7.40** That's a life-saving effect for the fat cell itself. However, the consequence now is that there's more and more inflammation throughout the body, all because of the really big fat cells. This is a problem that is called subacute inflammation. So it's not like the person's suffering from an overt illness. But if you measure inflammatory markers in their blood, they're significantly higher than in a lean person, all because the fat cells

have just become aggressive at releasing these pro-inflammatory proteins, not to promote an immune response, but to promote the growth of new blood vessels.

- 7.41** But as a result, the rest of the body now has an additional insulin resistance signal, because you'll recall, when we talked about insulin resistance in the previous lecture, inflammation was a primary cause of insulin resistance. So I've mentioned now that insulin is a key signal that promotes fat growth. Now, just as an interesting point in this topic, you are familiar, of course, with type 1 diabetes, which we've not talked about. Type 1 diabetes is a disease of the immune system destroying the beta cells of the pancreas. And the beta cells are the cells that make insulin.
- 7.42** So we have a person who must get insulin through a syringe. And the typical advice to a type 1 diabetic, who now must take a needle of insulin for the rest of their lives all the time, they are told to rotate their injection sites. "Do one injection here on the leg. Do one injection on this part of the belly. Do one injection on this part of the flank of the back." Rotate the sites, because if they don't, if all of that insulin continues to go to the same spot, it induces such dramatic growth that the person starts to have these lumps of fat.
- 7.43** So if the person is not rotating sites, they get these big balls of fat that are just protruding from the skin. It looks like they have tumors, but it's just big balls of fat because all of that insulin being injected only into that site consistently has resulted in the exaggerated growth of the fat cells. And we know, based on conducting adipose biopsies, actually pulling out needle pieces of fat, if you look at the fat at the site of a chronic insulin injection and compare the fat cells, just a few centimeters over to fat cells that haven't been getting the insulin injection, you see that 100- to 1,000-fold difference in size.
- 7.44** Insulin promotes hypertrophy, and it promotes it substantially. So there is another layer of nuance and complexity at this point in the conversation. And I want to use this as an opportunity to bring in this new idea called the personal fat threshold. The personal fat threshold is this idea that, based on a person's ability to make more or less fat cells, to turn on hyperplasia or not, they're going to reach a point where their fat cells get too big, and they can no longer store fat in a healthy way, because that, after all, is what fat tissue is for.
- 7.45** Fat tissue serves a fundamental purpose. It allows us to store more fat, to store the energy in our fat tissue rather than storing it in things like our muscle or our liver, what's called ectopic fat deposition. So when we reach this point of maximum dimension through hypertrophy, you could say that a person has reached their threshold. And any effort to try to store more energy, more fat, is pushing them over into this realm of insulin resistance.
- 7.46** Now, there is a fascinating ethnic disparity in the ability to store fat. And so different races or ethnicities have a different propensity for more or less hyperplasia. Case in

point, there was a study published a number of years ago that took Caucasian European men and compared them with South Asian or Indian men, and found men that were the exact same fatness, exact same amount of fat on their body, same percentage body fat. And then compared, they were both, all groups, the Caucasian, European, the Indian, South Asian fellows, exact same fatness, age, matched for fatness, matched for age and habits and everything.

- 7.47 And they found that, even though, when they were controlled for overall fat mass, the South Asian men tended to have much, much higher rates of insulin resistance and associated problems. So they were the same fatness, and yet they had all of these metabolic problems that come from insulin resistance, like prediabetes and hypertension and fatty liver disease, all of these derivative problems from insulin resistance that we've not talked about, but I do talk about in my other lecture series of why we get sick.
- 7.48 Now, what they found then, when they actually did fat biopsies across these ethnicities, these two different ethnic groups, is that the Caucasian men tended to have more hyperplasia, so more fat cells, but smaller. But it was the same total fat mass in the South Asian men who had fewer fat cells, but much, much larger. So their fat cells were substantially larger than the fat cells in the Caucasian men. But in some, they had the same amount of fat mass.
- 7.49 Based on these two different populations, who would you say has the higher threshold? So who could store more fat and yet stay metabolically healthy, despite that fat mass? Of course, it was the Caucasian group. Different ethnicities have a different propensity to store fat in a healthy way, which is basically the threshold. So that is the personal fat threshold. And again, the threshold is not met by a certain mass of fat, but rather the size of each individual fat cell.
- 7.50 So that is the question. When we want to understand the impact of fat tissue on overall health, it really is a matter of how big are the fat cells. So let's transition into that topic as we start to wrap up the lecture on adipose physiology and the adipose system. What is the signal that tells fat cells to expand? Well, there are two signals. Now, this is a matter of debate, but you are fortunate to be getting this from an expert on the topic. So there are two essential signals.
- 7.51 Some, and I would say naive individuals, say that it's purely a function of calories. It is simply energy. If there's more energy coming in than going out, so count your calories and monitor your steps and your physical activity, and if there's more coming in than you're determining going out, then you're going to gain fat. So that's one component, though, and it is only one of two. You must have the other component, which, as I've alluded to, is elevated insulin. In fact, let me make this exceptionally clear. There is no fat growth unless insulin is elevated.

- 7.52** Now, back at my lab, in the metabolism research lab that I run, we are growing fat cells very often, in little petri dishes, a classic experiment and experimental model. So we have fat cells that we've laid on the bottom of this little petri dish. And then within the medium, which is the liquid that the fat cells are bathed in, we have plenty of calories. There is fats in that medium, and there is glucose in that medium, which are the two sources to fuel the growth of fat cells.
- 7.53** And yet, the fat cells stay skinny, for lack of a better word. They are not big at all. They're normal, teeny little cells, indistinguishable from any other cell type, until we do one thing. We spike in insulin. When we put insulin into that little bath, that soupy culture medium, now all of a sudden, the fat cells know what to do with the energy, which is take it in and store it. And so if we look 24 hours later, they're suddenly bigger and juicier with a big fat blob in the middle of it. If we look 24 hours later, they're even bigger still, and then even bigger still.
- 7.54** And for the next few days, they've finally got the message, which is, "You want me to store this. I didn't know, until insulin told me to." But that is such a fundamental and important lesson to understand metabolism and just to appreciate the relevance of the endocrine system. But when it comes to energy, a cell doesn't know what it's supposed to do with that energy. It needs a hormone to tell it what to do. And no hormone can match the relevance of insulin in this process.
- 7.55** So in order for a fat cell to grow, you both need the signal to tell it to grow, namely elevated insulin. And then you need sufficient energy or calories to fuel the growth. So insulin tells the fat cell to grow, and the calories enable the growth. Now, what if then you took a scenario where calories are abundant, indeed in excess, but the insulin is very, very low? What happens then? I just got done saying definitively that, if insulin is low, a fat cell cannot grow. And so how do we reconcile this? What happens to all of that energy?
- 7.56** After all, Professor Bikman, the laws of thermodynamics are immutable. I agree, they are immutable, but we have to account for the energy. This topic became very interesting to me when I stumbled on the work of two legendary scientists. And what I found from their work was that, in the absence of an insulin signal, the body will find ways to waste the energy. And the two famous scientists—they're scientists, even though they're famous to me, most people would never have heard of them—Elliot P. Joslin is the godfather of modern endocrinology, and Francis G. Benedict is the godfather of our understanding of metabolic rate.
- 7.57** Both of them are still, in fact, discussed in either the Benedict equation, as an estimate of metabolic rate, or the Joslin Diabetes Center in the East coast in the United States, a famous endocrinology clinic. So their influence still lingers. These two geniuses came together to explore the metabolism in people with severe diabetes, which we would just call type 1. They didn't call it type 1 or type 2 at the time.

- 7.58** They found that people with type 1 diabetes—and remember, the fundamental problem in type 1 is that they have no insulin—their metabolic rate was 20% higher than it should have been based on their body size. And they found, in follow-up research, other scientists found that if you gave the type 1 diabetic a load of insulin, metabolic rate slowed down that 20% and went right back to where you'd expect it. Now, in some instances, people with type 2 diabetes are given insulin, and the same thing happens. If you bump up their insulin, metabolic rate slows down.
- 7.59** Importantly, the same thing happens even in non-diabetics. You can take humans that are non-diabetic, give them two meals that are identical in calorie number, but they differ in their ratio of fats to carbohydrates. And the high-carb version elicits a bigger insulin spike, and they have a more substantial reduction in their metabolic rate. In fact, as you compare the high-carb versus the low-carb, but matched in calorie number, at the end of the day, the low-carb-eating group will have had an increase of about 300 calories a day in their total energy expended.
- 7.60** And at the same time, if a person has low insulin, they're making lots of ketones because they're burning so much fat, and they start emitting the ketones in their breath and in their urine. And every ketone has a caloric value roughly similar to glucose. Now the person in a low insulin state has both an increased metabolic rate, accounting for the energy being burned, and they are now just wasting energy in the form of just dumping ketones from their body. Remember, each ketone has a caloric value.
- 7.61** This fact is so obvious to people with type 1 diabetes that some type 1 diabetics are tempted to abuse it. There's an eating disorder called diabulimia, where the type 1 diabetic has learned that they can eat whatever they want. They don't have to starve themselves. They don't have to vomit the food up. They can eat whatever they want and enjoy the entire sensation of getting it to their stomach and digesting it. All that satisfaction. And all they need to do is not inject themselves with a syringe of insulin. They will stay as skinny as they want.
- 7.62** Diabulimia, imagine the temptation to do this. Now, of course, there's a metabolic storm going on in their body. Their glucose levels are 10 times too high. Their ketone levels go into the realm of making the body very acidic. So they're very sick, and it will be very lethal, ultimately. But they'll be as thin as they want, simply because they've learned, "I can eat any amount of calories. And if I just underdose my insulin, I'll be as skinny as I want." That is the ultimate proof just indicating the unavoidable role of insulin in telling the body what to do with the energy.
- 7.63** And the lower insulin gets, the more impossible it becomes for a fat cell to stay big. Indeed, it must shrink and it cannot gain, regardless of the amount of calories consumed. OK, now, the final topic I want to mention is, in helping us understand the importance of body fat, is just mentioning liposuction. Now, liposuction is a surgical

procedure that, of course, is born from our hatred of body fat. We curse our body fat, and we don't want it.

- 7.64** The problem with liposuction is that it takes out the most metabolically benign type of, indeed, the most metabolically healthy fat. The best place to store fat is the subcutaneous fat, the pinchable, jiggable type of fat. That's the best place to store it. It's healthy. It's healthy to store it there, which is why women will uniformly be both fatter than their male counterparts, but healthier. Because they store fat almost exclusively in the subcutaneous fat depot of the buttocks and hips, whereas men will tend to store more fat viscerally, which is more problematic.
- 7.65** The problem with liposuction is that it will not touch any fat depot except the subcutaneous fat. That's the fat that is going in and literally yanked from the body. Now, remember what I said earlier. If the metabolic problems that come from fat mass were just purely a function of the fat mass itself, then removing fat mass from the body would be a benefit. But I didn't say that. If you recall, I said that the problem is the size of the fat cells. It's not the total mass of fat. And so as you start sucking out body fat, you're not changing the size of any of the fat cells.
- 7.66** And indeed, if the person hasn't changed their dietary habits, if they took out 10 pounds of fat, the body still wanted to store that fat because that fat was there because of the insulin being spiked and the sufficient calories to fuel the fat growth. So they start gaining fat back in other places. So if a woman has gone in and removed fat from her buttocks and hips, she starts to gain it at her belly. This is well known. But suffice it to say, regardless of any fat regain, what happens is no improvement in any clinical outcome.
- 7.67** So someone who's undergone liposuction, they may be 10 or 20 pounds lighter than they were before they went in. They may look a little better in their skimpy bathing suits, but not a single clinical outcome has improved. They are just as susceptible, if not more so, to metabolic complications than they were before. Because now, as you have removed fat tissue, fat cells, remember, you don't make new fat cells. So the remaining fat cells are forced to carry a larger metabolic burden.
- 7.68** And so you end up inadvertently, while you're a little leaner, your risk of metabolic disease and insulin resistance has actually gone up because all of the remaining fat cells, as I said, have to carry a larger metabolic burden and undergo more readily a degree of hypertrophy in the absence of being able to share the burden with other fat cells.
- 7.69** That concludes the adipose tissue system. As we have gone through the factors that determine how the body is storing fat, where the body is storing fat, and some of the products of that fat storage, and the intricate differences between the different places and different types of fat storage, white to brown, and then the different depots, you

are much more informed of this oft-overlooked part of the physiological systems, and I thank you for your attention.